## Hydrogen Radial Wavefunctions

The Hydrogen atom is special because it has electronic states and properties that scale with the principal, n, and orbital angular momentum, $\ell$ quantum numbers, in a simple and global way. This is "structure" that is more than a collection of unrelated facts. The H atom serves as our model for "electronic structure" of many-electron atoms, molecules, and possibly solids.

By showing how E, $\langle r^{\sigma} \rangle$ (size and shapes), $\langle n\lambda | r | \nu' \ell' \rangle$ (general matrix elements) scale with the n and $\ell$ quantum numbers, it tells us the kind of behavior to look for and how to understand it in more complex systems.

* as a perturbation on the Hydrogen atom (quantum defects)
* as a hint of relationships that are useful for extrapolation, assignment, and for recognizing when something behaves differently from naive expectations.

TODAY

0. A warning.
1. Simplified Radial Equation − a trick to make it look exactly like the 1D problems we have already solved.
2. Boundary conditions as $r \to 0$ and $r \to \infty$
3. qualitative features of $R_{n\ell}(r)$
4. n-scaling of $\langle r^{\sigma} \rangle$ , where $\sigma$ is an integer
5. mathematical form of $R_{n\ell}(r)$
6. regular and irregular Coulomb functions

**Warning**: This first lecture on the Hydrogen atom is not what you expect. Most textbooks treat the H atom as an exactly solved problem. The solution is based on standard techniques for solving differential equations and the solutions are expressed in terms of named and tabulated special function. The exact and analytic solution for the H atom is presented as a unique, beautiful gem to be admired and stored in a safe place, out of view and out of use. What I present here is based on techniques and approximations discussed in previous lectures. My goal is to arm you with methods to answer questions that apply to many-electron atoms, molecules, and to quantum number "scaling-rules" that provide a framework for describing intra-species and inter-quantum-state patterns of observable properties. These quantum number scaling ideas are mostly based on the stationary phase approximation (in what local region of space does an integral accumulate to its final value, what does the WKB approximation tell you about wavefunction amplitudes and node-spacing in this special region) and perturbation theory. These are techniques and ideas that you will use throughout your career to build insights that will amaze your colleagues.

For *every* central force problem

$$\hat{\mathbf{H}} = \left[ \frac{\hat{\mathbf{p}}_r^2}{2\mu} + \frac{\hat{\ell}^2}{2\mu r^2} \right] + V(r)$$

*We know that $\hat{\mathbf{H}}, \hat{\ell}^2$, and $\hat{\ell}_z$ commute, so spherical harmonics, $Y_\ell^m(\theta,\phi)$, are eigenfunctions of $\hat{\mathbf{H}}$ with eigenvalues $\hbar^2\ell(\ell+1)$.

*$\psi(r,\theta,\phi) = R(r)Y_\ell^m(\theta,\phi)$

*trial form for separation of variables in $\psi$

$$\hat{\mathbf{H}}\psi = \left( \frac{\mathbf{p}_r^2}{2\mu} + \frac{\ell^2}{2\mu r^2} + V(r) \right) Y_\ell^m(\theta,\phi) R(r) = E\psi$$

Operate on the $Y_\ell^m(\theta,\phi)$ angular wavefunction and move it through to left:

$$\hat{\mathbf{H}}\psi = Y_\ell^m(\theta,\phi) \left( \frac{\mathbf{p}_r^2}{2\mu} + \underbrace{\frac{\hbar^2\ell(\ell+1)}{2\mu r^2} + V(r)}_{V_\ell(r)} \right) R(r) = E\psi$$

So we can take $Y_\ell^m(\theta,\phi)$ out of the Schrödinger Equation and we are left with a 1-D radial equation where the only remaining hint of the angular part is the $\ell$-dependence of $V_\ell(r)$, the *effective potential energy function*.

Since the differential equation depends on $\ell$, R(r) must also depend on $\ell$, thus $R_{n\ell}(r)$ is the radial part of $\psi$, and it will generally be an explicit function of two (not 3) quantum numbers, $n$ and $\ell$.

Usually $n$ specifies the number of radial nodes and $\ell$ the number of angular nodes, but a special numbering convention for Hydrogen (and hydrogenic ions) causes a slight distortion of this rule.

The radial equation, when the explicit differential operator form of $\mathbf{p}_r^2$ is derived and inserted, has the form

$$\left\{ \underbrace{\left[ -\frac{\hbar^2}{2\mu}\frac{1}{r}\frac{d^2}{dr^2}r \right]}_{\mathbf{T}_r} + \underbrace{\left[ \frac{\hbar^2 \ell(\ell+1)}{2\mu r^2} + V(r) \right]}_{\mathbf{V}_\ell} \right\} R_{n\ell}(r) = E_{n\ell} R_{n\ell}(r)$$

It is customary and informative to simplify this equation by replacing $R_{n\ell}(r)$ by $\frac{1}{\mathrm{r}} u_{n\ell}(r)$

$$R_{n\ell}(r) = \frac{1}{r} u_{n\ell}(r)$$

* the resulting equation looks simpler
* the resulting volume element looks simpler
* the behavior as $\mathrm{r} \to 0$ seems more familiar

Insert $\frac{1}{r} u_{n\ell}(r)$ in place of $R_{n\ell}(r)$ and then multiply through <u>on left</u> by $r$, we get

$$\left[ -\frac{\hbar^2}{2\mu}\frac{d^2}{dr^2} + \frac{\hbar^2 \ell(\ell+1)}{2\mu r^2} + V(r) - E_{n\ell} \right] u_{n\ell}(r) = 0$$

This looks like ordinary 1-D Schrödinger Equation.
Boundary condition:

$u_{n\ell}(r) \to 0$ as $\mathrm{r} \to 0$

<u>WHY? Because for all $\ell > 0$, $V_\ell(0) \to \infty$.</u>

exactly as if $V(r) = \infty$ for $r \le 0$, but of course $r < 0$ is impossible, so we had better be careful about behavior of $u_{n\ell}(r)$ and $R_{n\ell}(r)$ as $r \to 0$

Note also that the volume element is $dV = r^2 \sin\theta dr d\theta d\phi$

and the r-dependence of the integrand is $R^*_{n'\ell'} R_{n\ell}\, r^2 dr = u^*_{n'\ell'}(r) u_{n\ell}(r) dr$

the $r^2$ factor is cancelled. So the volume element looks just as in 1-D problems

Return to special situation as $r \to 0$.

Why do we care? It turns out that only s-orbitals have $R_{ns}(0) \neq 0$ and that in ESR one measures "Fermi-contact" hyperfine structure which is related to the spin-density at specific nuclei. It is a direct measure of the *ns* atomic orbital character in each molecular orbital!

CTDL, p. 781: What is the worst possible divergence of $R_{n\ell}(r)$ as $r \to 0$?

For $r \to 0$, $R_{n\ell}(r)$ will be dominated by $r^s$ where the exponent $|s|$ is as small as possible. This is the most strongly divergent part of $R_{n\ell}(r)$, which is all we need to be concerned with as $r \to 0$.

Let $R_{n\ell} \sim Cr^s$, where this is a good approximation as $r \to 0$.

Plug this definition into the Schrödinger Equation

$$\frac{d^2}{dr^2} r R_{n\ell}(r) = \frac{d^2}{dr^2} C r^{s+1} = (s+1)(s) C r^{s-1}$$

$$\mathbf{T}_r = -\frac{\hbar^2}{2\mu} \frac{1}{r} \frac{d^2}{dr^2}$$

$$\widehat{\mathbf{H}} R_{n\ell}(r) = -\frac{\hbar^2}{2\mu} C(s+1)(s) r^{s-2} + \frac{\hbar^2 \ell(\ell+1)}{2\mu} C r^{s-2} + V(r) C r^s - E_{n\ell} C r^s = 0$$

if $V(r) \propto \frac{1}{r}$

[As $r \to 0$ $V(r)$ rarely diverges more rapidly than $1/r$, thus $V(r) C r^s$ gives $r^{s-1}$.]*

Then, in the limit $r \to 0$, the coefficients of the $r^{s-2}$ term (i.e. the most rapidly divergent term) must be = 0

$$-(s+1)s + \ell(\ell+1) = 0$$

*This excludes the stronger divergence of the centrifugal barrier term in $V_\ell(r)$, which is included separately.

Satisfied if $s = \ell$ or $s = -(\ell + 1)$

explore the second possibility:

$$s(s+1) = (-\ell - 1)(-\ell - 1 + 1) = -(\ell + 1)(-\ell) = \ell\,(\ell + 1)$$

If $s = \ell$, the $r \to 0$ form of $R_{n\ell}(r) = cr^{\ell}$ which is OK for all $\ell$.
If $s = -(\ell + 1)$, the $r \to 0$ form of $R_{n\ell}(r) = cr^{-(\ell+1)}$ diverges for all $r$.

In other words $R_{n\ell}(r) \to \underbrace{r^{\ell}}_{\text{well behaved at } r \to 0}$ OR (if $s = -(\ell+1)$) $\underbrace{\frac{1}{r^{\ell+1}}}_{\text{disaster even if } \ell = 0}$ as $r \to 0$

Actually, both of these possibilities satisfy the differential equation in the limit $r \to 0$ for $V(r) = \frac{1}{r}$ (known as the Coulomb — or H atom Hamiltonian), but the one that diverges as $r \to 0$ cannot satisfy the $r \to 0$ boundary condition for the H atom, but it turns out we will need both types of solution.

** "Regular" and "Irregular Coulomb" wavefunctions – we will return to these later in the context of Quantum Defect Theory.

So, for now, we insist that

$$R_{n\ell}(r) \to r^{\ell} \quad \text{as} \quad r \to 0$$

$R_{ns}(0) \neq 0$ special situation for $R_{ns}$(r)

$R_{n\ell>0}(0) = 0$

$u_{n\ell}(0) = 0$ for all $\ell$

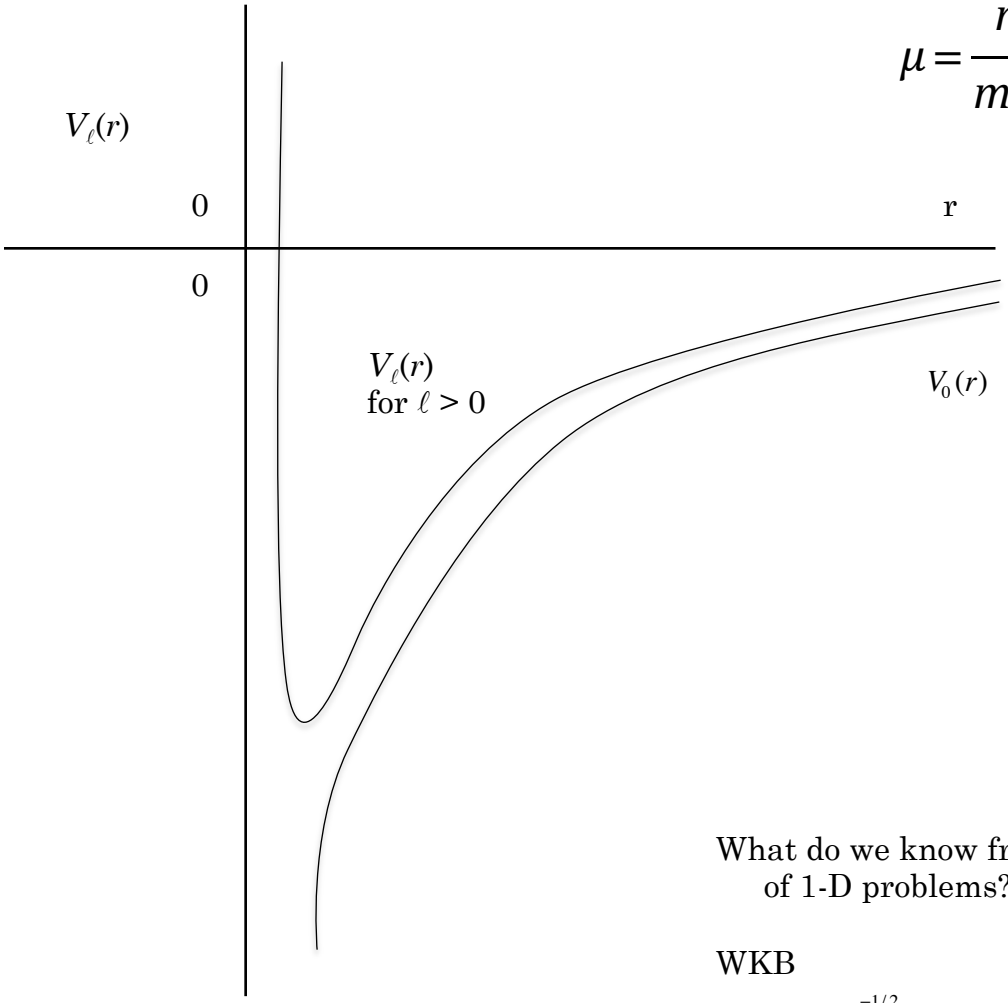
(no special case for $u_{ns}(r)$!)

For Hydrogen

$$V_\ell(r) = +\frac{\hbar^2 \ell(\ell+1)}{2\mu r^2} - \frac{e^2}{r}$$

$$e^2 \equiv \frac{q^2}{4\pi\varepsilon_0}$$

$$\mu = \frac{m_e m_p}{m_e + m_p} \approx m_e$$




**ASK QUESTIONS**:
What are the qualitative shapes of $u_{n\ell}(r)$ in inner-most and outermost lobes?

What do we know from our study of 1-D problems?

WKB

$\Psi_{\text{envelope}} \underset{\sim}{\propto} p^{-1/2}$

$\frac{\lambda}{2} = \frac{h}{2p(r)}$

# of nodes, placement of nodes, degeneracy, behavior at inner and outer turning points, location of inner and outer turning points

inner vs. outer part of $u_{n\ell}(r)$ — where is the extra $\frac{h}{2}$ of action $\left( J = \oint p(r)\,dr = h\left(n + \frac{1}{2}\right)\right)$ acquired (associated with tunneling into nonclassical region)?

$\left[\text{recall } \int_{r_<(E)}^{r_>(E)} p(r)\,dr = \frac{h}{2}(n + 1/2)\right]$

Find that $E_{n\ell} = -\frac{\Re}{n^2}$ $\Re_H = \frac{e^4 \mu_H}{2\hbar^2}$

$\mu_H = \frac{\left(m_{e^-}\right)\left(m_{p^+}\right)}{m_{e^-} + m_{p^+}} \simeq m_e$

At turning point $V_\ell(r_\pm) = E_{n\ell}$

$$-\frac{\Re}{n^2} = \frac{\hbar^2 \ell(\ell+1)}{2\mu r_\pm^2} - \frac{e^2}{r_\pm}$$

$$-\frac{\Re}{n^2} = \frac{\hbar^2 \ell(\ell+1) - r_\pm e^2 2\mu}{2\mu r_\pm^2}$$

$$-\frac{2\mu \Re r_\pm^2}{n^2} = \hbar^2 \ell(\ell+1) - r_\pm e^2 2\mu$$

Use Quadratic formula to find $r_\pm(n)$

Solve for $r_\pm$ as function of $n$ and $\ell$

$$r_\pm = a_0\left[n^2 \pm n\left(n^2 - \ell(\ell+1)\right)^{1/2}\right]$$

$$r_\pm = a_0 n^2\left[1 \pm \left(1 - \frac{\ell(\ell+1)}{n^2}\right)^{1/2}\right]$$

$a_0 = \frac{\hbar^2}{e^2 m_e}$ Bohr radius

when $\ell \ll n$, where are $r_+$ and $r_-$? (see table on page 28-12)

Use this equation for the turning points to construct qualitatively correct cartoons of $R_{n\ell}(r)$ in crucial regions.

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surprising systematic degeneracy

etc.

3s ———— 3p ———— 3d ————

2s ———— 2p ————

$E_{n\ell}$

1s ————

Because of the pattern of degeneracies, we use $n$ to label degenerate groups rather than number of radial nodes

$E_{n\ell} = -\frac{\Re}{n^2}$

hence $n$ is <u>not</u> # of radial nodes.

| orbital | # of radial nodes | |
|---|---|---|
| 1s | 0 | |
| 2s | 1 | |
| 2p | 0 | (because it is the lowest energy solution to the $\ell = 1$ equation) |
| 3s | 2 | |
| 3p | 1 | |
| 3d | 0 | |

$$\#\text{ radial nodes } = \left(n - \ell\right) - 1$$

$$\#\text{ angular nodal surfaces} = \ell$$

total # nodes is $n-1$. What is the degeneracy?

| $n$ | degeneracy |
|---|---|
| 1 | 1 |
| 2 | $1 + (2\ell + 1) = 4 = 2^2$ |
| 3 | $1 + 3 + 5 = 9 = 3^2$ |
| ⋮ | |
| $n$ | $n^2$ |

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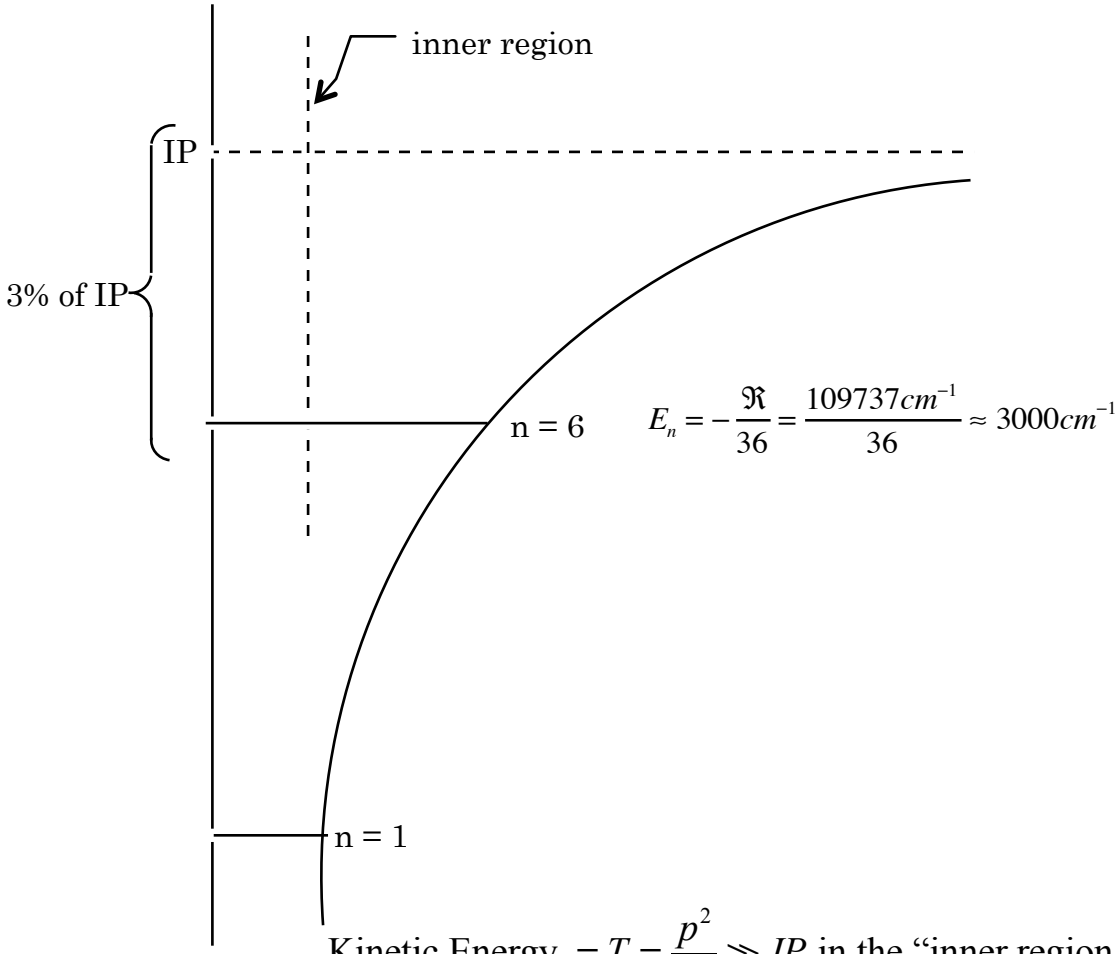
## $n$-scaling of $\langle r^\sigma \rangle$

| two limits | $\sigma < -1$ | vs. | $\sigma > 0$ |
|---|---|---|---|
| determined near: | inner turning point | | outer turning point |
| scaling: | $\sim n^{-3}$ | | Bohr model $r_{n\ell} = a_0 n^2$ |
| | (see argument for inner lobe on next page) | | $\langle r^\sigma \rangle \propto a_0^\sigma n^{2\sigma}$ see table on page 11 |

Expectation values of $r^\sigma$ vs. transition moments and off-diagonal matrix elements of $r^\sigma$. Stationary phase gives the essential insight!

↓

usually occurs in first (inner-most) lobe




$$E_n = -\frac{\Re}{36} = \frac{109737 cm^{-1}}{36} \approx 3000 cm^{-1}$$

Kinetic Energy $= T = \dfrac{p^2}{2\mu} \gg IP$ in the "inner region."

variation of $T$ from $n = 6$ to $n = \infty$ < 3%
variation of $p$ < 1.5%

deBroglie $\lambda \sim \dfrac{h}{p}$

$r = \dfrac{\lambda}{2}$ is the location of innermost node

} independent of $n$. Because $p$ is large and fractional change of $p$ vs. $n$ is negligible.

$e^-$ comes into core region fast and leaves fast — $\Delta t$ same for all n

fraction of time inside core?
Same as probability inside core.

$$\frac{\text{time inside}}{\text{one period}} = \frac{2(v/\lambda)^{-1}}{\left(\dfrac{h}{E_n - E_{n+1}}\right)}$$ $v$ is velocity

$$E_n = -\Re/n^2$$

$$E_{n+\delta/2} - E_{n-\delta/2} = \frac{2\delta\Re}{n^3}$$

$$\text{probability in inner lobe} = \frac{2\left[\dfrac{\dfrac{h}{p}}{v = p/m}\right]}{\dfrac{h}{2\Re/n^3}} \approx \frac{4m\Re}{p^2 n^3}$$

$\lambda$

$n$-independent (approximately for $n = 6$ to $n = \infty$)

probability of finding $e^-$ inside core $\mu\ n^{-3}$ !

fraction of time inside core region $\propto n^{-3}$

amplitude of $\psi_{n\ell} \propto n^{-3/2}$ inside core region

**This is the basis of all Rydberg scaling**

inner lobe { 1st node does not shift with n ; amplitude in first lobe scales as $n^{-3/2}$ } *Astonishingly* important!

All $n,n'$ matrix elements of $r^{\sigma}$, where $\sigma < -1$ scale as $\left(nn'\right)^{-3/2}$ ! what happens at $\sigma = -1$?

Some matrix elements scale this way even when $\sigma > 0$.

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McQuarrie, page 223

$$R_{n\ell}(r) = -\underbrace{\left[\frac{(n-\ell-1)!}{2n\left[(n+\ell)!\right]^3}\right]^{1/2}\left(\frac{2}{na_0}\right)^{\ell+3/2}}_{\text{normalization}} r^{\ell} e^{-r/na_0} L_{n+1}^{2\ell+1}\left(\frac{2}{na_0}\right)$$

$r^{\ell} e^{-r/na_0}$: exponential $\to 0$ as $r \to \infty$

$L_{n+1}^{2\ell+1}\left(\frac{2}{na_0}\right)$: associated Laguerre functions (polynominals)

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Regular and Irregular Coulomb functions ($E \le 0$)

$u_{n\ell}(r) \equiv rR(r)$ $\quad r \to 0$

increasing exponentially; $r \to \infty$ decreasing exponentially

regular $\quad f(E,\ell,r) \propto r^{\ell+1}$ (proper behavior as $r \to 0$)

$u(\nu,\ell,r)\sin \pi\nu - v(\nu,\ell,r)e^{i\pi\nu}$, which is an increasing exponential *except* when $\nu$ is a positive integer. Need some other way to satisfy $r \to \infty$ boundary condition when $\nu$ is not an integer

irregular $\quad g(E,\ell,r) \propto r^{-(\ell)}$ (bad news!)

$-u(\nu,\ell,r)\cos \pi\nu + v(\nu,\ell,r)\, e^{i\pi(\nu+1/2)}$, which blows up.

* $u(\nu,\ell,r)$ is an increasing exponential as $r \to \infty$
* $v(\nu,\ell,r)$ is a decreasing exponential as $r \to \infty$

(see Gallagher, Rydberg Atoms, page 16)

to behave properly as $r \to \infty$ we need to take linear combination of the f and g functions. Possible only at certain values of $\nu$ (we sometimes call this n*)

T.F. Gallagher, Rydberg Atoms, page 25

$\langle r \rangle \quad \frac{1}{2}\left[3n^2 - \ell(\ell+1)\right]$

$\langle r^2 \rangle \quad \frac{n^2}{2}\left[5n^2 + 1 - 3\ell(\ell+1)\right]$

$\langle 1/r \rangle \quad 1/n^2$

$\langle 1/r^2 \rangle \quad \frac{1}{n^3(\ell+1/2)}$

$\langle 1/r^3 \rangle \quad \frac{1}{n^3(\ell+1)(\ell+1/2)1}$

$\langle 1/r^4 \rangle \quad \frac{3n^2 - \ell(\ell+1)}{2n^5(\ell+3/2)(\ell+1)(\ell+1/2)\ell(\ell-1/2)}$

$\langle 1/r^6 \rangle \quad \frac{35n^4 - 5n^2\left[6\ell(\ell+1) - 5\right] + 3(\ell+2)(\ell+1)\ell(\ell-1)}{8n^7(\ell+5/2)(\ell+2)(\ell+3/2)(\ell+1)(\ell+1/2)\ell(\ell-1/2)(\ell-1)(\ell-3/2)}$

Note!

| | | |
|---|---|---|
| all $\langle r^\sigma \rangle$ | $\sigma < -1$ | scale as $n^{-3}$! |
| | $\sigma > 0$ | scale as $n^{2\sigma}$! |

**Outer and Inner Turning Points**

| Orbital | $r_+(n,\ell)$/Å | $r_-(n,\ell)$/Å |
|---|---|---|
| 1s | 1.058 | 0.00 |
| 2p | 3.612 | 0.624 |
| 3d | 7.508 | 2.012 |
| 4f | 12.696 | 4.232 |
| 5g | 19.137 | 7.313 |
| | | |
| 10s | 105.8 | 0.00 |
| 10p | 105.3 | 0.529 |
| 10d | 104.2 | 1.587 |
| 10f | 102.5 | 3.280 |

$$V_\ell\left(r_\pm\right) = -\frac{e^2}{r_\pm} + \frac{\hbar^2\ell(\ell+1)}{2\mu r_\pm^2} = E_{n\ell} = -\frac{\Re}{n^2}$$

in atomic units:

$$-\frac{1}{2n^2} = -\frac{1}{r_\pm} + \frac{\ell(\ell+1)}{2r_\pm^2}$$

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## Begin Many-e⁻ Atoms: Quantum Defect Theory

See MQDT Primer by Stephen Ross, pages 73-110 in Half Collision Resonance Phenomena in Molecules (AIP Conf. Proc. #225, M. Garciá-Sucre, G. Raseev, and S.C. Ross) 1991.

Last Time:

$$\text{turning points of } V_\ell(r) = -\frac{e^2}{r} + \frac{\hbar^2 \ell(\ell+1)}{2\mu r^2}$$

* $$r_\pm(n,\ell) = a_0 n^2 \left[1 \pm \left(1 - \frac{\ell(\ell+1)}{n^2}\right)^{1/2}\right] \approx a_0 n^2 \left[1 \pm 1 \mp \frac{\ell(\ell+1)}{2n^2}\right] \text{ for } \ell << n$$

* $u_{n\ell}(r) \equiv rR_{n\ell}(r)$ is dominated by a small lobe ($n$-independent nodal position) near the inner turning point, amplitude scales as $n^{-3/2}$, and a large lobe at the outer turning point (containing essentially all of the probability).

envelope
$u_{n\ell}(r) \propto p_{n\ell}(r)^{-1/2}$

* $$E_{n\ell} = IP - \frac{\Re}{n^2}$$

* nodes: $n - \ell - 1$ radial nodes
  $\ell$ angular nodes
  $n - 1$ total nodes
  $\bar{\lambda}/2$ gives spacing between radial nodes, $\lambda(x) = \dfrac{h}{p(x)}$

* expectation value scaling of $\langle r^\sigma \rangle$

| | |
|---|---|
| $\sigma < -1$ | $\propto n^{-3}$ |
| $\sigma \geq +1$ | $\propto n^{2\sigma}$ (see below) |
| $\sigma = -1$ | $\propto n^{-2}$ (H-atom energy levels) |

Geometric mean of expectation values of r for off-diagonal matrix elements:

$$\langle r \rangle_{n\ell} \propto n^2$$

$$\langle n\ell | r^\sigma | n'\ell' \rangle \propto \left[\left(r_{+n\ell}\right)^{1/2}\left(r_{+n'\ell'}\right)^{1/2}\right]^\sigma \approx \left[\left(n^2\right)^{1/2}\left(n'^2\right)^{1/2}\right]^\sigma$$

$$= \left(nn'\right)^\sigma \approx n^{2\sigma}$$

(when is $\langle r^n \rangle \approx \langle r \rangle^n$ ?)

TODAY

1. Many-e⁻ atom treated as a core plus an outer e⁻ that sees a partly shielded core as having a charge distribution Z(r).
2. $\ell$-dependent energy shifts → n-independent quantum defects $E_{n\ell} = IP - \dfrac{\Re}{\left(n - \mu_{n\ell}\right)^2}$ ($n \to \nu$)
3. energy shifts are actually phase shifts of $u_{n\ell}$(r) relative to $u_{n\ell}$(r) for H-atom
4. Rigorous QDT
   A. regular and irregular Coulomb functions *f,g* satisfy Hydrogen-like Schrödinger Equation <u>OUTSIDE</u> the core

   B. Boundary conditions at $r \to \infty$

   Noninteger values of $\nu = \left[-\dfrac{\Re}{E_{n\ell}}\right]^{1/2}$ require a sum of $f$ and $g$ Coulomb functions.

   Find the value of $\nu = n - \mu_\ell$ that satisfies the $r \to \infty$ boundary condition

   $\infty$ number of members in the series of $\nu$'s (effective principal quantum number) spaced by 1.000, $\therefore$ constant quantum defect

   C. $\pi\mu_\ell$ is a phase shift
   repeated patterns in each $\nu \to \nu + 1$ integer region of $\nu$

   D. Multi-channel QDT
   $\boldsymbol{\mu}$ matrices. The matrix $\boldsymbol{\mu}$ is a generalization of the quantum defect.

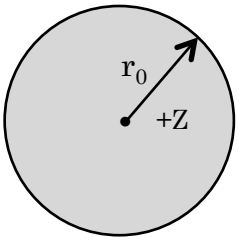
e⁻ colliding with core can also transfer energy and angular momentum to the core-e⁻

* "channels" rather than individual eigenstates

* focus on dynamics, but in a "black box" way. Dynamics happens within a restricted region of space. This region of space is *always* sampled, regardless of E, in the same way for every member of a Rydberg series. Everything is determined by the boundary conditions for the outgoing wave. SCATTERING THEORY rather than an EFFECTIVE HAMILTONIAN MODEL.

The goal here is to extract from a complicated many-body problem some regular features that will help in assigning, understanding, and modeling experimental data.

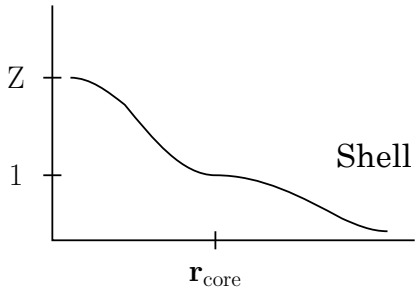
1. Many-e⁻ Atom




outside core e⁻ sees Z = +1
inside core e⁻ sees Z(r)

$e^-$

outer electron
(valence, Rydberg)




Shell Structure

ensures $n^{-3}$ rather than $n^{-2}$

as $r \to 0$, $\mathbf{H}^{(1)}$ diverges faster than $-e^2/r$ because Z(0) > 1.

2. $\ell$-dependent sampling of core
   high $\ell$ : see $\langle Z(r) \rangle \sim +1$
   low $\ell$ : see $\langle Z(r) \rangle = Z_\ell^{eff} >> 1$
   energy stabilization

$$\therefore E_{n\ell}^{(1)} = \langle n\ell | \mathrm{H}^{(1)} | n\ell \rangle \propto -\frac{|c_\ell|}{n^3}$$

$$\mathbf{H} = \mathbf{H}^{(0)} + \mathbf{H}^{(1)}$$

$$\mathbf{H}^{(1)} \approx -\frac{[Z(r)-1]e^2}{r}$$

$$\therefore E_{n\ell} = -\frac{\Re}{n^2} - \frac{|c_\ell|}{n^3} \approx \frac{-\Re}{(n-\mu_\ell)^2} \quad \therefore \quad \mu_\ell = \frac{|c_\ell|}{2\Re} << n$$

expand in Taylor series

call this $\nu$, "effective principal quantum number*"

so far we have focussed on $E_{n\ell}$

3. What does Z(r) do to $u_{n\ell}$(r)?

* outside core $e^-$ sees same $V_\ell(r)$ as Hydrogen atom
* must be same as $u_{n\ell}(r)$ for H except for phase shift inward (why inward?)
* all the unique stuff occurs inside the core — causes the phase shift.
  - nodal structure inside core is invariant with respect to n or E (the locations of the nodes are not dependent on $\nu$, but the amplitudes between nodes scale as $\nu^{-3/2}$)

Mulliken: "ontogeny recapitulates phylogeny"
intra-core nodal structure is $n$-independent
nodal structure encodes *all* $e^- \leftrightarrow$ nucleus dynamics!

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4. Do all of this more rigorously: Quantum Defect Theory

* regular Rydberg series, one series for each value of $\ell$
* $n$-scaling of inner lobe amplitude and of all matrix elements
* large quantum defects for small $\ell$
* entire Rydberg series and the associated ionization continuum ($e^-$ ejected in a specified $\ell$-partial wave) is a single entity

These are what we will obtain.

follow treatment by Ross, but not using atomic units:

redefine 0 of E $E_n = -\frac{\Re}{n^2}$ $\quad n = 1,2,\ldots$ for H, $E = 0$ at $n = \infty$

$$n = \left[-\frac{E_n}{\Re}\right]^{-1/2}$$

generalize to noninteger-$n$ for non-hydrogen: $\nu \equiv \left[-\frac{E_\nu}{\Re}\right]^{-1/2}$

use $\nu$ (effective principal quantum number, $\nu$) rather than E as a label for $u_{n\ell}(r)$

Schrödinger Equation for H (the "Coulomb Equation")

as a continuous variable rather than as an integer quantum number

$$\left\{-\frac{\hbar^2}{2\mu}\left[\frac{d^2}{dr^2} - \frac{\ell(\ell+1)}{r^2}\right] - \frac{e^2}{r} - \underbrace{E}_{E = -\frac{\Re}{\nu^2}}\right\} u_\ell(\nu, r) = 0$$

well known solutions:

2nd-order differential equation - two linearly independent solutions (at each $\ell$, $\nu$)

$$f_\ell(\nu, r) \to 0 \text{ as } r \to 0 \qquad \text{"regular"}$$

$$g_\ell(\nu, r) \to \infty \text{ as } r \to 0 \qquad \text{"irregular"}$$

of no use for Hydrogen, but it turns out that we need both $f$ and $g$ to satisfy $r \to \infty$ boundary condition when $\nu$ is non-integer

(We do not use f or g all the way in to $r \to 0$. We use them only outside some critical radius. So we are not concerned by the divergence of $g_\ell$ as $r \to 0$)

For the H atom, we have no use for $g_\ell(\nu, r)$ because it cannot satisfy the boundary condition as $r \to 0$

A. For many-e⁻ atoms, <u>beyond some critical</u> $r_0$, Schrödinger Equation is identical to that of H. The only difference is that we must treat the $r \to 0$ boundary condition differently.

Universal boundary conditions are $r \to \infty, u_\ell(\nu, r) \to 0$

*for* $E < 0, r \to \infty$, *asymptotic* forms for $f$ and $g$ are

$$f_\ell(\nu, r \to \infty) \to C(r)\sin[\pi(\nu - \ell)]e^{r/\nu}$$

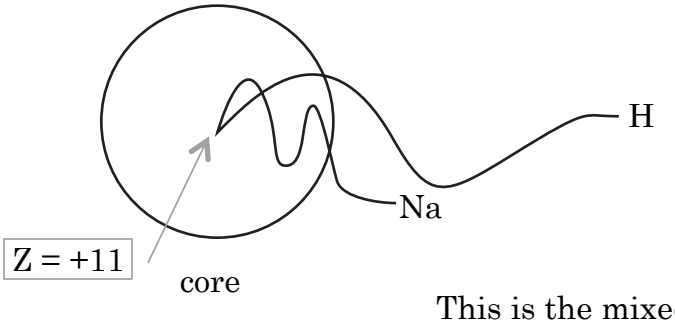
$$g_\ell(\nu, r \to \infty) \to -C(r)\cos[\pi(\nu - \ell)]e^{r/\nu}$$

$C(r) \to 0$ as $r \to \infty$

but $C(r)e^{r/\nu} \to \infty$ as $r \to \infty$, so the only way to satisfy the $r \to \infty$ boundary condition <u>for a pure</u> $u_\ell(\nu, r) = f_\ell(\nu, r)$ is for $(\nu - \ell)$=integer

└ regular Coulomb function

B. But we might want to use a mixture of $f_\ell$ and $g_\ell$ to deal with non-integer $(\nu - \ell)$, which we will need to deal with many-electron atoms.




For an Na atom, $u_\ell(\nu, r)$ emerges from the core with extra phase accumulation relative to the H atom. This corresponds to a "sucking in" of a hydrogenic function.

This is the mixed $f, g$ form of $u_\ell(\nu_s r)$ when $\nu - \ell$ is non-integer.

$$u_\ell(\nu, r) = \alpha f_\ell(\nu, r) - (1 - \alpha^2)^{1/2} g_\ell(\nu, r) \quad **$$

* invariance of intra-core nodal structure – amount of phase shift along a Rydberg series should be independent of $\nu$. [We expect this to be true.]

* mixing of 2 types of Coulomb functions is required in order to have noninteger $\nu$, yet still satisfy the $u_\ell(\nu, r) \to 0$ as the $r \to \infty$ boundary condition.

TRICK: $\left.\begin{aligned}\alpha &\equiv \cos(\pi\mu_\ell)\\ -(1-\alpha^2)^{1/2} &= -\sin(\pi\mu_\ell)\end{aligned}\right]$ plug this into the ** equation on page 29-5

$$\psi = u_\ell(v,r)\Phi_\ell(\text{CORE}) = \left[f_\ell(v,r)\cos(\pi\mu_\ell) - g_\ell(v,r)\sin(\pi\mu_\ell)\right]\Phi_\ell$$

plug in the asymptoptic forms for $f$ and $g$

$$\psi \Rightarrow \Big\{\underbrace{\sin[\pi(v-\ell)]}_{f_\ell(\nu,r)}\cos(\pi\mu_\ell) + \underbrace{\cos[\pi(v-\ell)]}_{-g_\ell(\nu,r)}\sin(\pi\mu_\ell)\Big\}C(r)e^{r/v}\Phi_\ell$$

the factor $\{\ \} \to 0$ as $r \to \infty$ is required. How?

$\{\ \} = 0$: $\sin\left[\pi(\nu-\ell)\right]\cos(\pi\mu_\ell) = -\cos\left[\pi(\nu-\ell)\right]\sin(\pi\mu_\ell)$

This requires that $\dfrac{\sin\left[\pi(\nu-\ell)\right]}{\cos\left[\pi(\nu-\ell)\right]} = -\dfrac{\sin(\pi\mu_\ell)}{\cos(\pi\mu_\ell)}$

constraint on $\nu$. What are <u>all</u> of the values of $\nu$ which are consistent with this constraint?

$\tan\left[\pi(\nu-\ell)\right] = -\tan(\pi\mu_\ell)$

$\tan\theta = -\tan(-\theta) = -\tan(-\theta + n'\pi)$ $n'$ integer ordinary trigonometry

let $\theta = \pi\mu_\ell$

$\therefore \tan[\pi(\nu-\ell)] = \tan(-\theta + n'\pi)$

thus $-\theta + n'\pi = \pi(\nu-\ell) \Rightarrow \theta = n'\pi - \pi(\nu-\ell)$

$\theta = \pi\mu_\ell$

$\downarrow$

but $\pi\mu_\ell = \pi(\underbrace{n' - \nu + \ell}_{n'+\ell=n})$

$\uparrow$

integer

$$\theta = \pi(n' - \nu + \ell)$$

$$n = n' + \ell,\ \mu_\ell = n - \nu$$

$$\boxed{\nu = n - \mu_\ell}$$

$\nu$ is smaller than integer-$n$ by constant term $\mu_\ell$.

This implies the existence of an infinite series of $\psi$ with values of integer $n$ for which $\psi \to 0$ as $r \to \infty$.

Get this infinite series of ν's, increasing in steps of 1, simply by specifying <u>one</u> ν-independent value of $\mu_\ell$!

All of the ν-dependence (E-dependence) of $\psi_\ell(\nu,r)$ is explicitly built into $f_\ell(\nu,r)$ and $g_\ell(\nu,r)$. $\mu_\ell$ describes the relative amounts of $f_\ell$ and $g_\ell$ in ψ. This amount of *f,g* mixing is determined when the $e^-$ leaves the core with the precise phase shift ensures that $\psi \to 0$ at $r \to \infty$.

C. How can we show that $\pi\mu_\ell$ is a phase shift?

The asymptotic form of ψ is

$$\psi \to \left\{\sin\left[\pi\left(\nu-\ell\right)\right]\cos\left(\pi\mu_\ell\right)+\cos\left[\pi\left(\nu-\ell\right)\right]\sin\left(\pi\mu_\ell\right)\right\}C(r)e^{r/\nu}$$

use the double angle formula: $\sin A\cos B+\sin B\cos A=\sin(A+B)$

$$\psi \to \left\{\sin\left[\pi\left(\nu-\ell\right)+\underline{\underline{\pi\mu_\ell}}\right]\right\}C(r)e^{r/\nu}$$

but $f_\ell\left(\nu,r\right)\to \sin\left[\pi\left(\nu-\ell\right)\right]C\left(r\right)e^{r/\nu}$

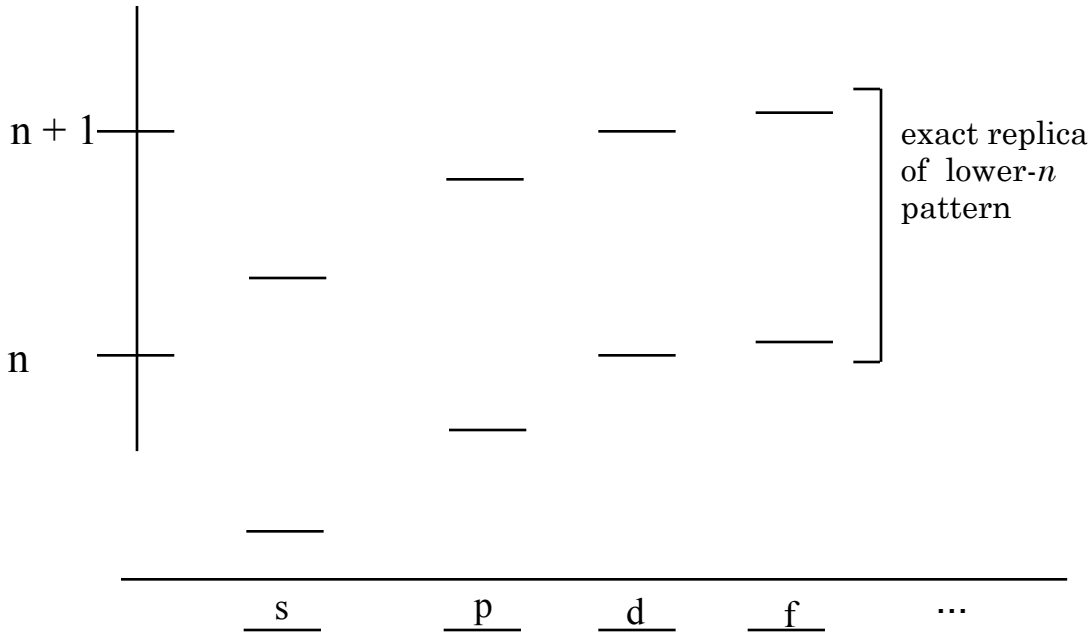
so this modified function is identical to the <u>regular</u> [i.e. $f_\ell(\nu,r)$] Coulomb function but with a $\pi\mu_\ell$ phase shift.

If $\mu_\ell > 0$, this corresponds to an <u>advance</u> of the phase of $u_\ell(\nu,r)$ relative to that for H. As expected, Ψ is sucked into the core by an amount $\pi\mu_\ell$ [+ an arbitrary number of 2π's] by the radial charge distribution Z(r) of the core.

$\pi\mu_\ell$ is the phase shift that occurs inside the core. It is the boundary condition at r = 0 shifted out to $r = r_0$. On the other hand, the $r \to \infty$ boundary conditions is satisfied by $\nu = n - \mu_\ell$ where *n* is an integer.

**the quantum defect!**

Small (replicated) region of the $n$-scaled energy level diagram:




$$\mu_s >> \mu_p > \mu_d > \mu_f \ldots \approx 0$$

<u>everything</u> is exactly repeated in each to + 1 region of

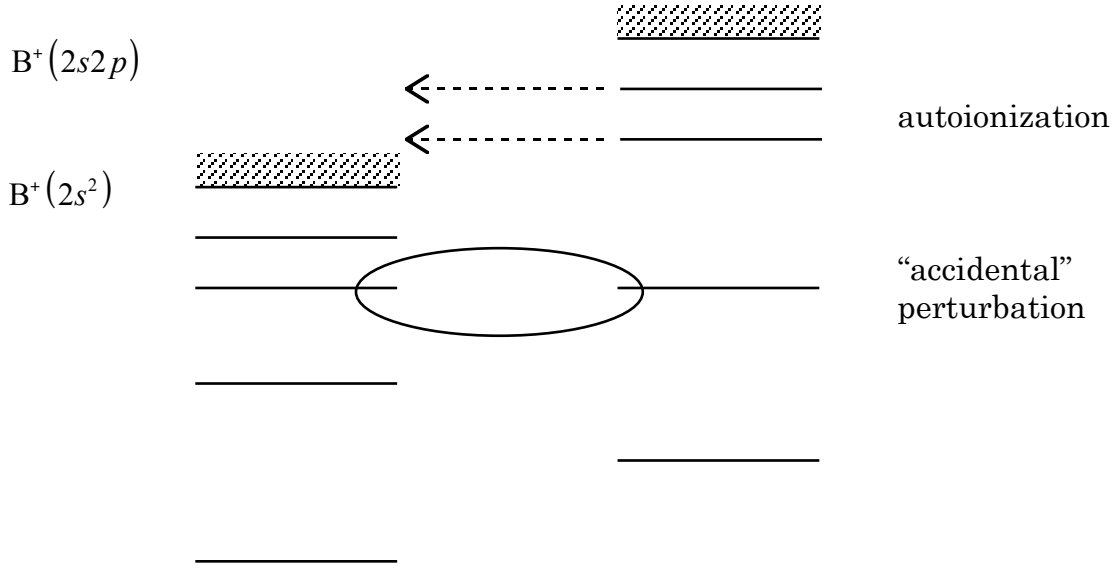
ν ν, not E, is the way to look at Rydberg "patterns"

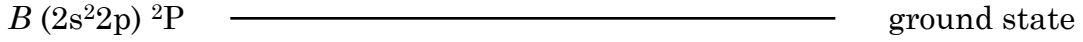
Finding the way to see a pattern is ALWAYS the route to both "assignment" and "insight".
**THIS COULD BE THE MOTTO FOR 5.73!**

$\mu_\ell$ decreases as $\ell$ increases because of the expected behavior of $Z^{\text{eff}}(r)$ as sampled in the presence of a centrifugal barrier $\propto \frac{\ell(\ell+1)}{2\mu r^2}$, which reduces the penetration of the electron inside the core.

As $\ell$ increases, the electron feels progressively less and less of the inside-core region in which $Z^{\text{eff}}(r) > 1$.

D. Inter-series interactions? Suppose you have B $2s^2 2p^1$ (B = Boron).




Separate series converging to two series limits: $B^+$ $2s^2$ and $B^+$ $2s2p$.
perturbations (accidental)
autoionization (systematic)

But there is a relationship between perturbation and auto-ionization matrix elements.

These inter-series interactions are all described by Multichannel Quantum Defect Theory

Replace $\mu_s, \mu_p, \mu_d$ etc.

by $3 \times 3\,\mu$ matrices, one for each symmetry of the ion:

$$\left[2s^2\left({}^1S\right) \otimes n_1\ell\right] \quad {}^2\ell$$

$$\left[2s2p\left({}^3P\right) \otimes n_2\ell \pm 1\right] \quad {}^2\ell$$

$$\left[2s2p\left({}^1P\right) \otimes n_3\ell \pm 1\right] \quad {}^2\ell$$

more complicated many-electron coupling problem

subject of the next few lectures.

Overall symmetry: **H** is totally symmetric.

Off-diagonal elements describe inter-channel interactions (exchange of angular momentum between Rydberg $e^-$ and core $e^-$s).

They describe what happens in a collision of the Rydberg $e^-$ with the ion-core. Does it change the state of the ion? Does it change the kinetic energy and/or angular momentum of the $e^-$? Unified picture of scattering at both negative E (bound states) and positive E.

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Next few lectures:

states of many-electron atoms

How to calculate matrix elements of many-electron (many Fermion) systems.

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5.73 Quantum Mechanics I
Fall 2018

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# Matrix Elements of Many-Electron Wavefunctions

Last time:

$v = \left[ -\frac{\Re}{E_{n,\ell}} \right]^{1/2}$ non-integer principal quantum number ($E_{\infty,\ell} = 0$)

$f_\ell(v,r)$, $g_\ell(v,r)$ — two linearly independent solutions to Schrödinger Equation are valid outside sphere of radius $r_0$

Need both *f* and *g* to satisfy boundary condition for $E \to 0$ as $r \to \infty$

$\nu = n - \mu_\ell \left( \text{non-integer } \nu \right)$

$\pi\mu_\ell$ is phase shift of $f_\ell(\nu,r)$

Infinite set of *integer-spaced* $\nu$-values that satisfy the $r \to \infty$ boundary condition

Wave emerges from the core with $\nu$-independent phase. Core transforms wave with correct $r \to 0$ limiting behavior into one that exits the sphere of radius $r_0$, which contains the core region, with $\pi\mu_\ell$ phase shift.

Core is sampled by a set of different $\ell$ values.

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Today:

Wavefunctions and Energy States of many-electron atoms: a magic decoder

1. (spin) orbitals → configurations → L-S states
2. electrons are Fermions → ψ must be "antisymmetrized": KEY PROBLEM
3. Slater determinants are antisymmetric with respect to *all* $e_i^-, e_j^-$ permutations
   - A. Normalization
   - B. Matrix Elements of one-$e^-$ Operators: e.g. $\mathbf{H}^{SO} = \sum_i a(r_i) \ell_i \cdot \mathbf{s}_i$
   - C. Matrix Elements of two-$e^-$ Operators: e.g. $\mathbf{H}^e = \sum_{i>j} e^2/\mathbf{r}_{ij}$ (a very strong "perturbation")

*next few lectures*

4. $H^{eff}$ in terms of $\underset{\text{orbital energy}}{\varepsilon_{n\ell}}$, $\underbrace{F^k, G^k}_{\text{Slater-Condon } 1/r_{ij}}$, $\underset{\text{spin-orbit}}{\zeta_{n\ell}}$ parameters

Page 31-9 is an example of what we will be able to do.
* Interpretable trends: Periodic Table
* Atomic energy levels: mysterious code — no atom-to-atom relationships evident without magic decoder ring.

Many-electron **H**

$$\mathbf{H} = \underbrace{\sum_{i=1}^{N} \mathbf{h}(i)}_{\mathbf{H}^{(0)}} + \underbrace{\sum_{i>j=1}^{N} \frac{e^2}{r_{ij}} + \sum_{i} a(r_i)\ell_i \cdot \mathbf{s}_i}_{\mathbf{H}^{(1)}}$$

sum of hydrogenic

$1-e^-$ terms: $-\frac{Z^2\Re}{n^2} = -\varepsilon_n$ (unshielded hydrogenic orbital energies)

How do we set up a matrix representation of this **H**?

take as much as possible out of $\langle \mathbf{H}^{(1)} \rangle$

$\mathbf{H}^{(0)}$ defines the basis set (complete, orthonormal, …)

$$\mathbf{H}^{(0)} = \sum_{i=1}^{N} \mathbf{h}(i) \leftrightarrow \left|\psi\right\rangle = \prod_{i=1}^{N} \left[\left|\phi_i(r_i)\right\rangle \left|sm_s(i)\right\rangle\right]$$

spatial part, spin part

"spin-orbital"

the $\phi_i(r_i)$'s could be hydrogenic or shielded-core Rydberg-like orbitals.

For two e⁻:

$$\left[\mathbf{h}(1)+\mathbf{h}(2)\right]\phi(1)\phi(2) = \left[\mathbf{h}(1)\phi(1)\right]\phi(2) + \left[\mathbf{h}(2)\phi(2)\right]\phi(1)$$

a constant with respect to **h**(1).

**H** as sum, E as sum, but get Ψ as product

Electronic Configuration: list of orbital occupancies

e.g. C $1s^2 2s^2 2p^2$ six e⁻

This is not sufficient to specify the state of a system

Several *L*,*S* terms arise from this configuration: e.g. $p^2 \rightarrow {}^1D, {}^3P, {}^1S$

$\vec{L} \equiv \sum_i \vec{\ell}_i$ $\quad$ $\vec{S} \equiv \sum_i \vec{s}_i$

$\vec{r}_{12} = \vec{r}_2 - \vec{r}_1$

$$\left|r_{12}\right| = \left[\left|r_1\right|^2 + \left|r_2\right|^2 - \underbrace{2r_1 \bullet r_2}\right]^{1/2}$$

vector with respect to $\ell_1$ and $\ell_2$; scalar with respect to L

We know that $\mathbf{L}^2, \mathbf{L}_z, \mathbf{S}^2, \mathbf{S}_z$ commute with $\mathbf{h}(i) + \frac{e^2}{r_{ij}}$

So we can use thse to bock diagonalize **H**

Note that although $\ell_i$ does not commute with $e^2/r_{ij}$, this is not a problem for $s_i^2$ and $s_{iz}$ because $\mathbf{h}(i)$ and $r_{ij}$ do not involve spin.

$r_{ij}$ destroys $\ell_i$ but does not destroy L!

How do we get eigenstates of $\mathbf{L}^2$, $\mathbf{L}_z$, $\mathbf{S}^2$, $\mathbf{S}_z$

either: I. Method of $M_L$, $M_S$ boxes
Advanced Inorganic Chemistry Course
Which L–S "terms" exist, but **not** the specific linear combinations of spin-orbital products that correspond to these terms.

II. Angular momentum coupling techniques:
3-j coefficients
ladders plus orthogonality
projection operators

We will return to this problem and approach it in all 3 angular momentum coupling ways.

One rigorous symmetry *must* be imposed:
**Pauli Exclusion Principle**: electrons are Fermions and therefore any acceptable wavefunction must be antisymmetric with respect to permutation of ANY pair of $e^-$

e.g. $|1,2\rangle = |\mu_1(1)\rangle|\mu_2(2)\rangle$ (orbitals; electrons)

$$\mathbf{P}_{12}|1,2\rangle = |\mu_1(2)\rangle|\mu_2(1)\rangle \equiv |2,1\rangle$$

$e^-$ are indistinguishable, $\therefore [\mathbf{H}, \mathbf{P}_{ij}] = 0$

$\therefore$ all $\psi$'s must belong to + or – eigenvalue of $\mathbf{P}_{ij}$ (note that $\mathbf{P}_{ij}^2 = \mathbf{I}$)

| | | |
|---|---|---|
| + | Boson | (integer spin) |
| – | Fermion | (1/2-integer spin) |

$$\psi_\pm = 2^{-1/2}\left[|1,2\rangle \pm |2,1\rangle\right]$$

$$\mathbf{P}_{12}\psi = 2^{-1/2}\left[|2,1\rangle \pm |1,2\rangle\right] = \pm\psi_\pm$$

$\psi_+$ bosons top sign

$\psi_-$ fermions bottom sign

generalize to 3 $e^-$? 3! combinations needed! Horrible!

* $\Psi$'s have N! terms (each is a product of N spin-orbitals)
* matrix elements have $(N!)^2$ additive terms! Even more horrible!

TRICK! Slater Determinants

$$\begin{vmatrix} \mu_1(1) & \mu_2(1) \\ \mu_1(2) & \mu_2(2) \end{vmatrix} = \mu_1(1)\mu_2(2) - \mu_1(2)\mu_2(1)$$

($e^-$ labels the columns; orbital labels the rows)

row is label for $e^-$
column is label for spin-orbital

$$\begin{vmatrix} u_1(1) & \cdots & u_n(1) \\ \vdots & \vdots & \vdots \\ u_1(n) & & \end{vmatrix}$$

You show that a 3 × 3 Slater determinant gives 6 additive product terms

Determinants are of N × N dimension

* N! additive terms in expansion of determinant
* determinant changes sign upon permutation of ANY two rows [$e^-$'s] or columns [spin-orbitals]
* determinant is zero if any two rows or columns are identical.
* determinant may be uniquely specified by its main diagonal

MUST SPECIFY IN ADVANCE A STANDARD ORDER IN WHICH THE SPIN-ORBITALS ARE TO BE LISTED ALONG THE MAIN DIAGONAL

e.g. $s\alpha,\ s\beta,\ p1\alpha,\ p1\beta,\ p0\alpha,\ p0\beta,\ p-1\alpha,\ p-1\beta,\ \ldots$

[or for $p^N$, suppress p in notation: $1\alpha1\beta0\alpha \rightarrow M_L = 2, M_S = 1/2$]

Need a fancy notation to demonstrate how Slater determinants are to be manipulated in evaluating matrix elements. This notation is meant to be forgotten as soon as it has served its immediate purpose here.

$$\left|\psi_N\right\rangle = (N!)^{-1/2}\sum_{\wp}(-1)^{p}\wp\left[\left|\mu_1(1)\right\rangle\ldots\left|\mu_N(N)\right\rangle\right]$$

# of binary permutations away from "standard order"

N! different $\wp$'s

$\wp$ is ONE prescription for rearranging the orbitals from the initially specified order

$\wp$ is product of several $\mathbf{P}_{ij}$'s or, more useful for proving theorems, a product of N factors $\mathbf{P}_i$ which specify whether the $i^{th}$ electron is to be left in the $i^{th}$ spin-orbital or transferred to some unspecified spin-orbital

$$\wp\left[\left|\mu_1(1)\right\rangle\ldots\left|\mu_N(N)\right\rangle\right]\prod_{i=1}^{N}\left|P_i\mu_i(i)\right\rangle$$

A. Normalization

Verify that $(N!)^{-1/2}$ is the correct normalization factor

$$\left\langle\psi_N\middle|\psi_N\right\rangle = (N!)^{-1}\sum_{\wp,\wp'}(-1)^{p+p'}\wp\left[\left\langle u_1(1)\right|\ldots\left\langle u_N(N)\right|\right]\wp'\left[\left|u_1(1)\right\rangle\ldots\left|u_N(N)\right\rangle\right].$$

Now rearrange into products of one-$e^-$ overlap integrals,

$$\left\langle\psi_N\middle|\psi_N\right\rangle = (N!)^{-1}\sum_{\wp,\wp'}(-1)^{p+p'}\prod_{i=1}^{N}\left\langle\mathbf{P}_i u_i(i)\middle|\mathbf{P}_i' u_i(i)\right\rangle.$$

The $\left|u_i\right\rangle$ are orthonormal.

$\left\langle u(i)\middle|u(j)\right\rangle$ has no meaning because the bra and ket must be associated with the same $e^-$

The only non-zero legal terms in $\sum_{\wp\,\wp'}$ are those where EACH $\mathbf{P}_i = \mathbf{P}_i'$

otherwise there will be at least 2 mis-matched bra-kets

$\left\langle u_i(k)\middle|u_j(k)\right\rangle\ldots\left\langle u_j(\ell)\middle|u_i(\ell)\right\rangle$

$= 0$ $= 0$

(Here the electron names match in each bra-ket, but the spin-orbital quantum numbers do not match.)

Thus it is necessary that $\wp = \wp', p = p', (-1)^{p+p'} = +1$ and

$$\langle \psi_N | \psi_N \rangle = (N!)^{-1} \sum_{\wp} \wp \left[ \langle \mu_1(1) | \mu_1(1) \rangle \ldots \langle \mu_n(N) | \mu_n(N) \rangle \right]$$

Each term in sum over $\wp$ gives +1, but there are N possibilities for $\mathbf{P}_1$, N – 1 possibilities for $\mathbf{P}_2$.

$\therefore$ N! possibilities for sum over $\wp$

$$\langle \psi_N | \psi_N \rangle = (N!)^{-1} \sum_{\wp}^{N} 1 = 1$$

Thus the assumed $(N!)^{-1/2}$ normalization factor is correct.

---

B. Matrix elements of one-electron operators

$$\mathbf{F} = \sum_i f(\mathbf{r}_i) \qquad \text{e.g. } \vec{\mathbf{L}} = \sum_i \vec{\ell}_i$$

$$|\psi_A\rangle \equiv (N!)^{-1/2} \sum_{\wp} (-1)^p \wp |a_1(1)\rangle \ldots |a_N(N)\rangle$$

$$|\psi_B\rangle \equiv (N!)^{-1/2} \sum_{\wp'} (-1)^{p'} \wp' |b_1(1)\rangle \ldots |b_N(N)\rangle$$

$$\langle \psi_A | \mathbf{F} | \psi_B \rangle = (N!)^{-1} \sum_{i,\wp,\wp'} (-1)^{p+p'} \wp \left[ \langle a_1(1) | \ldots \right] f(\mathbf{r}_i) \wp' \left[ |b_1(1)\rangle \ldots \right]$$

$$= (N!)^{-1} \sum_{i,\wp,\wp'} (-1)^{p+p'} \left[ \langle \mathbf{P}_1 a_1(1) | \mathbf{P}_1 b_1(1) \rangle \right]$$

$$\ldots \underbrace{\left[ \langle \mathbf{P}_i a_i(i) | f(\mathbf{r}_i) | \mathbf{P}'_i b_i(i) \rangle \right]}_{\text{special term because indices match for } e^- \text{ in both orbitals and the operator.}} \ldots \left[ \langle \mathbf{P}_N a_N(N) | \mathbf{P}'_N b_N(N) \rangle \right]$$

There is a product of N orbital matrix element factors in each term of sum. Of these, N–1 are orbital overlap integrals and only *one* involves the one-e$^-$ operator.

Consider the 3 possibilities for 1-e$^-$ operators: $\Delta$so = 0, =1 or > 1.

SELECTION RULE $\langle \Psi_A | \mathrm{F} | \Psi_B \rangle = 0$ if $|\psi_A\rangle$ and $|\psi_B\rangle$ differ by more than one spin-orbital (because at least one of the orbital overlap integrals would be zero)

two cases remain:

1. differ by one spin-orbital

$$\left.\begin{aligned} |\psi_A\rangle &= \|u_1(1)\ldots a_k(k)\ldots u_N(N)\| \\ |\psi_B\rangle &= \|u_1(1)\ldots b_k(k)\ldots u_N(N)\| \end{aligned}\right\}$$ the mismatched orbitals are in the same ($k^{th}$) position

Use $u_i$ to denote common spin-orbitals
Use $a_k$, $b_k$ to denote unique spin-orbitals

For this choice, all N $\mathbf{P}_i$ factors of each $\wp$ must be identical to all N factors of $\wp'$.

There is an additional requirement: $\wp$ must bring mismatched orbitals into i-th position so that they match up with the $f(\mathbf{r}_i)$ operator to give $\langle a_k(i) | f(\mathbf{r}_i) | b_k(i) \rangle$.

ANY OTHER ARRANGEMENT GIVES

$$\underbrace{\langle a_k(j) | b_k(j) \rangle}_{=0} \underbrace{\langle u_i(i) | f(\mathbf{r}_i) | u_i(i) \rangle}_{\neq 0} = 0$$

(N – 1)! ways of arranging the $e^-$ in the other N – 1 matched orbitals and there are N identical terms (in which the $e^-$ is in the privileged location) in the sum over $i$

$$\langle \psi_A | \mathbf{F} | \psi_B \rangle = (N!)^{-1} (N-1)! N \langle a_k | \mathbf{f} | b_k \rangle$$

from the $(N!)^{-1/2}$ normalization factor for each $|\psi\rangle$

If the order of spin-orbitals in $\psi_A$ or $\psi_B$ must be arranged away from the standard order in order to match the positions of $a_k$ and $b_k$, then we get an additional factor of $(-1)^p$ where p is the required number of binary permutations

$$\left\langle \psi_A \middle| \mathbf{F} \middle| \psi_B \right\rangle = (-1)^p \left\langle a_k \middle| \mathbf{f} \middle| b_k \right\rangle$$

for a difference of one spin-orbital

i.e. $A = \|12\ 5\ 7\|$

$$B = \|12\,3\,5\| = -\|12\,5\,3\|$$

(one binary permutation is required)

$$\left\langle \psi_A \middle| \mathbf{F} \middle| \psi_B \right\rangle = -\left\langle 7 \middle| \mathbf{f} \middle| 3 \right\rangle$$

2. $\psi_A = \psi_B$ Differ by zero spin-orbitals

$$\left\langle \psi_A \middle| \mathbf{F} \middle| \psi_A \right\rangle = (N!)^{-1} \sum_{i,\wp} \left[ \left\langle \mathbf{P}_i a_i(i) \middle| \mathrm{f}(\mathbf{r}_i) \middle| \mathbf{P}_i a_i(i) \right\rangle \right]$$

all other factors are =1

$N!$ identical terms from sum over $\wp$ [again $(N-1)!N$]

$$\left\langle \psi_A \middle| \mathbf{F} \middle| \psi_A \right\rangle = \sum_i \left\langle a_i \middle| f(\mathbf{r}_i) \middle| a_i \right\rangle$$

* Normalization and * 1-$e^-$ Operator **F** } comes out almost the same as naive expectation WITHOUT need for antisymmetrization!

what is new? only the factor of $(-1)^P$ for re-ordering $|\psi_B\rangle$ to match order in $|\psi_A\rangle$

Examples of $f^3$:

$$\psi = \|3\alpha 1\alpha - 2\alpha\| \qquad (\lambda = 3, \lambda = 1, \lambda = -2)$$

$$\langle \mathrm{L}_z \rangle = \hbar(3+1-2)$$

$$\langle \mathrm{L}_z \mathrm{S}_z \rangle = \hbar^2 \left(\tfrac{3}{2} + \tfrac{1}{2} - 1\right)$$

$$\mathrm{J}_+ \|3\alpha 1\alpha - 2\alpha\| = \mathrm{L}_+ \|3\alpha 1\alpha - 2\alpha\| + \mathrm{S}_+ \|3\alpha 1\alpha - 2\alpha\|$$

$$= \hbar\left[0 + 10^{1/2} \|3\alpha 2\alpha - 2\alpha\| + 10^{1/2} \|3\alpha 1\alpha - 1\alpha\| + 0 + 0 + 0\right]$$

**YOU VERIFY!**

next time **G***(i,j),* a 2-electron operator.

*updated August 27, 2020 @ 1:42 PM*

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# Matrix Elements of One-Electron, $F(i)$, and Two-Electron, $G(i,j)$, Operators

Last time: orbitals → configurations → <u>states</u> ("terms")
Fermions: Slater Determinants: Pauli Exclusion Principle

Compact Notation for Slater Determinant: $\|\text{main diagonal}\|$.

TODAY: 1. SLATER DETERMINANTAL MATRIX ELEMENTS

A. Normalization

B. $F(i)$: One-e$^-$ operator e.g. $\mathrm{H}^{SO} = \sum_i a(r_i)\,\ell_i \cdot \mathrm{s}_i$

C. $G(i,j)$: Two-e$^-$ operator e.g. $\mathrm{H}^{\mathrm{e}} = \sum_{i>j} e^2/\mathrm{r}_{ij}$

Recall: specify a standard order of spin-orbitals (because determinant changes sign upon every binary permutation)

Goal: make the inconvenience of Slater determinants almost vanish — matrix elements will be almost what you expect for simple non-antisymmetrized products of spin-orbitals.

pages 31-2,3,4 are repeat of pages 30-6, 7, 8

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A. Normalization: $|\psi_N\rangle = (N!)^{-1/2} \sum_{\wp} (-1)^p \wp[|u_1(1)\rangle \ldots |u_N(N)\rangle]$ anti-symmetrized product of spin-orbitals

verify that $(N!)^{-1/2}$ is the correct factor

$$\langle \psi_N | \psi_N \rangle = (N!)^{-1} \sum_{\wp,\wp'} (-1)^{p+p'} \wp\left[\langle u_1(1)| \ldots \langle u_N(N)|\right] \wp'\left[|u_1(1)\rangle \ldots |u_N(N)\rangle\right]$$

rearrange into products of one-e$^-$ overlap integrals:

$$= (N!)^{-1} \sum_{\wp,\wp'} (-1)^{p+p'} \prod_{i=1}^{N} \langle \mathrm{P}_i u_i | \mathrm{P}'_i u_i \rangle$$

* $\{|u_i\rangle\}$ are orthonormal

* $\langle u(i)|u(j)\rangle$ has no meaning because bra and ket must be associated with the SAME electron

* only non-zero LEGAL terms in $\sum_{\wp\,\wp'}$ are those where EACH $\mathbf{P}_i = \mathbf{P}'_i$,

otherwise get AT LEAST 2 MISMATCHED bra-kets

$\langle u_i(k)|u_j(k)\rangle \ldots \langle u_j(\ell)|u_i(\ell)\rangle$
$= 0$ $= 0$

(Here the electron names match in each bra-ket but the spin-orbital names do not match.)

Think of a one- or two-e$^-$ operator as a scheme for "dealing with" or "hiding" the small number of mismatched spin-orbitals.

Thus it is necessary that $\wp = \wp', p = p', (-1)^{p+p'} = +1$

and $\langle \psi_N | \psi_N \rangle = (N!)^{-1} \sum_{\wp} \wp \left[ \langle \mu_1(1) | \mu_1(1) \rangle \ldots \langle \mu_N(N) | \mu_N(N) \rangle \right].$
$= 1 \qquad = 1$

Each term in the sum over $\wp$ gives +1, but there are N possibilities for $\mathbf{P}_1$, N – 1 possibilities for $\mathbf{P}_2$, $N_2 - 2$ possibilities for $\mathbf{P}_3$ …

$\therefore N!$ possibilities for sum over $\wp$

$$\langle \psi_N | \psi_N \rangle = (N!)^{-1} \sum_{\wp} 1 = 1$$

Thus the assumed $(N!)^{-1/2}$ normalization factor is correct.

---

B. Matrix elements of one-electron operators

$$\mathbf{F} = \sum_i f(\mathbf{r}_i) \quad \text{e.g } \vec{\mathbf{L}} = \sum_i \vec{\ell}_i$$

$$|\psi_A\rangle \equiv (N!)^{-1/2} \sum_{\wp} (-1)^p \wp | a_1(1) \rangle \ldots | a_N(N) \rangle$$

$$|\psi_B\rangle \equiv (N!)^{-1/2} \sum_{\wp'} (-1)^{p'} \wp' | b_1(1) \rangle \ldots | b_N(N) \rangle$$

$$\langle \psi_A | \mathbf{F} | \psi_B \rangle = (N!)^{-1} \sum_{i,\wp,\wp'} (-1)^{p+p'} \wp \left[ \langle a_1(1) | \ldots \right] f(\mathbf{r}_i) \wp' \left[ | b_1(1) \rangle \ldots \right]$$

$$= (N!)^{-1} \sum_{i,\wp,\wp'} (-1)^{p+p'} \left[ \langle \mathbf{P}_1 a_1(1) | \mathbf{P}_1 b_1(1) \rangle \right]$$

$$\ldots \left[ \langle \mathbf{P}_i a_i(i) | f(\mathbf{r}_i) | \mathbf{P}'_i b_i(i) \rangle \right] \ldots \left[ \langle \mathbf{P}_N a_N(N) | \mathbf{P}'_N b_N(N) \rangle \right]$$

Product of N orbital matrix element factors in each term of sum. Of these, N–1 are orbital overlap integrals and only *one* involves the one-$e^-$ operator.

1. SELECTION RULE $\langle \Psi_A | \mathbf{F} | \Psi_B \rangle = 0$ if $|\psi_A\rangle$ and $|\psi_B\rangle$ differ by more than one spin-orbital (at least one of the orbital overlap integrals in the N-term product would be zero)

two cases remain:

2. $|\Psi_A\rangle$ and $|\Psi_B\rangle$ differ by one spin-orbital

$$\left.\begin{aligned} |\psi_A\rangle &= \|u_1(1)\ldots a_k(k)\ldots u_N(N)\| \\ |\psi_B\rangle &= \|u_1(1)\ldots b_k(k)\ldots u_N(N)\| \end{aligned}\right\}$$ the mismatched orbitals are in the same position

use $u_i$ to denote the in-common spin-orbitals
use $a_k$, $b_k \neq 0$ to denote unique spin-orbitals

For this choice, all N $\mathbf{P}_i$ factors of each $\wp$ must be identical to all N $\mathbf{P}'_i$ factors of $\wp'$

Additional requirement: $\wp$ must bring mismatched orbital pairs into i-th position so that they match up with the $f(\mathbf{r}_i)$ operator to give $\langle a_k(i)|f(\mathbf{r}_i)|b_k(i)\rangle$

ANY OTHER ARRANGEMENT GIVES

$$\underbrace{\langle a_k(\ell)|b_k(\ell)\rangle}_{=0}\underbrace{\langle u_i(i)|f(\mathbf{r}_i)|u_i(i)\rangle}_{\neq 0} = 0$$

(N – 1)! ways of arranging the $e^-$ in the other N – 1 matched orbital pairs, and there are N identical terms (in which the $e^-$ is in the privileged location) in the sum over $i$

$$\langle \psi_A | \mathbf{F} | \psi_B \rangle = (N!)^{-1}(N-1)!N\langle a_k | \mathbf{f} | b_k \rangle$$

If the order of spin-orbitals in $\Psi_A$ or $\Psi_B$ must be arranged away from the standard order in order to match the positions of $a_k$ and $b_k$, then we get an additional factor of $(-1)^p$ where $p$ is the number of binary permutations

$$\left\langle \psi_A \middle| \mathrm{F} \middle| \psi_B \right\rangle = (-1)^p \left\langle a_k \middle| f \middle| b_k \right\rangle$$

for a difference of one spin-orbital

i.e. $A = \|12\ 5\ 7\|$

$$B = \|12\ 3\ 5\| = -\|12\ 5\ 3\|$$

$$\left\langle \psi_A \middle| \mathbf{F} \middle| \psi_B \right\rangle = -\left\langle 7 \middle| \mathbf{f} \middle| 3 \right\rangle$$

a many-e⁻ integral is reduced to a single spin-orbital integral

3. $\Psi_A = \Psi_B$ Differ by zero spin-orbitals

$$\left\langle \psi_A \middle| \mathbf{F} \middle| \psi_A \right\rangle = (N!)^{-1} \sum_{i,\wp} \left[ \left\langle \mathbf{P}_i a_i(i) \middle| f(\mathbf{r}_i) \middle| \mathbf{P}_i a_i(i) \right\rangle \right]$$

all other factors are =1

$N!$ identical terms from sum over $\wp$ [again from $(N-1)!N$]

$$\left\langle \psi_A \middle| \mathbf{F} \middle| \psi_B \right\rangle = \sum_i \left\langle a_i \middle| f(\mathbf{r}_i) \middle| a_i \right\rangle$$

* Normalization and * 1-$e^-$ Operator **F** } comes out almost the same as naive expectation WITHOUT need to deal with antisymmetrization!

Examples of f³: $\psi = \|3\alpha 1\alpha - 2\alpha\|$

$$\left\langle \mathbf{L}_z \right\rangle = \hbar(3+1-2)$$

$$\left\langle \mathbf{L}_z \mathbf{S}_z \right\rangle = \hbar^2 \left(\tfrac{3}{2} + \tfrac{1}{2} - 1\right)$$

$$\mathbf{J}_+ \|3\alpha 1\alpha - 2\alpha\| = \mathbf{L}_+ \|3\alpha 1\alpha - 2\alpha\| + \mathbf{S}_+ \|3\alpha 1\alpha - 2\alpha\|$$

$$= \hbar\left[0 + 10^{1/2} \|3\alpha 2\alpha - 2\alpha\| + 10^{1/2} \|3\alpha 1\alpha - 1\alpha\| + 0 + 0 + 0\right]$$

# now $\mathbf{G}(i,j)$: A two-electron operator

C. $\mathbf{G}(i,j)$ : 4 cases
1. differ by more than 2 spin-orbitals: Matrix Element $\rightarrow$ 0
2. differ by 2 spin-orbitals: sum includes one pair of nonzero matrix elements
3. differ by 1 spin-orbital: sum over pairs of nonzero single spin-orbital matrix elements
4. expectation value : differ by 0 spin-orbitals: double sum over pairs of spin-orbital matrix elements

1. is obvious — only way to make up for orbital mismatch is to hide the mismatched orbitals in $\langle\,|g(i,j)|\,\rangle$ (rather than in an overlap integral). But one can only hide 2-mis-matched pairs in, e.g.

$$\left\langle a_i a_j \middle| g(i,j) \middle| b_i b_j \right\rangle$$

$$\left\langle \psi_A \middle| \mathbf{G}(i,j) \middle| \psi_B \right\rangle = 0$$ if $\psi_A$, $\psi_B$ differ by more than 2 spin-orbitals

2. differ by two pairs of spin-orbitals

$$\psi_A = \left\| u_1(1) \ldots a_1(i) \ldots a_2(j) \ldots u_N(N) \right\|$$

$$\psi_B = (-1)^p \left\| u_1(1) \ldots b_1(i) \ldots b_2(j) \ldots u_N(N) \right\|$$

$(-1)^p$: number of permutations needed to put $b_1$ and $b_2$ in the $i$ and $j$ positions

$$\left\langle \psi_A \middle| \mathbf{G} \middle| \psi_B \right\rangle = (N!)^{-1} \sum_{i>j} \sum_{\wp,\wp'} (-1)^{p+p'} \left[ \text{orthogonality integrals} \right] \times$$

$$\left[ \left\langle \mathbf{P}_i a_1(i) \middle| \left\langle \mathbf{P}_j a_2(j) \middle| g(i,j) \middle| \mathbf{P}'_i b_1(i) \right\rangle \middle| \mathbf{P}'_j b_2(j) \right\rangle \right]$$

* there are $(N-2)!$ ways of permuting the $N-2$ matched $u_k$ functions that are not filled with $e^-$ $i$ and $j$. Moreover these permutations must involve $\mathbf{P}_k = \mathbf{P}'_k$ (for all $k \neq i,j$).

* also $N(N-1)$ identical terms in sum over $i > j$

Thus there are $(N-2)!(N-1)N = N!$ identical terms in the $\sum_{i>j} \sum_{\wp,\wp'}$ sums.

But there still remain <u>two</u> possibilities:

1. $\wp = \wp'$ $\therefore p = p'$ and $\mathbf{P}_i = \mathbf{P}'_i$, $\mathbf{P}_j = \mathbf{P}'_j$
2. $\wp$ same as $\wp'$ <u>except</u> for $i, j$ pair where

THIS MEANS WE PUT THE $j^{th}$ $e^-$ in $P'_j$ where WE HAD PUT THE $i^{th}$ $e^-$ in $P_i$

$\left.\begin{matrix}\mathbf{P}_i = \mathbf{P}'_j \\ \mathbf{P}_j = \mathbf{P}'_i\end{matrix}\right\rangle \underbrace{(-1)^{p+p'}}_{} = -1$

The 2 $\wp$'s must differ by exactly one binary permutation

THUS: $\left\langle \psi_A \middle| \mathbf{G} \middle| \psi_B \right\rangle = \pm\left[\left\langle a_1(1)a_2(2)\middle|g(1,2)\middle|b_1(1)b_2(2)\right\rangle - \left\langle a_1(1)a_2(2)\middle|g(1,2)\middle|b_2(1)b_1(2)\right\rangle\right]$

For $\psi_A, \psi_B$ different by 2 spin-orbitals:

# of permutations needed to make $\psi_B$ match $\psi_A$ — no sign ambiguity if standard order is initially specified

3. $\psi_A, \psi_B$ differ by only one pair of spin-orbitals

You work this out

($a$, $b$ matched)

[($a$ matched with $u_n$), $u_n$ matched with $b$]

$$\left\langle \psi_A \middle| \mathbf{G} \middle| \psi_B \right\rangle = \pm \sum_{\substack{n \\ n\neq 1,2}} \left[\left\langle a(1)u_n(2)\middle|g(1,2)\middle|b(1)u_n(2)\right\rangle - \left\langle a(1)u_n(2)\middle|g(1,2)\middle|u_n(1)b(2)\right\rangle\right]$$

($a$ with $u_n$)

not arbitrary

4. differ by zero spin-orbitals : expectation value of $\mathbf{G}$

$$\left\langle \psi_A \middle| \mathbf{G} \middle| \psi_A \right\rangle = \sum_{n>m} \left[\left\langle u_n(1)u_m(2)\middle|g(1,2)\middle|u_n(1)u_m(2)\right\rangle - \left\langle u_n(1)u_m(2)\middle|g(1,2)\middle|u_m(1)u_n(2)\right\rangle\right]$$

| "DIRECT" | "EXCHANGE" |
|---|---|
| what we would expect without antisymmetrization | unexpected: consequence of antisymmetrization |

The ONLY real surprise that results from the anti-symmetrization requirement for two-electron operators is the presence of one additional term and some signs that would have no counterpart if anti-symmetrization had been ignored.

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SUMMARY

* antisymmetrize → Slater determinants
* matrix elements are only slightly more complicated than those of a simple product of spin-orbitals
  - signs due to permutation [away from Standard Order]
  - extra terms in matrix elements of $\mathbf{G}(i,j)$

---

Do some examples for $p^2$:

1. What $L,S$ terms belong to $p^2$ (Lecture #32: method of crossing out microstates)?
2. What is the correct linear combination of Slater determinants that corresponds to a specific $L$-$S$ term in either the $|\, JLSM_J\rangle$ or the $|\, LM_LSM_S\rangle$ basis set?

   Methods:
   - ladders plus orthogonality (Lecture #32)
   - $\mathbf{L}^2$ and $\mathbf{S}^2$ matrices (Lecture #33)
   - 3-j coefficients
3. $e^2/r_{ij} \rightarrow F^k(n\ell,n'\ell')$, $G^k(n\ell,n'\ell')$ "Slater-Condon" parameters
   Relative energies of $L$-$S$ terms expressed in terms of $F^k$ and $G^k$'s. (Lecture #34)
4. Matrix elements of $\mathbf{H}^{SO}$ (Lecture #35)

   - $\zeta(NLS)$: coupling constant for an $N$–$L$–$S$ state
   - $\zeta(NLS) \leftrightarrow \zeta_{n\ell}$: relation of $\zeta(NLS)$ to orbital integrals
   - full $\mathbf{H}^{SO}$ in terms of $\zeta_{n\ell}$
   - periodic table explains the $\zeta_{n\ell}$, but the $\zeta(NLS)$ do not directly display periodicity. WHAT WE WANT IS ENCODED IN WHAT WE CAN OBSERVE

**EXAMPLES:**

Slater Determinant: $p^2 \qquad \left\|1\alpha 1\beta\right\| \rightarrow {}^1D \quad M_L = 2, M_S = 0$

$$\left\langle \mathbf{L}_z \right\rangle = \left\langle \left\|1\alpha 1\beta\right\| \mathbf{L}_z \left\|1\alpha 1\beta\right\| \right\rangle$$
$$= \hbar[1+1] = 2\hbar$$
$$\left\langle \mathbf{S}_z \right\rangle = \hbar\left[\frac{1}{2} + \left(-\frac{1}{2}\right)\right] = 0\hbar$$

$\mathbf{L}_z^2$ tricky! $\quad \mathbf{L}_z^2 = \underbrace{\sum_i \ell_{z_i}^2}_{\text{1–e}^- \text{ operator}} + \underbrace{\sum_{i \neq j} \ell_{z_i} \ell_{z_j}}_{\text{2–e}^- \text{ operator}}$

Easier to do this by applying $\mathbf{L}_z = \sum_i \ell_{z_i}$ *twice*

The only 2 terms in sum are 1,2 & 2,1

$$\left\langle \mathbf{L}_z^2 \right\rangle = \hbar^2\left[\ell_{z_1}^2 + \ell_{z_2}^2\right] + \sum_{i \neq j}\left[\left\langle 1\alpha 1\beta \middle| \ell_{z_i}\ell_{z_j} \middle| 1\alpha 1\beta \right\rangle - \left\langle 1\alpha 1\beta \middle| \ell_{z_i}\ell_{z_j} \middle| 1\beta 1\alpha \right\rangle\right]$$

from spin-mismatch

$$= 2\hbar^2 + \hbar^2\left[1^2 + 1^2 - 0 - 0 = 4\hbar^2 \text{ as expected}\right]$$

$$\left\langle \mathbf{S}_z^2 \right\rangle = \hbar^2\left[\frac{1}{4} + \frac{1}{4}\right] + \hbar^2\left[-\frac{1}{4} - \frac{1}{4} - 0 - 0\right] = 0\hbar^2 \text{ as expected}$$

($\downarrow \alpha\beta$ $\downarrow \beta\alpha$)

$$\mathbf{L}^2 - \mathbf{L}_z^2 = \mathbf{L}_\mathbf{x}^2 + \mathbf{L}_\mathbf{y}^2 = \frac{1}{2}\left(\mathbf{L}_+\mathbf{L}_- + \mathbf{L}_-\mathbf{L}_+\right)$$

$$\mathbf{L}^2 = \frac{1}{2}\left(\mathbf{L}_+\mathbf{L}_- + \mathbf{L}_-\mathbf{L}_+\right) + \mathbf{L}_\mathbf{z}^2$$

Can you show that $\left\langle \mathbf{L}^2 \right\rangle = \hbar^2 6$ for $\left\|1\alpha 1\beta\right\|$ of $p^2$?

## Patterns of Lowest-Lying States: "Aufbau" for adults!

| Atom | C | N | O |
|---|---|---|---|
| lowest config. | $1s^2 2s^2 2p^2$ | $1s^2 2s^2 2p^3$ | $1s^2 2s^2 2p^4$ |
| L-S terms | $^1S$, $^1D$, $^3P$ | $^4S$, $^2P$, $^2D$ | $^1S$, $^1D$, $^3P$ |
| Lowest Term | $^3P_0$ | $^4S_{3/2}$ | $^3P_2$ |
| | ("regular") | (no fine structure) | ("inverted") |

| | **C** | **N** | **O** |
|---|---|---|---|
| excitation<br>$2p \leftarrow 2s$ | $2s2p^3$<br>$^{5,3}S$, $^{3,1}P$, $^{3,1}D$<br>$[^5S_2]$ | $2s2p^4$<br>$^{4,2}P$, $^2D$, $^2S$<br>$[^4P_{5/2}]$ | $2s2p^5$<br>$^{3,1}P$<br>$[^3P_2]$ |
| $3s \leftarrow 2p$ | $2s^2 2p^2\ 3s$<br>$^{3,1}P$<br>$[^3P_0]$ | $2s^2 2p^2\ 3s$<br>$^{4,2}P$, $^2D$, $^2S$<br>$[^4P_{1/2}]$ | $2s^2 2p^3\ 3s$<br>$^5S$, $^3$(D,P,S)<br>$[^5S_2]$ |
| $3d \leftarrow 2p$ | $2s^2 2p\ 3d$<br>$^{3,1}$(F,D,P)<br>$[^3F_2]$ | $2s^2 2p^2\ 3d$<br>$^2$(G,F,D,D,D,P,P,S),<br>$^4$(F,D,P)<br>$[^4P_{1/5}]$ | $2s^2 2p^3\ 3d$<br>$^5D$, $^3$(GFFDDDPPS),<br>$^1$(GFFDDPPS)<br>$[^5D_0]$ |

## Energy Levels of Lowest and First Excited Configurations

| | C | | N | | O | |
|---|---|---|---|---|---|---|
| | $^1P_1$ | 119878 cm$^{-1}$ | | | | |
| | $^3S_1$ | 105800 | | | | |
| | $^1D_2$ | 97878 | | | | |
| | IP($^2P_{1/2}$) | 90878 | | | | |
| | $^3P_{2,1,0}$ | 75256 | | | $^1P_1$ | 189837 |
| | $^3D_3$ | 64088 | IP($^3P_0$) | 117345 | $^3P_0$ | 126266 |
| | $^3D_2$ | 64093 | $^4P_{1/2}$ | 88173 | $^3P_1$ | 126340 |
| | $^3D_1$ | 64092 | $^4P_{3/2}$ | 88153 | $^3P_2$ | 126266 |
| | $^5S_2$ | 33735 | $^4P_{5/2}$ | 88110 | IP($^4S_{3/2}$) | 109827 |
| | $^1S_0$ | 21648 | | | $^1S_0$ | 33794 |
| | $^2D_2$ | 10194 | $^2P_{3/2,1/2}$ | 28840 | $^1D_2$ | 15868 |
| | $^3P_2$ | 44 | $^2D_{3/2}$ | 19231 | $^2P_0$ | 226 |
| Lowest Config-uration | $^3P_1$ | 16 | $^2D_{5/2}$ | 19223 | $^2P_1$ | 158 |
| | $^3P_0$ | 0 | $^4S_{3/2}$ | 0 | $^2P_2$ | 0 |

How does one begin to assign an atomic spectrum? You need predictions of the *L-S-J* symmetry of the electronic ground state (because all absorption spectra will originate from this ground state) and the symmetries of the upper states of the lowest energy allowed transitions to the expected lowest excited electronic configurations. This should be simple, right?

The energy levels for C I, N I, and O I (the I denotes neutral, II denotes singly charged, III...) are tabulated in Atomic Energy Levels as Derived from the Analysis of Optical Spectra (NSRDS-NBS 35, volume 1) are available on-line. There are an enormous number of assigned energy levels listed in the tables. One thing should be immediately clear: there are no obvious patterns. The energy levels that belong to the same electronic configuration are often widely spaced. *Where is the Periodic Table in all of this apparent complexity?*

The spin-orbit constants, $z_{2p}$, for the 2p orbital in C, N, and O are: 29.0, 53.7, and 151 cm-1 respectively (Lefebvre-Brion and Field, The Spectra and Dynamics of Diatomic Molecules, page 316, Elsevier, 2004). These spin-orbit constants clearly exhibit the qualitative effects implicit in the Periodic Table.

Where is the **periodic table** in all of this? The electronic spectra and the energy level diagrams for C, N, and O atoms are complicated and, most importantly, there are no apparent relationships among them. You need a *magic decoder* to assign the spectra and another *magic decoder* to discover how the spectra relate to **periodicity**. The intra-configurational splittings due to matrix elements of the $1/r_{ij}$ inter-electronic repulsion term are enormous (in some cases comparable to the ionization energy) and do not follow any obvious pattern. The matrix elements of $1/r_{ij}$ are expressed in terms of Slater-Condon parameters ($F_k$ and $G_k$). The two-electron $F_k$ and $G_k$ terms, one-electron orbital energies, and one-electron spin-orbit constants all exhibit recognizable and interpretable periodicity. To me, this is staggeringly beautiful and a large part of the reason I chose electronic spectroscopy as the core of my career research.

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# Configuration and Resultant L-S-J "Terms" (States)

Last time: Matrix elements of Slater determinantal wavefunctions
Normalization: $(N!)^{-1/2}$
$\mathbf{F}(i)$: selection rule ($\Delta$s-o $\leq 1$), sign depending on order
$\mathbf{G}(i,j)$: selection rule ($\Delta$s-o $\leq 2$), two additive terms with opposite signs

TODAY: Configuration → which L-S terms? → L-S basis states → matrix elements

- Method of crossing out $M_L$, $M_S$ boxes
- Ladders plus orthogonality
- Many worked out examples that will not be covered in lecture.

KEY IDEAS:

* $1/r_{ij}$ destroys spin-orbital labels as good quantum numbers.
* Configuration splits into widely spaced L-S-J "terms."
* $\sum_{i>j} 1/r_{ij}$ is a *scalar operator* with respect to **L, S,** and **J**, thus matrix elements are independent of $M_L$, $M_S$, and $M_J$.
* Configuration generates all possible $M_L$, $M_S$ components of each L-S term.
* It can't matter which $M_L$, $M_S$ component we use to evaluate the $1/r_{ij}$ matrix elements
* Method of microstates and boxes: Book-keeping for which L-S states are present, organizes the algebra to find eigenstates of $L^2$ and $S^2$, as basis for "sum rule" method (Lecture #33).

Longer term goals: represent "electronic structure" in terms of properties of atomic orbitals

1. Configuration → L,S terms
2. Derive correct linear combination of Slater determinants for each L,S term: several methods
3. $1/r_{ij}$ matrix elements → $F_k$, $G_k$ Slater-Condon parameters, Slater sum rule trick
4. $\mathbf{H}^{SO}$ Spin-Orbit
   * $\zeta(NLS)$ — coupling constant for each $L$-$S$ term in an electronic configuration
   * $\zeta(NLS) \leftrightarrow \zeta_{n\ell}$ a single spin-orbit orbital integral for the entire configuration
   * full $H^{SO}$ matrix in terms of $\zeta_{n\ell}$
5. Stark effect, Zeeman effect, optical transitions
6. transition strengths

$$\langle n\ell \| r \| n'\ell+1 \rangle \quad (\text{matrix elements of } \vec{\mathbf{r}}, \text{ magnetic g-values})$$

There are a vastly smaller number of orbital parameters than the number of electronic states. The periodic table provides a basis for rationalization of orbital parameters (dependence on atomic number and on number of electrons.) Intuition vs. numerical description.

Which L-S terms belong to $(nf)^2$

* shorthand notation for spin-orbitals

  $n\ell m_\ell \alpha/\beta$ e.g. 4f3α , could suppress 4 and f

($\|$main diagonal$\|$ represents Slater determinant, $|\ \rangle\|\ \rangle$…represents simple product of spin-orbitals)

* standard order (to get signs internally consistent): for f spin-orbitals

  3α3β2α2β … -3α -3β is my standard order for f ($\ell = 3$)

$(2\ell+1)(2s+1) = (7)(2) = 14$ spin-orbitals

* which Slater determinants are nonzero and distinct (i.e., not identical when spin-orbitals are permuted to a different ordering)?

$f^2$ - take any 2 s-o's and list in *standard order*

$\|2\alpha 0\alpha\|$ is OK, but $\|0\alpha 2\alpha\|$ is not in standard order, and $\|2\beta 2\beta\| = 0$.

How many nonzero and distinct Slater determinants are there for $f^2$?

$$\left.\begin{array}{l}14 \text{ spin-orbitals} \\ 2 \text{ identical electrons}\end{array}\right] \frac{14\cdot 13}{2} = \mathbf{91} \quad \textbf{Slater determinants!}$$

$$\text{general } (n\ell)^p : \prod_{n\ell} \frac{\left[2(2\ell+1)\right]!}{\left[2(2\ell+1)-p\right]!}\frac{1}{p!}$$

put p indistinguishable $e^-$ *and* $2(2\ell+1)$ –p indistinguishable "holes" into $2(2\ell+1)$ boxes

subshell : one such factor for each subshell

How to generate all 91 linear combinations of Slater determinants that correspond to the 91 possible $|LM_L SM_S\rangle$ basis states that arise from $f^2$? Next lecture.

all of these are labor intensive

* ladders plus orthogonality ← you know how to do this
* construct and diagonalize $\mathbf{L}^2$ and $\mathbf{S}^2$ matrices
* projection operators
* 3-j, 6-j, 9-j coefficients ← you should begin to know this

Sometimes all we want to know is "which L-S terms"?
[WHY? $1/r_{ij}$ is scalar with respect to **L,S**, and **J**, thus eigenenergies are independent of $M_L$, $M_S$ and $M_J$.]

EASY because we can read $\mathbf{L}_z = \sum_i \ell_{1z}$ and $\mathbf{S}_z = \sum_i \mathbf{s}_{1z}$ directly from the spin-orbital labels.

$$L_z \left|\left|2\alpha 1\beta\right|\right| = \sum_{\substack{i=1 \\ M_L = 3}}^{\tilde{}} \ell_{iz} \left|\left|2\alpha 1\beta\right|\right| = \hbar\left[2+1\right]\left|\left|2\alpha 1\beta\right|\right|$$

$M_L$ is sum of $m_\ell$'s
$M_S$ is sum of $m_S$'s

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NONLECTURE

What about $\mathbf{L}^2$? Can do this in either of two ways:

* as below (very cumbersome)
* $\mathbf{L}^2 = \mathbf{L}_z^2 + (1/2)(\mathbf{L}_+\mathbf{L}_- + \mathbf{L}_-\mathbf{L}_+)$ [separately apply each $1e^-$ operator rather than treat entire operator as a $2e^-$ operator.]

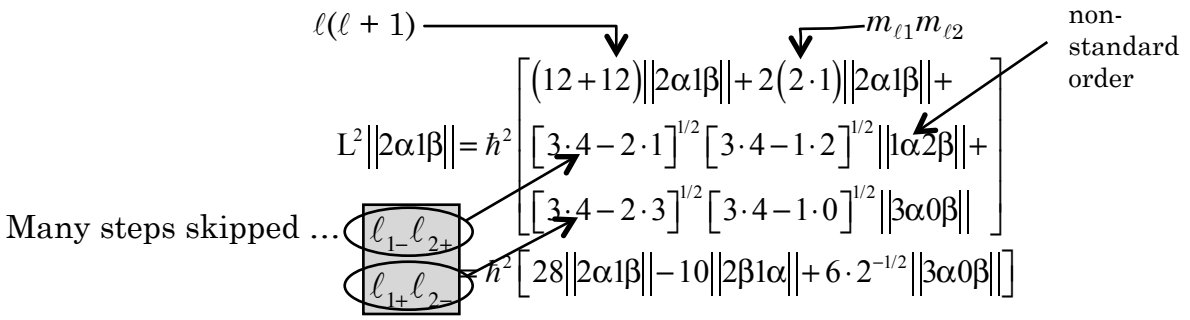
very laborious because $\mathbf{L}^2 = \sum_{i,j} \ell_i \cdot \ell_j = \underbrace{\sum \ell_i^2}_{\text{one e}^-} + 2\underbrace{\sum_{i>j} \ell_i \ell_j}_{\text{two-e}^-}$

$$\mathbf{L}^2 \left|\left|2\alpha 1\beta\right|\right| \neq \sum_i \hbar^2 \ell_i \left(\ell_i + 1\right) \left|\left|2\alpha 1\beta\right|\right| \qquad \ell_i = 3 \text{ for } f$$

WORK OUT $\mathbf{L}^2$ matrix for $M_L = 3$, $M_S = 0$ block of $f^2$ for future reference

$$\mathbf{L}^2 = \sum_{i,j} \ell_i \cdot \ell_j = \underbrace{\sum_i \left[\ell_i^2\right]}_{\Delta\ell = 0,\ \Delta M_\ell = 0} + 2\sum_{i>j} \left[\underbrace{\ell_{iz}\ell_{jz}}_{\Delta\ell = 0,\ \Delta M_\ell = 0} + \underbrace{\frac{1}{2}\left(\ell_{i+}\ell_{j-} + \ell_{i-}\ell_{j+}\right)}_{\substack{\Delta\ell = 0,\ \Delta M_\ell = 0 \\ \Delta m_{\ell 1} = -\Delta m_{\ell 2} = \pm 1}}\right]$$

all are $\Delta M_S = \Delta m_{s_1} = \Delta m_{s_2} = 0$




All of the 12, 21 exchange-type matrix elements are 0 because of $m_s$ mismatch.

e.g. $\left\langle 2\alpha(1)1\beta(1) \right|$ space only operator $\left| 1\beta(1)2\alpha(2) \right\rangle = 0$

Recall $\pm(\langle 12|G|12\rangle - \langle 12|G|21\rangle)$ for $2e^-$ operator.
We get:

$$\mathbf{L}^2\|2\beta 1\alpha\| = \hbar^2\left[28\|2\beta 1\alpha\| - 10\|2\alpha 1\beta\| + \left(12\cdot 2^{-1/2}\right)\|3\beta 0\alpha\|\right]$$

$$\mathbf{L}^2\|3\alpha 0\beta\| = \hbar^2\left[(24+3\cdot 0)\|3\alpha 0\beta\| + \left(12\cdot 2^{-1/2}\right)\|2\alpha 1\beta\|\right]$$

$$\mathbf{L}^2\|3\beta 0\alpha\| = \hbar^2\left[24\|3\beta 0\alpha\| + \left(12\cdot 2^{-1/2}\right)\|2\beta 1\alpha\|\right]$$

$$L^2 = \hbar^2 \begin{array}{c} \\ \|3\alpha 0\beta\| \\ \|2\alpha 1\beta\| \\ -\|1\alpha 2\beta\| \\ -\|0\alpha 3\beta\| \end{array} \begin{pmatrix} 24 & 6\cdot 2^{1/2} & 0 & 0 \\ 6\cdot 2^{1/2} & 28 & -10 & 0 \\ 0 & -10 & 28 & 6\cdot 2^{1/2} \\ 0 & 0 & 6\cdot 2^{1/2} & 24 \end{pmatrix}$$

[the bottom two Slater determinants are intentionally out of standard order to display effects of decreasing values of $m_\ell(1)$ and increasing values of $m_\ell(2)$.]

$\left|LSM_L M_S\right\rangle = \left|5130\right\rangle$

Example: found eigenvalues and eigenvectors of this block $M_L = 3$, $M_S = 0$ of $f^2$

$$\frac{L^2}{\hbar^2}\left[3^{-1/2}\left\|3\alpha 0\beta\right\| + 3^{-1/2}\left\|3\beta 0\alpha\right\| + 6^{-1/2}\left\|2\alpha 1\beta\right\| + 6^{-1/2}\left\|2\beta 1\alpha\right\|\right] = 30[\ ] \quad (L = 5)$$

$$\frac{L^2}{\hbar^2}\left[6^{-1/2}\left\|3\alpha 0\beta\right\| + 6^{-1/2}\left\|3\beta 0\alpha\right\| - 3^{-1/2}\left\|2\alpha 1\beta\right\| - 3^{-1/2}\left\|2\beta 1\alpha\right\|\right] = 12[\ ] \quad (L = 3)$$

$$\frac{L^2}{\hbar^2}\left[11^{-1/2}\left\|3\alpha 0\beta\right\| - 11^{-1/2}\left\|3\beta 0\alpha\right\| + 3\cdot 22^{-1/2}\left\|2\alpha 1\beta\right\| - 3\cdot 22^{-1/2}\left\|2\beta 1\alpha\right\|\right] = 42[\ ] \quad (L = 6)$$

$$\frac{L^2}{\hbar^2}\left[3\cdot 22^{-1/2}\left\|3\alpha 0\beta\right\| - 3\cdot 22^{-1/2}\left\|3\beta 0\alpha\right\| - 11^{-1/2}\left\|2\alpha 1\beta\right\| + 11^{-1/2}\left\|2\beta 1\alpha\right\|\right] = 20[\ ] \quad (L = 4)$$

(Note how easy it is to see that normalization is correct.) Look at the sum of the squares of each coefficient!

a lot of algebra is not presented here! (especially the derivation of the 4 eigenvectors)

* each Slater basis state gets "used up" [sum of squares of that basis set is 1]
* the first 2 eigenfunctions are in the form: $\alpha\ \beta\ +\ \beta\ \alpha \rightarrow S = 1$
* the second 2 eigenfunctions are in the form: $\alpha\ \beta\ -\ \beta\ \alpha \rightarrow S = 0$

You could prove these S = 1 and S = 0 results by applying $\mathbf{S}^2$ to above eigenfunctions of $\mathbf{L}^2$.

We have obtained $\left|LSM_L M_S\right\rangle = \left|5130\right\rangle, \left|3130\right\rangle, \left|6030\right\rangle$, and $\left|4030\right\rangle$ eigenstates.

END OF NON-LECTURE

Non-lecture pages were intended to show that applying $\mathbf{L}^2$ and $\mathbf{S}^2$ to Slater determinants is laborious — much more so than applying $\mathbf{L}_z$ and $\mathbf{S}_z$.

This is one reason why we use the "crossing out $M_L$, $M_S$ microstates" method to figure out which L,S states must be considered. Often this is sufficient — and it can be the basis for some shortcut tricks!

The $M_L$, $M_S$ method works because:

* each configuration generates the full (2L + 1) (2S + 1) manifold of $M_L$, $M_S$ states associated with each L,S term. Why? If you have one $|M_L M_S\rangle$ member of $|LM_L SM_S\rangle$ you can generate all of the others for that L,S using $\mathbf{L}_\pm$ and $\mathbf{S}_\pm$ operators.
* This must be true because, starting with $M_L = L$, $M_S = S$, $\mathbf{L}_-$ and $\mathbf{S}_-$ can be used to generate all $M_L$, $M_S$ components of the full L,S term without the need to go outside the specific configuration.

$M_L$, $M_S$ method

| $M_S$ | $M_L = L_{MAX}$ | $L_{MAX} - 1$ | $L_{MAX} - 2$ | … | 0 |
|---|---|---|---|---|---|
| $M_S = S_{MAX}$ | list all Slater determinants | | | | |
| $S - 1$ | | | | | |
| 0 | | | | | |

$S_{MAX} = (\#\text{ of } e^-)/2.$

No need to include negative values of $M_S$ or $M_L$. Why? They are accessed by $\mathbf{L}_- + \mathbf{S}_-$ and contain no new information.

$f^2$

| $M_S$ \ $M_L$ | 6(I) | 5(H) | 4(G) | 3(F) | 2(D) | 1(P) | 0(S) |
|---|---|---|---|---|---|---|---|
| 1 | ~~$\|3\alpha 3\alpha\|$~~ | $\|3\alpha 2\alpha\|$ | $\|3\alpha 1\alpha\|$ | $\|3\alpha 0\alpha\|$ | $\|2\alpha 0\alpha\|$ | $\|3\alpha - 2\alpha\|$ | $\|3\alpha - 3\alpha\|$ |
| exclusively S = 1 states | | | ~~$\|2\alpha 2\alpha\|$~~ | $\|2\alpha 1\alpha\|$ | $\|3\alpha - 1\alpha\|$ | $\|2\alpha - 1\alpha\|$ | $\|2\alpha - 2\alpha\|$ |
| | | | | | ~~$\|1\alpha 1\alpha\|$~~ | $\|1\alpha 0\alpha\|$ | $\|1\alpha - 1\alpha\|$ |
| | | | | | | | ~~$\|0\alpha 0\alpha\|$~~ |
| 0 | $\|3\alpha 3\beta\|$ | $\|3\alpha 2\beta\|$ | $\|3\alpha 1\beta\|$ | $\|3\alpha 0\beta\|$ | $\|2\alpha 0\beta\|$ | $\|3\alpha - 2\beta\|$ | $\|3\alpha - 3\beta\|$ |
| | | $\|3\beta 2\alpha\|$ | $\|3\beta 1\alpha\|$ | $\|3\beta 0\alpha\|$ | $\|2\beta 0\alpha\|$ | $\|3\beta - 2\alpha\|$ | $\|3\beta - 3\alpha\|$ |
| Both S = 1 and S = 0 states | | | $\|2\alpha 2\beta\|$ | $\|2\alpha 1\beta\|$ | $\|3\alpha - 1\beta\|$ | $\|2\alpha - 1\beta\|$ | $\|2\alpha - 2\beta\|$ |
| | | | | $\|2\beta 1\alpha\|$ | $\|3\beta - 1\alpha\|$ | $\|2\beta - 1\alpha\|$ | $\|2\beta - 2\alpha\|$ |
| | | | | | $\|1\alpha 1\beta\|$ | $\|1\alpha 0\beta\|$ | $\|1\alpha - 1\beta\|$ |
| | | | | | | $\|1\beta 0\alpha\|$ | $\|1\beta - 1\alpha\|$ |
| | | | | | | | $\|0\alpha 0\beta\|$ |

Slaters for $f^2$

Need not include $M_S < 0$ or $M_L < 0$ because these are identical to the $M_L > 0$ and $M_S > 0$ quadrant.

Notice that as you go down by 1 in $M_L$, the number of Slater determinants in each $M_L$, $M_S$ box increases only by 1. This is a prerequisite for using the **L**_ plus orthogonality method! This useful simplicity does not occur as you go down a column in $M_S$.

This convenient situation does not occur for $d^3$ or $f^3$. Why? Because there can be more than one *L-S* term of a specified symmetry. For example, for $d^2$ there are $^1S$, $^3P$, $^1D$, $^3F$, $^1G$ terms, but for $d^3$ there are $^2P$, $^4P$, **two** $^2D$, $^2F$, $^4F$, $^2G$ and $^2H$ terms.

No J by convention

| | S | P | D | F | G | H | I | K |
|---|---|---|---|---|---|---|---|---|
| L = | 0, | 1, | 2, | 3, | 4, | 5, | 6, | 7 |

Start in extreme $M_L$, $M_S$ corner — This generally contains only one Slater determinant

$L = M_{L_{MAX}}$ , $S = M_{S_{MAX}}$ so we have one of the L - S terms

This L-S term includes one of each $M_L$, $M_S$ in the range

$$-L \le M_L \le L$$
$$-S \le M_S \le S$$

This means this L-S term will "use up" the equivalent of one Slater determinant in each $M_L$,$M_S$ box.

Bookkeeping: cross out one Slater determinant, any one, from each relevant $M_L$,$M_S$ box.

Now repeat, again starting at the remaining extreme $M_L$,$M_S$ corner

| | | | |
|---|---|---|---|
| etc. | * $^1$I | $1 \times 13$ | = 13 |
| | * $^3$H | $3 \times 11$ | = 33 |
| | * $^1$G | $1 \times 9$ | = 9 |
| | * $^3$F | $3 \times 7$ | = 21 |
| | * $^1$D | $1 \times 5$ | = 5 |
| | * $^3$P | $3 \times 3$ | = 9 |
| | * $^1$S | $1 \times 1$ | = 1 |
| | | | 91 |

as required! [It is a good idea to use this total degeneracy of the configuration as a check.]

Since there is only one Slater determinant in the $M_L = 5$, $M_S = 1$ box, generate all triplets by repeated application of $\mathbf{L}_-$ to $||3\alpha\, 2\alpha||$ (plus orthogonality) and generate all singlets by $\mathbf{L}_-$ on $\|3\alpha\, 3\beta\|$. Many orthogonalization steps are needed! Especially for singlets. Need to use $\mathbf{S}_-$ also.

Before illustrating the ladders plus orthogonality method, it is useful to show some patterns and list some valuable tricks.

The most difficult cases are $(n\ell)^m$ where m = 2, 3,… $2\ell$.

Easy to combine n$\ell$ with n′$\ell'$ because no need for special bookkeeping.

| $\ell$ | $(n\ell)^2$ | $(n\ell)^3$ |
|---|---|---|
| s | $^1S$ | — |
| p | $^1D$, $^3P$, $^1S$ | $^4S$, $^2D$, $^2P$ |
| d | $^1G$, $^3F$, $^1D$, $^3P$, $^1S$ | $^2H$, $^2G$, $^2F$, $^4F$, $^2D(2)$, $^4P$, $^2P$ |
| f | $^1I$, $^3H$, $^1G$, $^3F$, $^1D$, $^3P$, $^1S$ | |
| | A simple, memorable pattern | Rather complicated |

Get the same L-S states for 2 and 3 "holes" (e.g. $p^4 \leftrightarrow p^2$, $d^3 \leftrightarrow d^7$) instead of electrons.

Also $\left(n\ell\right)^2 n'\ell' \rightarrow \left[n\ell^{2\ 2S+1}L\right] \otimes \left(^2\ell'\right) = {}^{(2S+2,\text{ and } 2S)}\left(L+\ell', L+\ell'-1, \cdots \left|L-\ell'\right|\right)$

Simple vector coupling of the $n'\ell'$ electron to the two-electron $n\ell^{2\ 2S+1}$L term. No Pauli exclusion because $n'\ell'$ is distinguished from $n\ell$.

When the $e^-$ are in distinct subshells (different values of $\ell$ and $n$), there is no need to be as careful about the exclusion principle.

Ladders plus Orthogonality Method

$f^2$ example

Start with 2 extreme UNIQUE states

1. $\left|{}^3\mathrm{H}\ M_L = 5, M_s = 1\right\rangle = \|3\alpha 2\alpha\|$

Use this to generate all triplet states by applying $\mathbf{L}_-$ repeatedly and using orthogonality when necessary. Note that # of determinants in each $M_L$,$M_S$=1 box increases no faster than in steps of 1.

To get to ${}^3$P, must not only apply orthogonality several times, but must follow each L state down to the $M_L$ = 1 box!

2. To get singlets, start with the unique $|{}^1\mathrm{I}\ M_L = 6, M_s = 0\rangle$ state.

Again, as $\mathbf{L}_-$ takes us to successively lower-$M_L$ boxes, # of determinants increases in steps of 1. But some of these steps are due to triplets with $M_S = 0$. Need to step triplets down into $M_S = 0$ territory using $\mathbf{S}_-$ once. Lots more orthogonality steps, lots more trails being followed. AWFUL, but do-able.

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Nonlecture

$$\left|{}^3\mathrm{H}\ M_L M_S\right\rangle =$$

$$\mathbf{L}_-\left|{}^3\mathrm{H}\ 51\right\rangle = \sum_i \ell_{i^-}\|3\alpha 2\alpha\|$$

$$\hbar\left[5\cdot 6 - 5\cdot 4\right]^{1/2}\left|{}^3\mathrm{H}\ 41\right\rangle = \hbar\left[3\cdot 4 - 3\cdot 2\right]^{1/2}\|2\alpha 2\alpha\| + \hbar\left(3\cdot 4 - 2\cdot 1\right)^{1/2}\|3\alpha 1\alpha\|$$

($\|2\alpha 2\alpha\| \to 0$)

$$\left|{}^3\mathrm{H}\ 41\right\rangle = \|3\alpha 1\alpha\| \qquad \text{big surprise!}$$

$$\mathbf{L}_-\left|{}^3\mathrm{H}\ 41\right\rangle = \Sigma\ell_{i^-}\|3\alpha 1\alpha\|$$

$$\left|{}^3\mathrm{H}\ 31\right\rangle = \left(1/3\right)^{1/2}\|2\alpha 1\alpha\| + \left(2/3\right)^{1/2}\|3\alpha 0\alpha\|$$

orthogonality: $$\left|{}^3\mathrm{F}\ 31\right\rangle = \left(\frac{2}{3}\right)^{1/2}\|2\alpha 1\alpha\| - \left(\frac{1}{3}\right)^{1/2}\|3\alpha 0\alpha\|$$

and so on, to get all $|\,{}^3\mathrm{L}\ \mathrm{L}\ 1\rangle$ ($M_L$, $M_S$) many-electron functions

$M_S = 0$

Try a detour into singlet territory, and then check for self-consistency.

$$\mathbf{S}_- \left| \; {}^3\mathrm{F} \quad 31 \; \right\rangle = \sum_i \mathbf{s}_{i^-} \left[ \left(\frac{2}{3}\right)^{1/2} \left|\left|2\alpha 1\alpha\right|\right| - \left(\frac{1}{3}\right)^{1/2} \left|\left|3\alpha 0\alpha\right|\right| \right]$$

(by orthogonality with $\left| {}^3\mathrm{H} \quad 31 \right\rangle$)

$$\hbar\left[1\cdot 2 - 1\cdot 0\right]^{1/2} \left| \; {}^3\mathrm{F} \quad 30 \; \right\rangle = \hbar \left[ \left(\frac{2}{3}\right)^{1/2} \left[ \frac{1}{2}\cdot\frac{3}{2} - \frac{1}{2}\left(-\frac{1}{2}\right) \right]^{1/2} \left( \left|\left|2\beta 1\alpha\right|\right| + \left|\left|2\alpha 1\beta\right|\right| \right) \right.$$

$$\left. - \left(\frac{1}{3}\right)^{1/2} \left[1\right]^{1/2} \left( \left|\left|3\beta 0\alpha\right|\right| + \left|\left|3\alpha 0\beta\right|\right| \right) \right]$$

this factor is always 1 or 0

$$\left| {}^3\mathrm{F}\, 30 \right\rangle = \left(\frac{1}{3}\right)^{1/2} \left( \left|\left|2\beta 1\alpha\right|\right| + \left|\left|2\alpha 1\beta\right|\right| \right) - \left(\frac{1}{6}\right)^{1/2} \left( \left|\left|3\beta 0\alpha\right|\right| + \left|\left|3\alpha 0\beta\right|\right| \right)$$

$$\mathbf{S}_- \left| {}^3\mathrm{H}\, 31 \right\rangle = \sum_i s_{i^-} \left[ \left(\frac{1}{3}\right)^{1/2} \left|\left|2\alpha 1\alpha\right|\right| + \left(\frac{2}{3}\right)^{1/2} \left|\left|3\alpha 0\alpha\right|\right| \right]$$

$$\left| {}^3\mathrm{H}\, 30 \right\rangle = \left(\frac{1}{6}\right)^{1/2} \left( \left|\left|2\beta 1\alpha\right|\right| + \left|\left|2\alpha 1\beta\right|\right| \right) + \left(\frac{1}{3}\right)^{1/2} \left( \left|\left|3\beta 0\alpha\right|\right| + \left|\left|3\alpha 0\beta\right|\right| \right)$$

There are 4 Slater determinants in the $M_L = 3$, $M_S = 0$ box. We can't find the other two singlet linear combinations uniquely without using $\mathbf{L}_-$ on the extreme singlets.

$$L_- \left| \; {}^1\mathrm{I} \quad 60 \; \right\rangle = \Sigma \ell_{i^-} \left|\left|3\alpha 3\beta\right|\right|$$

$$\hbar\left[6\cdot 7 - 6\cdot 5\right]^{1/2} \left| \; {}^1\mathrm{I} \quad 50 \; \right\rangle = \hbar\left[3\cdot 4 - 3\cdot 2\right]^{1/2} \left( \left|\left|2\alpha 3\beta\right|\right| + \left|\left|3\alpha 2\beta\right|\right| \right)$$

wrong order: must change sign to reverse order

$$\left| \; {}^1\mathrm{I} \quad 50 \; \right\rangle = \left(\frac{1}{2}\right)^{1/2} \left[ \left|\left|3\alpha 2\beta\right|\right| - \left|\left|3\beta 2\alpha\right|\right| \right] \quad \text{orthogonality} \quad \left| \; {}^3\mathrm{H} \quad 50 \; \right\rangle = \left(\frac{1}{2}\right)^{1/2} \left[ \left|\left|3\alpha 2\beta\right|\right| + \left|\left|3\beta 2\alpha\right|\right| \right]$$

$$L_- \left| \; {}^1\mathrm{I} \quad 50 \; \right\rangle = \Sigma \ell_{i^-} \left(\frac{1}{2}\right)^{1/2} \left[ \left|\left|3\alpha 2\beta\right|\right| - \left|\left|3\beta 2\alpha\right|\right| \right]$$

wrong order

$$\left|{}^1\mathrm{I}\,40\right\rangle=\left(\frac{1}{44}\right)^{1/2}\left[(10)^{1/2}\left(\left|\left|3\alpha1\beta\right|\right|-\left|\left|3\beta1\alpha\right|\right|\right)+6^{1/2}\left(\left|\left|2\alpha2\beta\right|\right|-\left|\left|2\beta2\alpha\right|\right|\right)\right]$$

$$\left|{}^1\mathrm{I}\,40\right\rangle=\left(\frac{5}{22}\right)^{1/2}\left[\left(\left|\left|3\alpha1\beta\right|\right|-\left|\left|3\beta1\alpha\right|\right|\right)+\left(\frac{6}{11}\right)\left|\left|2\alpha2\beta\right|\right|\right]$$

$$\left|{}^3\mathrm{H}\,40\right\rangle=\left(\frac{1}{20}\right)^{1/2}\left[(6)^{1/2}\left(\left|\left|2\alpha2\beta\right|\right|+\left|\left|2\beta2\alpha\right|\right|\right)+10^{1/2}\left(\left|\left|3\alpha1\beta\right|\right|+\left|\left|3\beta1\alpha\right|\right|\right)\right]$$

wrong order

$$\left|{}^3\mathrm{H}\,40\right\rangle=\left(\frac{1}{2}\right)^{1/2}\left(\left|\left|3\alpha1\beta\right|\right|+\left|\left|3\beta1\alpha\right|\right|\right)$$

orthogonality

$$\left|{}^1\mathrm{G}\,40\right\rangle=\left(\frac{3}{11}\right)^{1/2}\left[\left(\left|\left|3\alpha1\beta\right|\right|-\left|\left|3\beta1\alpha\right|\right|\right)-\left(\frac{5}{11}\right)^{1/2}\left|\left|2\alpha2\beta\right|\right|\right]$$

At last we are ready to enter the $M_L = 3$, $M_S = 0$ block!

It is clear that if we apply $\mathbf{L}_-$ to $|{}^3\mathrm{H}\ 40\rangle$, we will get the same form that we already derived starting from $|{}^3\mathrm{H}\ 51\rangle$ Instead, let's lower $|{}^1\mathrm{I}\ 40\rangle$.

$$\mathbf{L}_-\left|{}^1\mathrm{I}\ \ 40\right\rangle=\sum_i \ell_{i^-}\left\{\left(\frac{5}{22}\right)^{1/2}\left[\left|\left|3\alpha1\beta\right|\right|-\left|\left|1\beta3\alpha\right|\right|\right]+\left(\frac{6}{11}\right)^{1/2}\left|\left|2\alpha2\beta\right|\right|\right\}$$

$$\left|{}^1\mathrm{I}\ \ 30\right\rangle=(30)^{1/2}\left\{\left(\frac{5}{22}\right)^{1/2}(6)^{1/2}\left(\left|\left|2\alpha1\beta\right|\right|-\left|\left|2\beta1\alpha\right|\right|\right)+\left(\frac{5}{22}\right)^{1/2}(12)^{1/2}\left(\left|\left|3\alpha0\beta\right|\right|-\left|\left|3\beta0\alpha\right|\right|\right)\right.$$
$$\left.+\left(\frac{6}{11}\right)^{1/2}(10)^{1/2}\left(\left|\left|2\alpha1\beta\right|\right|-\left|\left|2\beta1\alpha\right|\right|\right)\right\}$$

$$\left|{}^1\mathrm{I}\ \ 30\right\rangle=\left[\left(\frac{1}{22}\right)^{1/2}+\left(\frac{4}{22}\right)^{1/2}\right]\left(\left|\left|2\alpha1\beta\right|\right|-\left|\left|2\beta1\alpha\right|\right|\right)+\left(\frac{2}{22}\right)^{1/2}\left(\left|\left|3\alpha0\beta\right|\right|-\left|\left|3\beta0\alpha\right|\right|\right)$$

$$\left|{}^1\mathrm{I}\quad 30\right\rangle=\left(\frac{9}{22}\right)^{1/2}\left(\left\|2\alpha1\beta\right\|-\left\|2\beta1\alpha\right\|\right)+\left(\frac{2}{22}\right)^{1/2}\left(\left\|3\alpha0\beta\right\|-\left\|3\beta0\alpha\right\|\right)$$

Finally, by orthogonality:

IMPORTANT →

$$\left|{}^1\mathrm{G}\quad 30\right\rangle=-\left(\frac{1}{11}\right)^{1/2}\left(\left\|2\alpha1\beta\right\|-\left\|2\beta1\alpha\right\|\right)+\left(\frac{9}{22}\right)^{1/2}\left(\left\|3\alpha0\beta\right\|-\left\|3\beta0\alpha\right\|\right)$$

Does this match what one would get from $\mathbf{L}_-\,|{}^1G\ 40\rangle$?

$$\mathbf{L}_- \left|{}^1\mathrm{G}\ \ 40\right\rangle = \sum_i \ell_{i^-} \left\{ \left(\frac{3}{11}\right)^{1/2} \left[ \left|\left|3\alpha 1\beta\right|\right| - \left|\left|1\beta 3\alpha\right|\right| \right] - \left(\frac{5}{11}\right)^{1/2} \left|\left|2\alpha 2\beta\right|\right| \right\}$$

$$\left|{}^1\mathrm{G}\ \ 30\right\rangle = (8)^{1/2} \left\{ \left(\frac{5}{11}\right)^{1/2} (6)^{1/2} \left( \left|\left|2\alpha 1\beta\right|\right| - \left|\left|2\beta 1\alpha\right|\right| \right) + \left(\frac{3}{11}\right)^{1/2} (12)^{1/2} \left( \left|\left|3\alpha 0\beta\right|\right| - \left|\left|3\beta 0\alpha\right|\right| \right) \right.$$
$$\left. - \left(\frac{5}{11}\right)^{1/2} (10)^{1/2} \left( \left|\left|2\alpha 1\beta\right|\right| - \left|\left|2\beta 1\alpha\right|\right| \right) \right\}$$

IMPORTANT →
$$\left|{}^1\mathrm{G}\ \ 30\right\rangle = -\left(\frac{1}{11}\right)^{1/2} \left( \left|\left|2\alpha 1\beta\right|\right| - \left|\left|2\beta 1\alpha\right|\right| \right) + \left(\frac{9}{22}\right)^{1/2} \left( \left|\left|3\alpha 0\beta\right|\right| - \left|\left|3\beta 0\alpha\right|\right| \right)$$

**checks!**

End of Non-Lecture

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As you see, this ladders plus orthogonality method is extremely laborious. There is a much better way!

** There are several patterns: singlets for $M_S = 0$ always have the form $(\alpha\beta - \beta\alpha)$ and $M_S = 0$ triplets always $(\alpha\beta + \beta\alpha)$.

This can be generalized for any value of S (page 151 of Hélène Lefebvre-Brion-Robert Field Perturbations 2004 book)
[Also M. Yamazaki, Sci. Rep. Kanezawa Univ. 8, 371 (1963).]

2. **Failure and Inconvenience of ladder method**
The ladder method is OK when you have a single target $|LM_L SM_S\rangle$ state, especially when it is near an edge of the $M_L$,$M_S$ box diagram. Essential that # of Slater determinants in each $M_L M_S$ box increases in steps of 1 as you step down in $M_L$ or $M_S$.

Fails when there are two L-S terms of same L and S in a given configuration. Then we must set up a 2 × 2 secular equation anyway.

e.g. $(nd)^3$ $^2\mathrm{H}, ^2\mathrm{G}, ^2\mathrm{F}, ^4\mathrm{F}, \boxed{^2\mathrm{D}(2),} ^4\mathrm{P}, ^2\mathrm{P}$

3. **$\mathbf{L}^2$ and $\mathbf{S}^2$ Matrix Method**
Another method is based on constructing $\mathbf{L}^2$ and $\mathbf{S}^2$ matrices in the Slater determinantal basis set. This is no cakewalk either (but this is easier)!

Since usually $S_{MAX} \ll L_{MAX}$ for a configuration when using $\mathbf{L}^2 + \mathbf{S}^2$ matrices method, it is best to start with the $\mathbf{S}^2$ matrix because it is simpler.

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# L-S Terms via $L^2$, $S^2$ and Projection

LAST TIME:

* method of $M_L$, $M_S$ boxes. [For $^3$L states, cross out boxes starting from both ($M_L$=L, $M_S$=1) and ($M_L$=L, $M_S$=0).]
* there must be a complete (2L + 1)(2S + 1) dimensionality for each L-S term [# of boxes]
* $n\ell^2$ pattern
* $(n\ell)^2$ $n'\ell'$ shortcut
* method of ladders plus orthogonality

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TODAY:

$\mathbf{L}^2$, $\mathbf{S}^2$ method to obtain $|LM_LSM_S\rangle$, especially for the $M_L$,$M_S$ boxes in which the method of ladders plus orthogonality is *most inconvenient*: e.g, $M_L = 0$, $M_S = 0$

* $\mathbf{L}^2 \rightarrow \mathbf{L}_+\mathbf{L}_-$ only for $M_L = 0$ block. Every L–S term in the configuration is represented in this most evil block.
* set up and diagonalize $\mathbf{S}^2$ — easy — by forming ± linear combinations
  (singlet and triplet)
  $\alpha\beta - \beta\alpha$ $\alpha\beta + \beta\alpha$
* transform $\mathbf{L}^2$ to singlet, triplet basis using the transformation that diagonalizes $\mathbf{S}^2$ (block diagonalization), then complete the diagonalization of $\mathbf{L}^2$ by knowing (from crossing out boxes method) the $\mathbf{L}^2$ eigenvalues: L(L + 1)

other, strong spin-orbit basis sets

Modern calculations use projection operators: designed to project away all unwanted parts of $\psi$ yet preserve normalization.

Look at the $M_L = 0$, $M_S = 0$ block of $f^2$ and construct all L – S basis states. All extant L-S terms of $f^2$ are represented once in the $M_L = M_S = 0$ block. Never try to get to this block by ladders and orthogonality!

$$\psi_1 = \|3\alpha - 3\beta\|$$
$$\psi_2 = \|3\beta - 3\alpha\|$$
$$\psi_3 = \|2\alpha - 2\beta\|$$
$$\psi_4 = \|2\beta - 2\alpha\|$$
$$\psi_5 = \|1\alpha - 1\beta\|$$
$$\psi_6 = \|1\beta - 1\alpha\|$$
$$\psi_7 = \|0\alpha 0\beta\|$$

This is for $f^2$. Do $d^2$ in lecture

Exploit cute trick that works especially well in the $M_L = 0$, $M_S = 0$ block because many otherwise awful terms vanish.

$$L^2 = L_z^2 + \frac{1}{2}\left(L_+L_- + L_-L_+\right) = L_z^2 + \frac{1}{2}\left(L_+L_- + L_+L_- - \left[L_+, L_-\right]\right)$$

$$\left[L_+, L_-\right] = 2\hbar L_z$$

$$L^2 = \underbrace{L_z^2 - \hbar L_z}_{\substack{\text{diagonal but} \\ \text{vanishes in} \\ M_L = 0}} + \underbrace{L_+L_-}_{\text{nondiagonal}} \qquad \text{(same as } L^2 = L_z^2 + \hbar L_z + L_-L_+\text{)}$$

So for the $M_L = 0$ block *only*, can replace $\mathbf{L}^2$ by $\mathbf{L}_+\mathbf{L}_-$ (or $\mathbf{L}_-\mathbf{L}_+$) and, for $M_S = 0$ only, replace $\mathbf{S}^2$ by $\mathbf{S}_+\mathbf{S}_-$ (or $\mathbf{S}_-\mathbf{S}_+$).

For $f^2$:

$$\mathbf{S}^2\left|\left|3\alpha-3\beta\right|\right| = \mathbf{S}_+\mathbf{S}_-\left|\left|3\alpha-3\beta\right|\right| = \mathbf{S}_+\left|\left|3\beta-3\beta\right|\right| = \left|\left|3\alpha-3\beta\right|\right| + \left|\left|3\beta-3\alpha\right|\right|$$

$$\mathbf{L}^2\left|\left|3\alpha-3\beta\right|\right| = \mathbf{L}_+\mathbf{L}_-\left|\left|3\alpha-3\beta\right|\right| = \mathbf{L}_+ 6^{1/2}\left|\left|2\alpha-3\beta\right|\right| =$$

ignore factors of $\hbar^2$

$$6^{1/2}\left[\left[12-6\right]^{1/2}\left|\left|3\alpha-3\beta\right|\right| + \left[12-6\right]^{1/2}\left|\left|2\alpha-2\beta\right|\right|\right]$$

$$= 6\left[\left|\left|3\alpha-3\beta\right|\right| + \left|\left|2\alpha-2\beta\right|\right|\right]$$

etc.

| | | |
|---|---|---|
| $\psi_1 = \|3\alpha-3\beta\|$ | $\mathbf{S}^2\psi_1 = \psi_1+\psi_2$ | $\mathbf{L}^2\psi_1 = 6\psi_1+6\psi_3$ |
| $\psi_2 = \|3\beta-3\alpha\|$ | $\mathbf{S}^2\psi_2 = \psi_1+\psi_2$ | $\mathbf{L}^2\psi_2 = 6\psi_2+6\psi_4$ |
| $\psi_3 = \|2\alpha-2\beta\|$ | $\mathbf{S}^2\psi_3 = \psi_3+\psi_4$ | $\mathbf{L}^2\psi_3 = 6\psi_1+16\psi_3+10\psi_5$ |
| $\psi_4 = \|2\beta-2\alpha\|$ | $\mathbf{S}^2\psi_4 = \psi_3+\psi_4$ | $\mathbf{L}^2\psi_4 = 6\psi_2+16\psi_4+10\psi_6$ |
| $\psi_5 = \|1\alpha-1\beta\|$ | $\mathbf{S}^2\psi_5 = \psi_5+\psi_6$ | $\mathbf{L}^2\psi_5 = 10\psi_3+22\psi_5+12\psi_7$ |
| $\psi_6 = \|1\beta-1\alpha\|$ | $\mathbf{S}^2\psi_6 = \psi_5+\psi_6$ | $\mathbf{L}^2\psi_6 = 10\psi_4+22\psi_6-12\psi_7$ |
| $\psi_7 = \|0\alpha-0\beta\|$ | $\mathbf{S}^2\psi_7 = 0$ | $\mathbf{L}^2\psi_7 = 12\psi_5-12\psi_6+24\psi_7$ |
| | all easy | require a bit more work |

Now we know, for $2e^-$, $\mathbf{S}^2$ can only have $2\hbar^2$ and $0\hbar^2$ eigenvalues (triplet and singlet)

diagonalize $\mathbf{S}^2$ by inspection

$t : \alpha\beta+\beta\alpha$
$s : \alpha\beta-\beta\alpha$

$$\psi_{1t} = 2^{-1/2}\left(\psi_1+\psi_2\right) \qquad \psi_{1s} = 2^{-1/2}\left(\psi_1-\psi_2\right)$$

$$\psi_{2t} = 2^{-1/2}\left(\psi_3+\psi_4\right) \qquad \psi_{2s} = 2^{-1/2}\left(\psi_3-\psi_4\right)$$

$$\psi_{3t} = 2^{-1/2}\left(\psi_5+\psi_6\right) \qquad \psi_{3s} = 2^{-1/2}\left(\psi_5-\psi_6\right)$$

$$\psi_{4s} = \psi_7$$ → This also has αβ–βα form

Confirm that these functions diagonalize $\mathbf{S}^2$ and give correct values of diagonal elements. Also, they give orthogonality for singlets with triplets.

a diagonal element: $\left\langle \psi_{1t} \middle| \mathbf{S}^2 \middle| \psi_{1t} \right\rangle = \frac{1}{2}\left\langle \left(\psi_1 + \psi_2\right) \middle| \mathbf{S}^2 \middle| \left(\psi_1 + \psi_2\right) \right\rangle$

$$= \frac{1}{2}\hbar^2 \left\langle \left(\psi_1 + \psi_2\right)\left(2\psi_1 + 2\psi_2\right) \right\rangle = \frac{1}{2}\hbar^2 (2+2) = 2\hbar^2$$

an off-diagonal element: $\left\langle \psi_{1t} \middle| \mathbf{S}^2 \middle| \psi_{1s} \right\rangle = \frac{1}{2}\left\langle \left(\psi_1 + \psi_2\right) \middle| \mathbf{S}^2 \middle| \left(\psi_1 - \psi_2\right) \right\rangle$

$$= \frac{1}{2}\hbar^2 \left\langle \left(\psi_1 + \psi_2\right)\left(\psi_1 + \psi_2 - \psi_1 - \psi_2\right) \right\rangle = 0 \text{ as expected}$$

also $\left\langle \psi_{1s} \middle| \mathbf{S}^2 \middle| \psi_{1s} \right\rangle = 0$

$$\mathbf{S}^2 = \hbar^2 \begin{pmatrix} 2 & & & & & & & \\ & 2 & & & & & \mathbf{0} & \\ & & 2 & & & & & \\ & & & 0 & & & & \\ & & & & 0 & & & \\ & & \mathbf{0} & & & 0 & & \\ & & & & & & 0 & \end{pmatrix} \begin{matrix} 1t \\ 2t \\ 3t \\ 1s \\ 2s \\ 3s \\ 4s \end{matrix}$$

What does $\mathbf{L}^2$ look like when expressed in the basis set that diagonalizes $\mathbf{S}^2$?

$$\mathbf{L}^2\psi_{1t} = 2^{-1/2}\hbar^2\left[6\psi_1 + 6\psi_3 + 6\psi_2 + 6\psi_4\right]$$

$$\left\langle \psi_{1t} \middle| \mathbf{L}^2 \middle| \psi_{1t} \right\rangle = \frac{1}{2}\hbar^2[6+6] = 6\hbar^2$$

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NONLECTURE

$$\left\langle \psi_{2t} \middle| \mathbf{L}^2 \middle| \psi_{1t} \right\rangle = \frac{1}{2}\hbar^2 \left\langle \psi_3 + \psi_4 \middle| 6\psi_1 + 6\psi_3 + 6\psi_2 + 6\psi_4 \right\rangle = \frac{1}{2}\hbar^2\left[6+6\right] = 6\hbar^2$$

$$\left\langle \psi_{2t} \middle| \mathbf{L}^2 \middle| \psi_{2t} \right\rangle = \frac{1}{2}\hbar^2\left[\left\langle \psi_3 + \psi_4 \middle| \mathbf{L}^2 \middle| \psi_3 + \psi_4 \right\rangle\right]$$

$$= \frac{1}{2}\hbar^2 \left\langle \psi_3 + \psi_4 \middle| 6\psi_1 + 16\psi_3 + 10\psi_5 + 6\psi_2 + 16\psi_4 + 10\psi_6 \right\rangle$$

$$= \frac{1}{2}\hbar^2\left(16+16\right) = 16\hbar^2$$

$$\mathbf{L}^2 = \hbar^2 \left(\begin{array}{ccc|cccc} 6 & 6 & 0 & & & & \\ 6 & 16 & 10 & & \mathbf{0} & & \\ 0 & 10 & 22 & & & & \\ \hline & & & 6 & 6 & 0 & 0 \\ & \mathbf{0} & & 6 & 16 & 10 & 0 \\ & & & 0 & 10 & 22 & 24\cdot 2^{-1/2} \\ & & & 0 & 0 & 24\cdot 2^{-1/2} & 24 \end{array}\right) \begin{array}{c} 1t \\ 2t \\ 3t \\ \hline 1s \\ 2s \\ 3s \\ 4s \end{array}$$

These 2 matrices are easier to diagonalize than the full 7 × 7 matrix, especially because we know the eigenvalues in advance!

But our goal is actually the eigen<u>vectors</u> not the eigen<u>values:</u>

$$\underline{\text{TRIPLETS}} \quad \mathrm{L}^2 \left| {}^3\mathrm{H}\ M_L = 0, M_S = 0 \right\rangle = \hbar^2 30 \left| \ {}^3\mathrm{H} \quad 00 \ \right\rangle$$

$$\begin{pmatrix} 6 & 6 & 0 \\ 6 & 16 & 10 \\ 0 & 10 & 22 \end{pmatrix} \begin{pmatrix} a \\ b \\ c \end{pmatrix} = 30 \begin{pmatrix} a \\ b \\ c \end{pmatrix}$$

$(\mathbf{L}^2)$ eigenvector equation

$$6a + 6b + 0c = 30a \qquad \rightarrow b = \frac{24}{6} a = 4a \qquad a = b/4$$

$$6a + 16b + 10c = 30b$$

$$0a + 10b + 22c = 30c \quad \rightarrow b = \frac{8}{10} c$$

$$1 = [a^2 + b^2 + c^2]^{1/2}$$

$$a = 42^{-1/2}$$

$$b = (8/21)^{1/2}$$

$$c = (25/42)^{1/2}$$

$$\left|{}^3\mathrm{H}\quad 00\right\rangle = 42^{-1/2}\psi_{1t} + (8/21)^{1/2}\psi_{2t} + (25/42)^{1/2}\psi_{3t}$$

Similarly,

$$\left|{}^3\mathrm{F}\quad 00\right\rangle = 3^{-1/2}\left(\psi_{1t} + \psi_{2t} - \psi_{3t}\right)$$

$$\left|{}^3\mathrm{P}\quad 00\right\rangle = -\left(\frac{9}{14}\right)^{-1/2}\psi_{1t} + \left(\frac{2}{7}\right)^{1/2}\psi_{2t} - 14^{-1/2}\psi_{3t}$$

Note that each $\psi_{nt}$ basis state gets completely "used up" and all eigenvectors are normalized and mutually orthogonal. You should verify both "used up" and orthogonality.

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Nonlecture: **Singlets**

$$\mathrm{L}^2\left|{}^1\mathrm{I}\quad 00\right\rangle = \hbar^2 42\left|{}^1\mathrm{I}\quad 00\right\rangle$$

$$\begin{pmatrix} 6 & 6 & 0 & 0 \\ 6 & 16 & 10 & 0 \\ 0 & 10 & 22 & 24\bullet 2^{-1/2} \\ 0 & 0 & 24\bullet 2^{-1/2} & 24 \end{pmatrix}\begin{pmatrix} a \\ b \\ c \\ d \end{pmatrix} = 42\begin{pmatrix} a \\ b \\ c \\ d \end{pmatrix}$$

$6a + 6b = 42a \qquad 6b = 36a \Rightarrow \qquad \boxed{a = b/6}$

$6a + 16b + 10c = 42b \qquad 6a + 10c = 26b \qquad 10c = 25b \qquad \boxed{c = \frac{5}{2}b}$

$24\cdot 2^{-1/2}c + 24d = 42d \qquad 24\cdot 2^{-1/2}c = 18d \qquad \boxed{d = \frac{10}{3\cdot 2^{1/2}}b}$

normalization: $\qquad 1 = b\left[\frac{1}{36} + 1 + \frac{25}{4} + \frac{50}{9}\right]^{1/2} \qquad \boxed{b = (6/77)^{1/2}}$

$$\left|{}^1\mathrm{I}\ 00\right\rangle = \frac{1}{6}\left(\frac{6}{77}\right)^{1/2}\psi_{1s} + \left(\frac{6}{77}\right)^{1/2}\psi_{2s} + \frac{5}{2}\left(\frac{6}{77}\right)^{1/2}\psi_{3s} + \frac{10}{3}\left(\frac{3}{77}\right)^{1/2}\psi_{4s}$$

A lot of algebra skipped here:

$$\left|{}^1\mathrm{G}\ 00\right\rangle = \left[\frac{9}{77}\right]^{1/2}\psi_{1s} + \left[\frac{49}{77}\right]^{1/2}\psi_{2s} + \left(\frac{1}{77}\right)^{1/2}\psi_{3s} - \left(\frac{18}{77}\right)^{1/2}\psi_{4s}$$

$$\left|{}^1\mathrm{D}\ 00\right\rangle = -\left(\frac{25}{42}\right)^{1/2}\psi_{1s} + 0\psi_{2s} + \left(\frac{9}{42}\right)^{1/2}\psi_{3s} - \left(\frac{8}{42}\right)^{1/2}\psi_{4s}$$

$$\left|{}^1\mathrm{S}\ 00\right\rangle = -\left(\frac{2}{7}\right)^{1/2}\psi_{1s} + \left(\frac{2}{7}\right)^{1/2}\psi_{2s} - \left(\frac{2}{7}\right)^{1/2}\psi_{3s} + \left(\frac{1}{7}\right)^{1/2}\psi_{4s}$$

Again note that each $\psi_{ns}$ is used up. You should verify orthogonality!

---

Two opposite strategies:

1. ladder down from extreme $M_L$, $M_S$
2. $\mathbf{L}^2 + \mathbf{S}^2$ matrices are large but easy to write out for $M_L = 0$ and $M_S = 0$ ONLY — could then ladder up from any $\mathbf{L}^2$, $\mathbf{S}^2$ eigenfunction (no need to use orthogonality because every L–S state is present in the $M_L = 0$, $M_S = 0$ block).

Before going to Projection Operators, look briefly at the problems associated with deriving 2 other kinds of basis states.

$m_j$

$|j\omega\ell s\rangle$

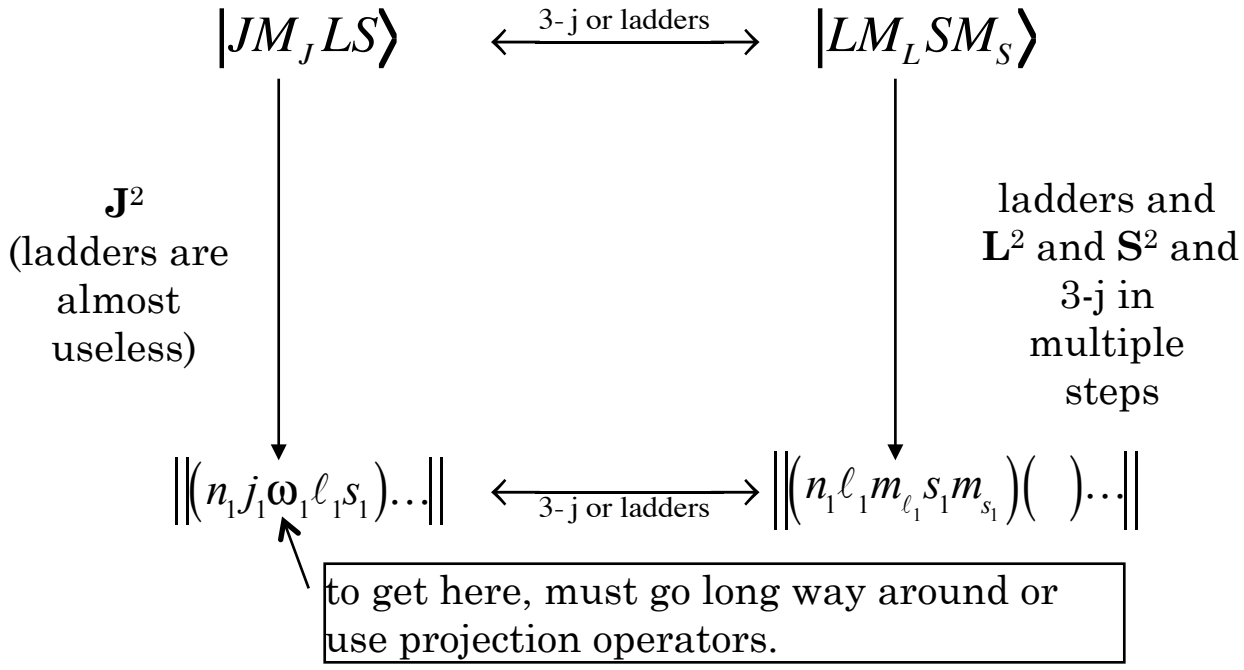
"coupled" orbitals — important for strong spin-orbit limit with HEAVY ATOMS.

($\mathbf{H}^{SO}$ is diagonal in $|j\omega\rangle$ and in $|JM_JLS\rangle$)

$\zeta_{n\ell} \gg$ energy separations between L-S terms

(all $\zeta_{n\ell}$ are $\geq 0$)

$|JM_JLS\rangle$ coupled many-electron L-S-J states.
Again — useful in strong spin-orbit limit



NONLECTURE

BOX METHOD FOR $|j\omega\ell s\rangle$ orbital basis: $(nf)^2$ example
No need to specify $\ell$ and $s$.

Standard Order: (7/2 7/2), (7/2 5/2), (7/2 3/2), (7/2 1/2), (7/2, -1/2), (7/2, -3/2), (7/2 -5/2), (7/2 -7/2), (5/2 5/2), (5/2 3/2), (5/2 1/2), (5/2 -1/2), (5/2 -3/2), (5/2 -5/2)

14 functions $\left|j_1\omega_1 j_2\omega_2\right\rangle$ List only Slater determinants with $M_J \geq 0$.

Suppress the /2's

| # | $M_J$ | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| (0) | 7 | ~~$\|7777\|$~~ | | | | | | | |
| (2) | 6 | $\|7775\|$ | $\|7755\|$ | | | | | | |
| (3) | 5 | $\|7773\|$ | $\|7753\|$ | $\|7555\|$ | | | | | |
| (6) | 4 | $\|7771\|$ | $\|7751\|$ | $\|7573\|$ | $\|7553\|$ | $\|7355\|$ | $\|5553\|$ | | |
| (7) | 3 | $\|777-1\|$ | $\|775-1\|$ | $\|7571\|$ | $\|7551\|$ | $\|7353\|$ | $\|5551\|$ | $\|7155\|$ | |
| (10) | 2 | $\|777-3\|$ | $\|775-3\|$ | $\|777-1\|$ | $\|775-1\|$ | $\|7371\|$ | $\|7351\|$ | $\|7153\|$ | $\|7-155\|$ |
| | | $\|555-1\|$ | $\|5351\|$ | | | | | | |
| (11) | 1 | $\|777-5\|$ | $\|775-5\|$ | $\|757-3\|$ | $\|755-3\|$ | $\|737-1\|$ | $\|735-1\|$ | $\|7151\|$ | $\|7-153\|$ |
| | | $\|7-355\|$ | $\|555-3\|$ | $\|535-1\|$ | | | | | |
| (13) | 0 | $\|777-7\|$ | $\|757-5\|$ | $\|755-5\|$ | $\|737-3\|$ | $\|735-3\|$ | $\|717-1\|$ | $\|715-1\|$ | ~~$\|7-171\|$~~ |
| | | $\|7-151\|$ | ~~$\|7-373\|$~~ | $\|7-353\|$ | $\|7-555\|$ | $\|555-5\|$ | $\|535-3\|$ | $\|515-1\|$ | |

AWFUL! The number of Slater determinants increases in steps larger than 1 as you move down from $M_J = J$. Ladders plus orthogonality cannot work.

Work in the 13 member $M_J = 0$ block

Worst possible one for ladders plus orthogonality.

$\mathbf{J}^2 = \underbrace{\mathbf{J}_z^2 - \hbar\mathbf{J}_z}_{\text{diagonal}} + \mathbf{J}_+\mathbf{J}_-$ [Hopeless to attempt to set up $\mathbf{L}^2$ and $\mathbf{S}^2$ matrices!] Why?

Dimension of Various J blocks:

| J = | Dimension = |
|---|---|
| 6 | 2 |
| 5 | 1 |
| 4 | 3 |
| 3 | 1 |
| 2 | 3 |
| 1 | 1 |
| 0 | 2 |

All blocks are manageable! Ladder up from $M_J = 0$.

coupled basis sets are convenient for $\mathbf{L} \cdot \mathbf{S}$ and $\ell_i \cdot \mathbf{s}_i$ (spin-orbit)

uncoupled basis sets are convenient for $(\mathbf{L}_z + 2\mathbf{S}_z)$ (Zeeman)

Either of the two many-electron basis sets is OK for $\frac{e^2}{r_{ij}}$.
The big problem for $e^2/r_{ij}$ is that it has
many off-diagonal matrix elements in the Slater determinantal basis set. These are extremely tedious to evaluate. The solution to this is the
"Slater Sum Rule" method.
It is based on the fact that the trace of a matrix is equal to the sum of its eigenvalues. This is true regardless of what representation is used to express the matrix.

SUM RULE METHOD: Diagonal matrix elements of $e^2/r_{ij}$ in the Slater determinantal basis set

NEXT TIME

NONLECTURE: Projection Operators

This is an alternative method to set up $|LM_L SM_S\rangle$ or $|\ JLSM_J\rangle$ basis sets in terms of either $n\ell m_\ell s m_s$ or $nj\omega\ell s$ spin-orbital Slater basis sets.

1. Work out $\mathbf{L}^2$ and $\mathbf{S}^2$ matrices for $n\ell m_\ell s m_s$ (or $\mathbf{J}^2$ for $nj\omega\ell s$). These matrices are block diagonal in $M_L, M_S$ (or $M_J$).

2. Construct an operator which, when applied to an arbitrary function, annihilates the undesired part of that function.

   e.g. annihilate L″ by $\left[\hat{\mathbf{L}}^2 - \hbar^2 L''(L''+1)\right]\Psi$

3. Modify the above operator so that it preserves the amplitude of the L′ component of ψ.

   e.g. annihilate L″ , and preserve amplitude of L′

$$\left[\frac{\hat{\mathbf{L}}^2 - \hbar^2 L''(L''+1)}{\hbar^2\left[L'(L'+1) - L''(L''+1)\right]}\right]\Psi \equiv \mathbf{P}\Psi$$

   show how this works by applying it to $\Psi = a\psi_{L'} + b\psi_{L''}$

$$\begin{aligned}\mathbf{P}(a\psi_{L'} + b\psi_{L''}) &= a\frac{L'(L'+1) - L''(L''+1)}{L'(L'+1) - L''(L''+1)}\psi_{L'} + b\frac{L''(L''+1) - L''(L''+1)}{L'(L'+1) - L''(L''+1)}\psi_{L''} \\ &= a\psi_{L'} + 0\psi_{L''}\end{aligned}$$

4. Now recognize that one can build a projection operator that annihilates all of the undesired L″ components by taking a product of operators like that in #3, one for each L″ .

$$\mathbf{P}_{L'} = \prod_{\text{all } L'' \neq L'} \frac{\hat{\mathbf{L}}^2 - \hbar^2 L''(L''+1)}{\hbar^2 L'(L'+1) - \hbar^2 L''(L''+1)}$$

5. Recognize that $\mathbf{P}_{L'}\psi = a_{L'}\psi_{L'}$, which is not normalized, because $a_{L'}$ is the amplitude of $\psi_{L'}$ in $\psi$. Get a normalized $\psi_{L'}$ by recognizing that $\langle \psi_{L'} | \Psi \rangle = a_{L'}$

$$\psi_{L'} = \frac{\mathbf{P}_{L'}\Psi}{\langle \psi_{L'} | \Psi \rangle}$$

This method is useful for dealing with $|JM_JLS\rangle$ in the $|j\omega\ell s\rangle$ orbital basis because there is no *simple* way of block diagonalizing $\mathbf{J}^2$ in terms of $\mathbf{L}^2$ and $\mathbf{S}^2$. It is only possible to block diagonalize $\mathbf{J}^2$ in terms of $M_J$.

Modern calculations will simply set up the $\mathbf{J}^2$,$\mathbf{J}_z$ matrix, diagonalize $\mathbf{J}^2$, and then discover to which eigenvalues of $\mathbf{L}^2$ and $\mathbf{S}^2$ each $\mathbf{J}^2$, $\mathbf{J}_z$ basis function belongs. In many cases two or more L–S terms will contain L–S–J components which belong to the same eigenvalue of $\mathbf{J}^2$.

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## $e^2/r_{ij}$ and Slater Sum Rule Method

LAST TIME:

1. The $\mathbf{L}^2$,$\mathbf{S}^2$ matrix method for setting up $|NLM_LSM_S\rangle$ many-electron basis states in terms of linear combination of Slater determinants $\mathbf{L}^2 \rightarrow \mathbf{L}_+\mathbf{L}_-$
   * $M_L = 0$, $M_S = 0$ block: $\mathbf{S}^2 \rightarrow \mathbf{S}_+\mathbf{S}_-$

   * diagonalize $\mathbf{S}^2$ (singlets and triplets)
   * diagonalize $\mathbf{L}^2$ in same basis that diagonalizes $\mathbf{S}^2$
   [Recall: to get matrix elements of $\mathbf{L}^2$, first evaluate $\mathbf{L}^2 \left\|\psi_i\right\rangle$ and then left multiply by $\left\langle\left\|\psi_j\right\|\right|$]
2. coupled representations $|\text{njw}\ell\text{s}\rangle$ and $|NJLSM_J\rangle$
3. Projection operators: automated projection of $\mathbf{L}^2$ eigenfunctions
   * remove unwanted $L''$ part
   * preserve normalization of wanted $L'$ part
   * remove overlap factor
   * easy to write computer program that automates the projection method

---

TODAY:

1. Slater Sum Rule Trick (based on trace invariance): MAIN IDEA OF LECTURE.

2. Evaluate $\sum_{i>j} e^2/r_{ij}$ matrix elements (tedious, but good for you)
   [$1/r_{ij}$ is a $2-e^-$ operator that involves spatial coordinates only, scalar with respect to **J, L,** and **S**].

   * *multipole expansion* of charge distribution due to "other electrons"
   * matrix element selection rules for $e^2/r_{ij}$ in both Slater determinantal and many-$e^-$ basis sets
   * Gaunt Coefficients ($c^k$) (tabulated) and Slater-Condon ($F^k$, $G^k$) Coulomb and Exchange parameters. Because of the sum rule, can evaluate most $\langle ab\left|\frac{1}{r_{ij}}\right|ab\rangle$ and $\langle ab\left|\frac{1}{r_{ij}}\right|ba\rangle$ type matrix elements and never need to evaluate $\langle ab\left|\frac{1}{r_{ij}}\right|cd\rangle$-type matrix elements except when the configuration includes two same-L,S terms.

3. Apply Sum Rule Method
4. Hund's 1st and 2nd Rules

1. Slater's Sum Rule Method

It is almost always possible to evaluate $e^2/r_{ij}$ matrix elements without solving for all $|LM_LSM_S\rangle$ basis states

* trace of any Hermitian matrix, expressed in ANY representation, is the sum of the eigenvalues of that matrix (thus invariant to unitary transformation)

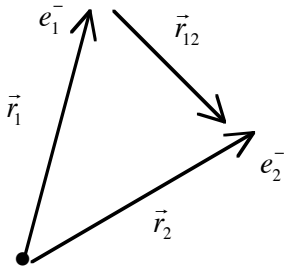
* $\sum_{i>j} e^2/r_{ij}$ and every scalar operator with respect to $\hat{J}$ (or $\hat{L}, \hat{S}$) has non-zero matrix elements *diagonal* in $J$ and $M_J$ (or $L$ and $M_L$) and *independent* of $M_J$ (or $M_L$,$M_S$)

[W-E Theorem: **J** is the GENERIC ANGULAR MOMENTUM with respect to which $e^2/r_{ij}$ is classified]

Recall from definition of $r_{12}$, that $e^2/r_{ij}$ is a scalar operator with respect to $\hat{\boldsymbol{J}}, \hat{\boldsymbol{L}}, \hat{\boldsymbol{S}}$ but not with respect to $\mathbf{j}_i$ or $\boldsymbol{\ell}_i$.

Interelectronic Repulsion: $\sum_{i>j} e^2/r_{ij}$

* destroys the single-electron orbital approximation $|n\ell\lambda\rangle$ for electronic structure calculations
* "correlation energy," "shielding"☐




$$e_1^- \text{ at } (r_1, \theta_1, \phi_1)$$

$$e_2^- \text{ at } (r_2, \theta_2, \phi_2)$$

$$\vec{r}_{12} = \vec{r}_2 - \vec{r}_1$$

Scalar with respect to **J**, **L**, **S**, $\mathbf{s}_i$ but not $\mathbf{j}_i$, $\boldsymbol{\ell}_i$. Do you know why?

$$\longrightarrow r_{12}^2 = r_1^2 - 2r_1 \cdot r_2 + r_2^2$$

$$r_{12} = \left[ r_1^2 + r_2^2 - 2|r_1||r_2|\cos(\vec{r}_1, \vec{r}_2) \right]^{1/2}$$

expand $r_{12}^{-1}$ as power series in $\left(\frac{r_<}{r_>}\right)$

where $r_<$ is the smaller of $|r_1|, |r_2|$

(integrals evaluated in 2 regions: $r_1 < r_2$, $r_2 < r_1$) the larger $r_i$ is seeing the multipoles of the smaller $r_j$

lengthy algebra [see Eyring, Walter, and Kimball, "Quantum Chemistry", pages 369-371 and, for relationship between Legendre polynomials and $Y_\ell^m(\theta,\phi)$, pages 52-59.]

will evaluate for orbitals occupied by i$^{th}$ e$^-$

2$^n$-pole moment (n=0 monopole, n=1 dipole,...)

multipole expansion

$$\frac{1}{r_{ij}} = \sum_{n=0}^{\infty} \sum_{m=-n}^{n} \left(\frac{4\pi}{2n+1}\right) \frac{r_<^n}{r_>^{n+1}} Y_n^m(\theta_i,\phi_i)\left[Y_n^m(\theta_j,\phi_j)\right]^*$$

not principal q.n.!

convergent series

angular momenta magnitude n, projection m

scalar product of 2 angular momenta, one for i-th particle, one for j-th

$\frac{1}{r_>}\left(\frac{r_<}{r_>}\right)^n$

* converts m to –m

An n-pole charge distribution is an n-th rank tensor with 2n+1 components.

No dependence on electron spin, so $1/r_{ij}$ is scalar with respect to S, $s_i$, $s_j$.

$$\left[Y_n^m(\theta_i,\phi_i) = \left\langle \theta_i,\phi_i \middle| \ell_i = \underbrace{n, m_{\ell_i} = m}_{\text{indices of } Y_n^m} \right\rangle\right]$$

The reason for this rather complicated looking expansion is that it is well suited for integrals over atomic orbitals which are expressed in terms of $r_i$, $\theta_i$, $\phi_i$, which are the coordinates of the i-th e⁻ with respect to the center of symmetry (nucleus) rather than the other e⁻. It enables use of atomic orbital basis states. Otherwise the $1/r_{ij}$ integrals would be nightmares.

$$Y_n^m(\theta,\phi) = \langle \theta,\phi | n = \ell, m = m_\ell \rangle$$

**Selection rules** for matrix elements:

$$\langle \ell_i m_i | Y_n^m | \ell'_i m'_i \rangle \langle \ell_j m_j | Y_n^m | \ell'_j m'_j \rangle$$

not principal q. n.

$$\text{orbitals} \begin{cases} |\Delta \ell_i| \le n, \quad \Delta m_{\ell_i} = m, \quad \Delta m_{s_i} = 0 \\ |\Delta \ell_j| \le n, \quad \Delta m_{\ell_j} = -m, \quad \Delta m_{s_j} = 0 \end{cases}$$

term in multipole expansion

triangle rule, $|\ell_i - \ell'_i| \le n \le \ell_i + \ell'_i$

(non-zero for steps in $n$ of an even number because of parity)

! overall: $\Delta L = 0$, $\Delta S = 0$, $\Delta M_L = 0$, $\Delta M_S = 0$, and independent of $M_L$, $M_S$. Can use any $M_L$, $M_S$ Slater determinant from the box diagram.

It is also clear how to evaluate the angular factors of the atomic orbital matrix elements using 3-j coefficients. Special tables of "Gaunt Coefficients" (also Condon and Shortley pages 178-179, Golding, page 41).

general $1/r_{12}$ matrix element (non-zero matrix elements of the $1/r_{12}$ operator follow the $\Delta$s-o = 0, 1, and 2 spin-orbital selection rule for change in spin-orbitals)

$$\left\langle \left\| \overset{e_2^-}{\overbrace{ab\left\| 1/r_{12} \right\| cd}} \right\| \right\rangle = \left\langle ab\left|\frac{1}{r_{12}}\right|cd\right\rangle - \left\langle ab\left|\frac{1}{r_{12}}\right|dc\right\rangle$$

$e_1^-$

$$\left\langle ab\left|\frac{1}{r_{12}}\right|cd\right\rangle = \underbrace{\delta\left(m_{s_a}, m_{s_c}\right)\delta\left(m_{s_b}, m_{s_d}\right)}_{\text{** } 1/r_{12} \text{ does not operate on spin coodinates } \text{**}} \underbrace{\delta\left(m_{\ell_a} + m_{\ell_b}, m_{\ell_c} + m_{\ell_d}\right)}_{\substack{1/r_{12} \text{ scalar with respect to } \hat{L}_{12} = \hat{\ell}_1 + \hat{\ell}_2 \\ \text{(can't change } M_L\text{)}}} \times$$

$$\underbrace{\sum_{k=0}^{\infty}}_{\substack{\text{tensor rank for product of AOs occupied by } e^- \#1 \text{ must be same as for } e^- \#2 \text{ for scalar product of two n-th rank tensors}}} \underbrace{c^k\left(\ell_a m_{\ell a}, \ell_c m_{\ell c}\right)}_{e_1^-} \underbrace{c^k\left(\ell_b m_{\ell b}, \ell_d m_{\ell d}\right)}_{e_2^-} \times \underbrace{R^k\left(n_a \ell_a n_b \ell_b n_c \ell_c n_d \ell_d\right)}_{\substack{\text{radial factor} \\ \text{A "reduced matrix element" because all m quantum numbers are gone.}}}$$

GAUNT COEFFICIENTS — ANGULAR FACTOR OF INTEGRAL

$$\underbrace{c^k\left(\ell m_\ell, \ell' m_{\ell'}\right)}_{\text{tabulated}} \equiv \left[\frac{2\ell'+1}{2\ell+1}\right]^{1/2} \underbrace{A_{000}^{k\ell\ell'} A_{m_\ell - m'_\ell, m'_\ell - m_\ell}^{k\ell'\ell}}_{\substack{\text{Clebsch-Gordan coefficients that result from integral over the product of three spherical harmonics — one from operator, two from orbitals}}}$$

triangle rule: $\left|\ell - \ell'\right| \le k \le \ell + \ell'$

$\ell + \ell' + k =$ even $\quad \left(\text{from properties of } A_{000}^{k\ell\ell'}\right)$ (including parity)

restrictions on k and m: $e_1^-$ integral $m_{\ell_1} + m = m'_{\ell_1}$

$$\left\langle n_1 \ell_1 m_{\ell_1} \left| Y_k^m \right| n'_1 \ell'_1 m'_{\ell_1} \right\rangle$$

triangle rule

For intra-configuration matrix elements, $R^k(abcd)$ has an especially simple form (because the same one or two orbitals appear in both the bra and in the ket).

$$\left.\begin{aligned} R^k(ab,ab) &\equiv F^k(a,b) \\ R^k(ab,ba) &\equiv G^k(a,b) \end{aligned}\right\} \text{“Slater – Condon” parameters}$$

(these are reduced matrix elements dependent only on $\ell_a$, $\ell_b$, $\ell_c$, $\ell_d$ and not on any of the $m_\ell$ quantum numbers.) All *L-S* states that belong to the same configuration are expressed in terms of the same set of $F^k$, $G^k$ parameters.

$$\left\langle \|ab\| \left| \frac{e^2}{r_{12}} \right| \|ab\| \right\rangle = J(a,b) - \delta(m_{s_a}, m_{s_b}) K(a,b)$$

spins must match or K term will vanish

DIRECT EXCHANGE

(This is how singlet and triplet states have different E even though $1/r_{ij}$ does not operate on the spin factor.)

$$J(a,b) \equiv \left\langle ab \left| \frac{e^2}{r_{12}} \right| ab \right\rangle = \sum_{k=0}^{\infty} c^k\left(\ell_a m_{\ell_a}, \ell_a m_{\ell_a}\right) c^k\left(\ell_b m_{\ell_b}, \ell_b m_{\ell_b}\right) \times F^k\left(n_a \ell_a, n_b \ell_b\right)$$

$a^k\left(\ell_a m_{\ell_a}, \ell_b m_{\ell_b}\right)$ $\left[\iint a^*(1) a(1) \hat{\mathbf{O}}\mathbf{p} b^*(2) b(2) d\tau_1 \, d\tau_2\right]$

*classical* charge distributions

$$K(a,b) \equiv \left\langle ab \left| \frac{e^2}{r_{12}} \right| ba \right\rangle = \delta\left(m_{s_a}, m_{s_b}\right) \sum_{k=0}^{\infty} \left[c^k\left(\ell_a m_{\ell_a}, \ell_b m_{\ell_b}\right)\right]^2 G^k\left(n_a \ell_a, n_b \ell_b\right)$$

$b^k\left(\ell_a m_{\ell_a}, \ell_b m_{\ell_b}\right)$

$\left[\iint a^*(1) b(1) \hat{\mathbf{O}}\mathbf{p} a(2) b^*(2) d\tau_1 \, d\tau_2\right]$

something not *classical!*

for special cases, such as $nd^2$, we have the simplified result that $n_a\ell_a = n_b\ell_b$ and $F^k = G^k$

Now we are ready to set up tables of $c^k$ (or, more conveniently, $a^k$ and $b^k$) to evaluate the $e^2/r_{ij}$ matrix.

Easy example: $nf^2$ (recall that $L$-$S$ terms of $f^2$ are $^1I$, $^3H$, $^1G$, $^3F$, $^1D$, $^3P$, $^1S$)

$$\left.\begin{aligned}\left|{}^1I\ 60\right\rangle &= \|3\alpha 3\beta\| \\ \left|{}^3H\ 51\right\rangle &= \|3\alpha 2\alpha\|\end{aligned}\right\}$$

$^1I$ and $^3H$ are the only $L$-$S$ states from the $f^2$ configuration that are represented by a single Slater determinant — extremes of the $M_L$,$M_S$ box diagram.
[You really do not want to calculate off-diagonal matrix elements of a two-electron operator if you can help it!]

Since $e^2/r_{ij}$ is a scalar operator with respect to $\hat{\mathbf{L}}$, $\hat{\mathbf{S}}$, $\hat{\mathbf{J}}$, matrix elements are $M_L$, $M_S$, and $M_J$ independent — so we can use *any* $M_L$,$M_S$ component to evaluate the matrix element — whichever is most convenient!

$$\left\langle {}^1I\left|\frac{e^2}{r_{12}}\right|{}^1I\right\rangle = \sum_{k=0,2,4,6} c^k(\underbrace{33}_{\ell},\underbrace{33}_{m_\ell})\,c^k(33,33)F^k(nf,nf) - \underbrace{\delta(\alpha,\beta)}_{=0}\sum_k \left[c^k(33,33)\right]^2 G^k(nf,nf)$$

$2\ell$

one spin is $\alpha$, other spin is $\beta$

$F^k(nf^2)$

$$= \sum_{k=0,2,4,6} \left[c^k(33,33)\right]^2 F^k(nf,nf)$$

$$\left\langle {}^3H\left|\frac{e^2}{r_{12}}\right|{}^3H\right\rangle = \sum_{k=0,2,4,6} \left\{\left[c^k(\underbrace{33}_{\alpha},\underbrace{33}_{\alpha})\,c^k(\underbrace{32}_{\beta},\underbrace{32}_{\beta})\right]F^k(nf,nf) - \left[c^k(\underbrace{33}_{\alpha},\underbrace{32}_{\beta})\right]^2 G^k(nf,nf)\right\}$$

both spins $\alpha$

$F^k(nf^2)$

Use table of $c^k$ in Golding (page 41)/C&S handout (C&S page 179).

Note that $[1/(7361 \cdot 64)]^{1/2}$ is implicit after the first entry for $f^2$, $k = 6$.
Here is where everyone makes mistakes!

| | k = 0 | 2 | 4 | 6 |
|---|---|---|---|---|
| $c^k(33,33)$ | 1 | –1/3 | 1/11 | $-[1/7361\cdot64]^{1/2}$ |
| $c^k(32,32)$ | 1 | 0 | –7/33 | $+[36/7361\cdot64]^{1/2}$ |
| $c^k(33,32)$ | 0 | +1/3 | $-30^{1/2}/33$ | $+[7/7361\cdot64]^{1/2}$ |
| $D_k$ | 1 | 225 | $1089 = 33^2$ | $7361\cdot64$ |

convenient factor

C&S Table: the number listed goes inside the SQRT replacing the numerator in the first row.

$D_k$ is a factor that simplifies the expressions. Each term has the form $F^k/D_k$. Call this ratio $F_k$ [notice $F^k$ vs. $F_k$]. Get simpler looking expressions when you replace $F^k$ by $D_kF_k$ ($D_k$ appears in denominators of $c^k$ as $[\ldots/D_k]^{1/2}$ )

Always have the product of two factors of $c^k$. Thus $F^k$ gets divided by $D_k$ to yield $F_k$.

$$\left\langle {}^1I\left|\frac{e^2}{r_{12}}\right|{}^1I\right\rangle = F^0 + \left(\frac{1}{9}\right)F^2 + \left(\frac{1}{121}\right)F^4 + \left(\frac{1}{7361\cdot 64}\right)F^6$$

$$= F_0 + 25F_2 + 9F_4 + F_6$$

$$\left\langle {}^3H\left|\frac{e^2}{r_{12}}\right|{}^3H\right\rangle = F^0 + \left[\left(-\frac{1}{3}\right)(0) - (1/3)^2\right]F^2 + \left[\left(\frac{1}{11}\right)\left(\frac{-7}{33}\right) - \frac{30}{33\cdot 33}\right]F^4 + \left[\frac{-6-7}{7361\cdot 64}\right]F^6$$

$$= F^0 - \frac{1}{9}F^2 - \frac{51}{(33)^2}F^4 \frac{-13}{7361\cdot 64}F^6$$

$$= F_0 - 25F_2 - 51F_4 - 13F_6$$

A lot of bookkeeping, but it's possible to learn how to use tables of $c^k$, $a^k$, $b^k$, and $D_k$, except it is much more work for $f^3$ than for $f^2$ (but the job is not yet complete for the $L$-$S$ terms beyond ${}^1I$ and ${}^3H$!

---

## SUM RULE METHOD:

Basic idea is that the sum of all the diagonal elements in the single Slater determinant basis set within an $M_L$, $M_S$ box is equal to the sum of the eigenvalues!

Look at the $M_L = 3, M_S = 1$ box: $||3\alpha 0\alpha||$ and $||2\alpha 1\alpha||$. This box generates $\left|{}^3H\,31\right\rangle$ and $\left|{}^3F\,31\right\rangle$, but the trace is $E({}^3H) + E({}^3F)$ and we already know $E({}^3H)$!

So $$E({}^1I) = \langle||3\alpha 3\beta||\rangle$$

$$E({}^3H) = \langle||3\alpha 2\alpha||\rangle$$

$$E({}^3F) = \langle||3\alpha 0\alpha||\rangle + \langle||2\alpha 1\alpha||\rangle - E({}^3H)$$

$$E({}^1G) = \langle||3\alpha 1\beta||\rangle + \langle||3\beta 1\alpha||\rangle + \langle||2\alpha 2\beta||\rangle - E({}^1I) - E({}^3H)$$

$$E({}^1D) = \langle||3\alpha -1\beta||\rangle + \langle||3\beta -1\alpha||\rangle + \langle||2\alpha 0\beta||\rangle + \langle||2\beta 0\alpha||\rangle$$
$$+ \langle||1\alpha 1\beta||\rangle - E({}^1I) - E({}^1G) - E({}^3H) - E({}^3F)$$

$$E({}^3P) = \langle||3\alpha -2\alpha||\rangle + \langle||2\alpha -1\alpha||\rangle + \langle||1\alpha 0\alpha||\rangle - E({}^3H) - E({}^3F)$$

$$E({}^1S) = \text{sum of seven } \langle||\ ||\rangle \text{ – sum of six } E({}^{2S+1}L)$$

This seems exceptionally laborious, but it is much easier than:

* generating each $|LM_L = L \;\; SM_S = S\rangle$ eigen-state as an explicit linear combination of Slater determinants
* then calculating matrix elements of $e^2/r_{ij}$, because there are many nonzero off-diagonal matrix elements between Slater determinants in the same $M_L, M_S$ box.

Here is the final result for the energies of all of the $(nf)^2\,{}^{2S+1}L$ terms:

$$E = E^{(0)} + E^{(1)} + E^{(2)}$$

$E^{(0)}$ = sum of orbital energies from $\mathbf{h}^{(0)} = -\dfrac{Z^2R}{n^2} = \varepsilon_{n\ell}$

Bare nucleus hydrogenic orbital energy — or partly shielded by filled shells.

$$E^{(1)} = \underbrace{\left\langle e^2/r_{ij} \right\rangle}_{\text{ready now}} + \underbrace{\left\langle \mathbf{H}^{\text{SO}} \right\rangle}_{\substack{\text{next} \\ \text{lecture}}}$$

$$E^{(2)} = \left(\text{intra configuraional spin-orbit}\right) + \left(\text{inter configurational } e^2/r_{ij}\right)$$

Configuration Interaction

For $nf^2$

| | shielded by all filled subshells | shielded by same subshell | | | | |
|---|---|---|---|---|---|---|
| ${}^1\text{I}$ | $2\varepsilon_{\text{nf}}$ | $+ F_0(nf^2)$ | $+ 25F_2(nf^2)$ | $+ 9F_4(nf^2)$ | $+ F_6(nf^2)$ | |
| ${}^3\text{H}$ | $2\varepsilon_{\text{nf}}$ | $+ F_0$ | $- 25F_2$ | $- 51F_4$ | $- 13F_6$ | ← lowest |
| ${}^1\text{G}$ | $2\varepsilon_{\text{nf}}$ | $+ F_0$ | $- 30F_2$ | $+ 97F_4$ | $+ 78F_6$ | |
| ${}^3\text{F}$ | $2\varepsilon_{\text{nf}}$ | $+ F_0$ | $- 10F_2$ | $- 33F_4$ | $- 286F_6$ | ← might have been lowest |
| ${}^1\text{D}$ | $2\varepsilon_{\text{nf}}$ | $+ F_0$ | $+ 19F_2$ | $- 99F_4$ | $+ 715F_6$ | |
| ${}^3\text{P}$ | $2\varepsilon_{\text{nf}}$ | $+ F_0$ | $+ 45F_2$ | $+ 33F_4$ | $- 1287F_6$ | ← also might have been lowest |
| ${}^1\text{S}$ | $2\varepsilon_{\text{nf}}$ | $+ F_0$ | $+ 60F_2$ | $+ 198F_4$ | $+ 1716F_6$ | |

shielded-core configurational energy

intraconfiguration $L$-$S$ term splittings

(there is NO center of Gravity Rule for degeneracy weighted $L$-$S$ terms)

Now it is easy to show that all $F_k$'s are > 0 and $F_k \gg F_{k+2}$ etc. (by roughly a factor of 10 per step in k).

From this we get an **empirical rule** (<u>empirical</u> because we expect that contributions to $E(L,S)$ from $F_4$ and $F_6$ can be ignored).

Lowest $E$ of all $L$–$S$ terms is the one with

* MAXIMUM $S$
* of those with Maximum $S$, lowest is the one with MAXIMUM $L$.

# These are <u>Hund's</u> **first** and **second** (of three) <u>rules</u>.

Note also that Hund's rules make no predictions about the energy order of $L$-$S$ terms <u>except</u> for the identity of the single, lowest energy $L$-$S$ term.

**Non-Lecture**

There are several interesting problems also solved by this $e^2/r_{ij}$ formalism.

1. The energy splittings between and the Slater determinantal characters of two or more $L,S$ terms of the same $L$ and S that belong to the same electronic configuration

   e.g. $d^3 \rightarrow$ two $^2D$ terms
   see pages 47-50 of Golding for $2 \times 2$ secular determinant for $^2D$ of $d^3$

2. matrix elements of $e^2/r_{ij}$ between same–$L,S$ terms that belong to two different configurations

   e.g. $nd^2$ $^1S,^3P,^1D,^3F,^1G$

   ndn′d $\left\{\begin{matrix} ^1S,^3P,^1D,^3F,^1G \\ ^3S,^1P,^3D,^1F,^3G \end{matrix}\right\}$ no Pauli restrictions

choose any pair of orthogonal combinations of Slaters. What you choose determines the values of the off-diagonal matrix elements but not the eigen-energies

So, for L-S terms that belong to the nd2 configurations, there will be

$^1S \sim {}^1S$

$^3P \sim {}^3P$

$^1D \sim {}^1D$

$^3F \sim {}^3F$

$^1G \sim {}^1G$

interconfigurational interaction matrix elements and each of these 5 interaction matrix elements will NOT be of the same magnitude. There will be different Configuration Interaction energy shifts for the various $L$-$S$ terms in a configuration.

Knowing the single configuration expected pattern of $L$-$S$ states (energies and other properties) enables detection of local inter-configuration perturbations. Predicted patterns are EVERYTHING to an experimentalist!

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## SPIN - ORBIT: Many-Electron ς(N, L, S) ↔ Single-Orbital $\varsigma_{n\ell}$ Coupling Constants

LAST TIME: $\sum_{i \geq j} e^2/r_{ij}$

- death of orbital picture
- expansion of $1/r_{ij}$: multipoles, integrals over AOs in nucleus-centered coordinates
- SELECTION RULES: orbital and many-e⁻ basis sets
- Gaunt Coefficients: $a^k, b^k, c^k$ [products of 3-j coefficients]
- Slater-Condon ($F^k$, $G^k$)⟶($F_k$, $G_k$)Parameters
- Sum Rule Method - avoid necessity to derive:
  - *eigenvectors
  - * off-diagonal elements in Slater basis
- Hund's 1st and 2nd Rules → predict lowest L-S term of configuration based on rapid decrease in size of $F_k$ as k increases

---

A single-configuration pattern of assigned L-S terms.

For $nf^2$ there are 91 Slater determinants. To use the sum rule method to get the relative energies of all 7 L-S terms ($^1$I, $^3$H, $^1$G, $^3$F, $^1$D, $^3$P, $^1$S), it turns out it is necessary to compute 28 *diagonal* matrix elements in the $f^2$ Slater determinantal basis set. This is wonderful because it is unnecessary to compute any off-diagonal matrix elements (see page 34-8). These L-S term energies are expressed in terms of $F_0$, $F_2$, $F_4$, and $F_6$ (and $G_k$) Slater-Condon parameters.

---

TODAY:

A. General Importance of spin-orbit term

$\mathbf{H}^{SO} = \sum_i a(r_i)\hat{\ell}_1\hat{\mathbf{s}}_1$ $\quad 1-e^-$ operator

B. *Trick*: replace 1–e⁻ operator by more convenient $\zeta(N,L,S)\mathbf{L}\cdot\mathbf{S}$ for ΔS = 0, ΔL = 0 special case matrix elements

C. *Pattern*: Landé Interval Rule (Patterns are for breaking! Broken patterns provide information about "dark" states)

D. $\mathbf{H}^{SO}$ matrix elements in Slater Determinantal Basis Set

* another operator replacement

* A single-orbital integral is the most fundamental control parameter: $n,\ell$-scaling of $\zeta_{n\ell}$

* $\zeta_{n\ell} \leftrightarrow \zeta(N, L, S)$ relationship between a single-orbital coupling constant and that for a specific $L$-$S$ state

* off-diagonal spin-orbit matrix elements: $\Delta L \neq 0$, $\Delta S \neq 0$, $\Delta J = 0$.

PATTERN BREAKING

next time → Hund's 3rd Rule and Lande $g_J$–factors

A. Importance of spin-orbit

1. $\mathbf{H}^{SO}$ produces diagnostically significant "fine structure"
   CONFIGURATIONAL ASSIGNMENTS (based on which L-S terms are present and the ± signs of spin-orbit splittings)
   L,S term assignments are based on
   PATTERNS:* # components
   * sign of pattern (largest splitting on top or on bottom)
   * statistical weight (2J + 1) of lowest vs. highest energy component
   * overall magnitude of splitting

2. for heavy atoms, $\mathbf{H}^{SO}$ is responsible for such large splittings and off-diagonal interactions that L-S terms "vanish", ΔS selection rules are violated. "Inter-System Crossing (ISC)".

   Need to "deperturb" to recover $F_k$, $G_k$ inter-electronic $(1/r_{ij})$parameters which should vary smoothly from atom to atom (isoelectronic series) (shielding rules). Periodicity! The PERIODIC TABLE

3. Spin-forbidden transitions provide energy linkages between manifolds of states with different values of S. "InterSystem Crossing (ISC)", e.g. Hg $^3P_1 \leftarrow {}^1S_0$ 254nm

4. Non-textbook Zeeman tuning coefficients (clues about unobserved "dark" states) finer detail to be used after S-O patterns reduce the possibilities that must be considered.

**Atoms, Molecules, Quantum Dots, solids: in an electronic transition, light acts on a single $e^-$ and operates exclusively on the spatial part of $\psi \rightarrow$ spin-flips are forbidden except when $H^{SO}$ mixes states of different S — forbidden transitions "borrow" intensity from allowed transitions. In the time-domain: a short pulse prepares, at t = 0, a non-eigenstate that is a pure $\Delta S = 0$ excitation (and $\Delta\ell = \pm 1$) basis state. The "pluck"!**

Language: the name of each eigenstate is based on its dominant (i.e., "nominal") character. It is the name of the dominant basis state. Use of the same name for both *eigenstate* and *basis state* is a source of confusion.

B. Operator Replacement for $\mathbf{H}^{SO}$

$$\mathbf{H}^{SO} = \sum_i a(r_i)\boldsymbol{\ell}_i \bullet \mathbf{s}_i$$ a one-e⁻ operator

Wigner-Eckart Theorem for a vector operator — operator replacement for special cases where only $\Delta J = 0$ matrix elements are considered.

$$\underbrace{\left\langle JM_J \left| \hat{\mathbf{A}} \right| JM_J' \right\rangle}_{\Delta J = 0} = \underbrace{\left\langle J \left\| \hat{\mathbf{A}} \right\| J \right\rangle}_{\substack{\text{proportionality constant} \\ \text{reduced matrix element}}} \underbrace{\left\langle JM_J \left| \hat{\mathbf{J}} \right| JM_J' \right\rangle}_{\substack{\text{matrix element of} \\ \text{corresponding component of } \hat{\mathbf{J}}}}$$

CTDL p. 1054 use projection theorem: $\mathbf{V}_{\|} = \dfrac{\langle \mathbf{J} \bullet \mathbf{V} \rangle}{\langle \mathbf{J}^2 \rangle} \mathbf{J}$.

Especially useful when $\mathbf{V}$ is an angular momentum that is included in $\mathbf{J}$.

useful trick for $\mathbf{H}^{SO}$

$a(r_i)\boldsymbol{\ell}_i$ vector with respect to $\mathbf{L}$
$\mathbf{s}_i$ vector with respect to $\mathbf{S}$

Special case of $\Delta L = 0$, $\Delta S = 0$ matrix elements in $|NLM_L SM_S\rangle$ basis set

$$\mathbf{H}^{SO} = \sum_i a(r_i)\boldsymbol{\ell}_i \cdot \mathbf{s}_i \xrightarrow{\text{configuration label}} \underbrace{\zeta(N,L,S)\mathbf{L}\cdot\mathbf{S}}_{\substack{\text{operator} \\ \text{replacement!}}}$$

* $\zeta(N,L,S) \equiv \sum_i \left\langle L \left\| a(r_i)\boldsymbol{\ell}_i \right\| L \right\rangle \left\langle S \left\| \mathbf{s}_i \right\| S \right\rangle$
* a different spin-orbit coupling constant for EACH L-S term of the N configuration (loss of simplicity)
* convenient because it is easy to evaluate matrix elements of $\mathbf{L}\cdot\mathbf{S}$ without having to resort to use of the Slater determinantal basis set

[ASIDE: $a(r_i)\boldsymbol{\ell}_i$ and $\mathbf{s}_i$ are both vectors with respect to $\mathbf{J}$, thus $\mathbf{H}^{SO}$ is scalar with respect to $\mathbf{J}$, hence matrix elements in the $\left| NJLSM_J \right\rangle$ basis set have 3 special characteristics: $\Delta J = 0, \Delta M_J = 0$, and independent of $M_J$.]

CAUTION: the $\mathbf{L}\cdot\mathbf{S}$ operator seems to imply a $\Delta S = 0$ selection rule, but we *assumed* $\Delta S = 0$ in deriving the simplified form of $\mathbf{H}^{SO}$: $\zeta\mathbf{L}\cdot\mathbf{S}$

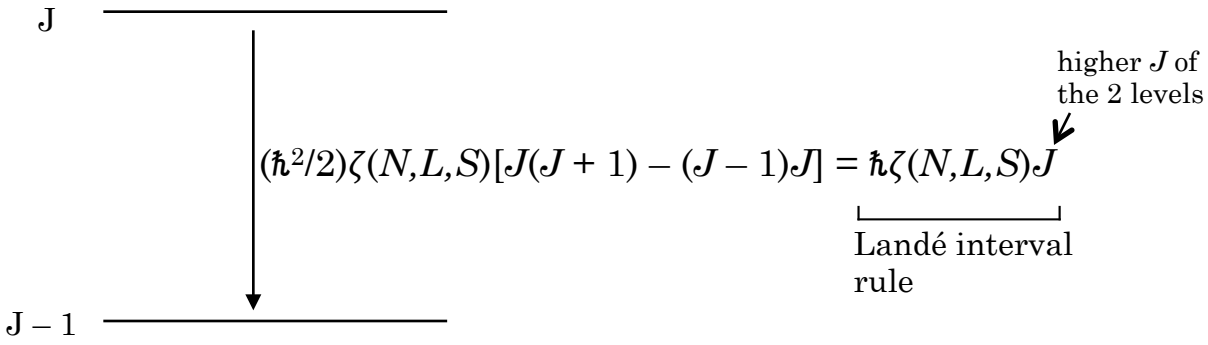
C. Landé Interval Rule (useful for recognizing and assigning an isolated L-S term)

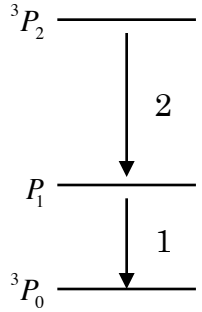
$$\mathbf{J} = \mathbf{L} + \mathbf{S} \qquad \mathbf{J}^2 = \mathbf{L}^2 + \mathbf{S}^2 + 2\mathbf{L}\cdot\mathbf{S}$$

$$\mathbf{L}\cdot\mathbf{S} = \frac{\mathbf{J}^2 - \mathbf{L}^2 - \mathbf{S}^2}{2}$$

$$\left\langle NJLSM_J \left| \mathbf{H}^{SO} \right| NJLSM_J \right\rangle = \frac{\hbar^2}{2}\underbrace{\zeta(N,L,S)}_{\substack{\text{can be positive,}\\ \text{zero, or negative}}}\left[J(J+1) - L(L+1) - S(S+1)\right]$$

So, within an L-S term, $\mathbf{H}^{SO}$ causes splitting into 2S+1 (or 2L+1 if S > L) components.




* spacing is ∝ to larger of 2 J's
* largest spacing locates largest J
* large spacing on top: "regular" L-S term
bottom: "inverted" L-S term

[Convince yourself that you are able to tell $^3$D from $^3$P…]

regular $^3$P [2 intervals: 2:1]

$^3$D 3:2 $^3$F 4:3
$^4$P 5/2:3/2 $^4$D 7/2:5/2:3/2 (4 components)
(3 components)

Easy to show that the degeneracy–weighted average (each $J$-component has degeneracy $2J + 1$) spin-orbit energy of a multiplet = 0 (easiest to show from the trace of the $\mathbf{H}^{SO}$ in $|LM_LSM_S\rangle$basis). Off-diagonal elements (between same-J components of different L-S states) do not affect the trace of $\mathbf{H}^{SO}$.

$$\sum_{J=|L-S|}^{L+S}(2J+1)E_J = 0$$

The interval rule plus the number of J-components of a multiplet determine the values of both L and S.
[$^4$P 5:3, 2 intervals; $^2$P 1 interval, $^4$D 7:5:3, 3 intervals]

D. Matrix Elements of $\mathbf{H}^{SO}$ in Slater Determinantal Basis Set

GOALS:
* $\Delta S \neq 0$ matrix elements, $\Delta L \neq 0$ matrix elements
* relationships between $\underbrace{\zeta(N,L,S)}_{\text{L-S term}}$ and $\underbrace{\zeta_{n\ell}}_{\text{orbital}}$
* excluding interconfigurational $\mathbf{H}^{SO}$ matrix elements

---

NONLECTURE: alternative operator replacement for $\mathbf{H}^{SO}$ that is appropriate for orbital matrix elements

$$\mathbf{H}^{SO} = \sum_i a(r_i)\boldsymbol{\ell}_i \cdot \mathbf{s}_i$$

replace $a(r_i)\boldsymbol{\ell}_i$ by $\zeta_{n\ell}\boldsymbol{\ell}_i$ using completeness:

$$\left\langle n'\ell'm'_\ell sm'_s \middle| \mathbf{H}^{SO} \middle| n\ell m_\ell s m_s \right\rangle =$$

$$\sum_i \sum_{"} \left\langle n'\ell'm'_\ell sm'_s \middle| a(r_i) \middle| \underbrace{n''\ell''m''_\ell sm''_s \rangle\langle n''\ell''m''_\ell sm''_s}_{\text{completeness}} \middle| \boldsymbol{\ell}_i \bullet \mathbf{s}_i \middle| n\ell m_\ell s m_s \right\rangle$$

$a(r_i)$ is scalar with respect to $\mathbf{s}_i \rightarrow m'_s = m''_s$ and value is $m'_s$-independent

$a(r_i)$ *i*s scalar with respect to $\boldsymbol{\ell}_i \rightarrow \ell' = \ell'', m'_\ell = m''_\ell,$ and value is $m'_\ell$-independent

$\boldsymbol{\ell}_i$ can't change $\ell$ in $\left|\ell m_\ell\right\rangle \rightarrow \ell'' = \ell$

$\boldsymbol{\ell}_i \cdot \mathbf{s}_i$ does not operate on radial part of $\psi \rightarrow n'' = n$

thus $\left\langle n'\ell'm'_\ell sm'_s \middle| \mathbf{H}^{SO} \middle| n\ell m_\ell s m_s \right\rangle = \delta_{\ell'\ell} \left\langle n'\ell' \middle|\middle| a(r_i) \middle|\middle| n\ell \right\rangle \left\langle \ell m'_\ell sm'_s \middle| \boldsymbol{\ell} \cdot \mathbf{s} \middle| \ell m_\ell s m_s \right\rangle$

$$\left\langle n'\ell \middle|\middle| a(r_i) \middle|\middle| n\ell \right\rangle = \underbrace{(n'n)^{-3/2}}_{\substack{\text{Rydberg scaling for} \\ \text{inner part of orbital}}} \zeta^\circ_\ell = \left(\frac{n'}{n}\right)^{-3/2} \underbrace{\zeta_{n\ell}}_{\propto n^{-3}}$$

$$\text{so, for } n' = n, \quad \left\langle n\ell \middle|\middle| a(r_i) \middle|\middle| n\ell \right\rangle = \zeta_{n\ell} = n^{-3}\zeta^\circ_\ell$$

Spin-orbit scaling for all members of a Rydberg series.

---

This reduction of $\mathbf{H}^{SO}$ shows that, for atoms, $\mathbf{H}^{SO}$ acts exclusively within a configuration except for interconfigurational matrix elements where the two configurations differ by a single spin-orbital of the same value of $\ell$: $\underbrace{n\ell \leftrightarrow n'\ell}_{\text{same } \ell}$

---

Examples

A is a single Slater determinant

$$\left\langle A\left|\mathrm{H}^{\mathrm{SO}}\right|A\right\rangle = \sum_{\substack{k \\ \text{spin-orbitals}}} \left\langle u_k\left|a\left(r_k\right)\ell_k\cdot \mathrm{s}_k\right|u_k\right\rangle$$

$$= \sum_k \zeta_{n_k\ell_k}\left\langle \ell_k m_{\ell_k} s_k m_{s_k}\left|\underbrace{\ell\cdot \mathrm{s}}_{\substack{\text{diagonal element picks out } \ell_z\mathbf{s}_z}}\right|\ell_k m_{\ell_k} s_k m_{s_k}\right\rangle$$

$$= \hbar^2 \sum_{\substack{k \\ \text{spin-orbitals}}} \zeta_{n_k\ell_k} m_{\ell_k} m_{s_k}$$

if $\left|\mathrm{A}\right\rangle$ is also an eigenfunction of $\mathbf{L}^2, \mathbf{L}_z, \mathbf{S}^2,$ and $\mathbf{S}_z$ then

$$\left\langle NLM_L SM_S\left|\mathbf{H}^{SO}\right|NLM_L SM_S\right\rangle = \zeta(N,L,S)\hbar^2 M_L M_S$$

$$\zeta(N,L,S) = \frac{\sum_k \zeta_{n_k\ell_k} m_{\ell_k} m_{s_k}}{M_L M_S}$$

The matrix element is evaluated 2 ways in order to reduce a many-e⁻ spin-orbit coupling constant ($\varsigma$(N,L,S)) to a sum of one-e⁻ orbital coupling constants ($\varsigma_{n\ell}$)! This reveals the "periodicity" for which the periodic table is famous.

Example 1. $nf^2$ $^3$H uncoupled representation

uncoupled $\left|nf^2 \quad {}^3H \quad M_L = 5 \quad M_S = 1\right\rangle = \|\overbrace{3\alpha}^{3\times 1/2}\overbrace{2\alpha}^{2\times 1/2}\|$ Single Slater determinant!

$$\zeta\left(nf^2, {}^3H\right) = \frac{\zeta_{nf}\left[3(1/2)+2(1/2)\right]}{\underbrace{5\cdot 1}_{M_L M_S}} = \zeta_{nf}/2$$

(fill in the steps!)

Example 2. $nf^2$ in coupled representation

coupled $\left|\ nf^2 \quad {}^3H_6 \quad M_J = 6 \quad\right\rangle = ||3\alpha 2\alpha||$

Landé :

$$\left\langle nf^2\ {}^3H_6\ M_J = 6\middle|\mathrm{H^{SO}}\middle|nf^2\ {}^3H_6\ 6\right\rangle$$

$$= \frac{\hbar^2}{2}\left[J(J+1) - L(L+1) - S(S+1)\right]\zeta\left(nf^2, {}^3H\right)$$

($J(J+1)$: $6\cdot 7$; $L(L+1)$: $5\cdot 6$; $S(S+1)$: $1\cdot 2$)

$$= \hbar^2 5\zeta\left(nf^2, {}^3H\right) \text{ from many-e}^- \text{ form}$$

$$= \hbar^2\zeta_{nf}\left[3(1/2)+2(1/2)\right] \quad \text{from orbital form}$$

$$\therefore \underbrace{\zeta\left(nf^2, {}^3H\right)}_{\text{what you measure}} = \underbrace{\zeta_{nf}/2}_{\text{what you want to know}}$$

Example 3: $\zeta(nf^2, {}^3F)$

$^3$F is never expressed as a single Slater determinant for any value of ($M_L$,$M_S$)

Evaluate in either of 2 ways:

a. Obtain explicit linear combination of Slater determinants using ladders and orthogonality or using $\mathbf{L}^2$,$\mathbf{S}^2$ to get $\left|\mathrm{nf}^2 \quad {}^3F \quad M_L = 3 \quad M_S = 1\right\rangle$ [laborious].

b. Slater sum rule method [simple].

$M_L = 3, M_S = 1$ box: contains only $||3\alpha 0\alpha||, ||2\alpha 1\alpha||$

trace $\left\langle ||3\alpha 0\alpha|| \right\rangle + \left\langle ||2\alpha 1\alpha|| \right\rangle = E\left({}^3H\ M_L = 3, M_S = 1\right) + E\left({}^3F\ M_L = 3, M_S = 1\right)$

1st Matrix Element $\left\langle ||3\alpha 0\alpha|| \right\rangle = \left\langle ||3\alpha 0\alpha|| \middle| \mathrm{H^{SO}} \middle| ||3\alpha 0\alpha|| \right\rangle = \hbar^2\zeta_{nf}\left[\frac{3}{2} + 0\right]$

second matrix element $\left\langle \left\| 2\alpha 1\alpha \right\| \right\rangle = \left\langle \left\| 2\alpha 1\alpha \right\| \mathbf{H}^{\mathrm{SO}} \right\| \left| 2\alpha 1\alpha \right| \right\rangle = \hbar^2 \zeta_{nf} \left[ 1 + \frac{1}{2} \right]$

trace of $M_L = 3, M_S = 1$ box is $3\hbar^2\zeta_{nf}$

exploit sum rule $E\left( {}^3H\ M_L = 3\ M_S = 1 \right) = \left\langle {}^3H\ M_L = 3\ M_S = 1 \left| \mathrm{H}^{\mathrm{SO}} \right| {}^3H\ 31 \right\rangle = \zeta\left( nf^2, {}^3H \right) \hbar^2 3 \cdot 1$

but we already showed that $\zeta(nf^2, {}^3H) = \zeta_{n\ell}/2$

$$E\left( {}^3H\ M_L = 3, M_S = 1 \right) = \hbar^2 \zeta_{nf} \left( 3/2 \right)$$

$$\therefore E\left( {}^3F\ M_L = 3, M_S = 1 \right) = 3\hbar^2\zeta_{nf} - \left( 3/2 \right)\hbar^2\zeta_{nf} = \left( 3/2 \right)\hbar^2\zeta_{nf}$$

sum for the $M_L = 3$, $M_S = 1$ box

$$= \left\langle {}^3F\ 31 \left| \mathbf{H}^{\mathrm{SO}} \right| {}^3F\ 31 \right\rangle = \zeta\left( nf^2, {}^3F \right)\left( 3 \cdot 1 \right)\hbar^2$$

$$\therefore \zeta\left( nf^2, {}^3F \right) = \frac{1}{2}\zeta_{nf}$$

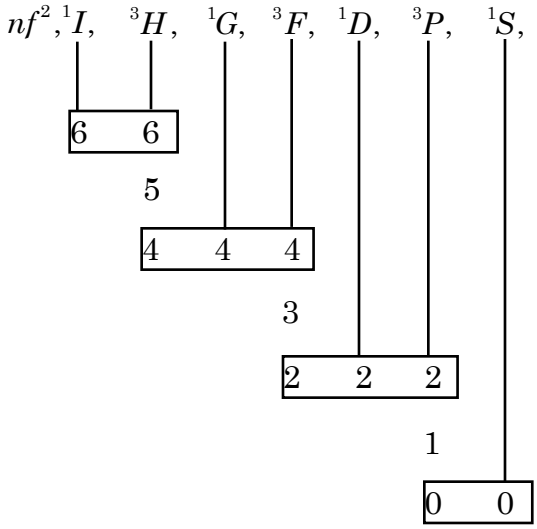
$\left( \text{actually would find, for } nf^2, \zeta\left( nf^2, {}^3L \right) = \frac{1}{2}\zeta_{nf} \text{ for all } L \right)$

[not true for all configurations]

---

We are not done. There are some $\Delta J = 0$ *off-diagonal in L,S matrix elements* among the L-S-J terms of the same configuration.

$\mathbf{H}^{\mathrm{SO}}$ is diagonal in J
must diagonalize two $3 \times 3$
and two $2 \times 2$ matrices.



set up the J = 6 matrix because it is simplest

$$\left|{}^1I_6 \;\; 6\right\rangle = \left\|3\alpha 3\beta\right\|$$

$$\left|{}^3H_6 \;\; 6\right\rangle = \left\|3\alpha 2\alpha\right\|$$

$$\left\langle {}^1I_6 \;\; 6\left|\mathbf{H}^{\mathrm{SO}}\right|{}^3H_6 \;\; 6\right\rangle = \left\langle \left\|3\alpha 3\beta\right\|\left|\mathbf{H}^{\mathrm{SO}}\right|\left\|3\alpha 2\alpha\right\|\right\rangle$$

Mismatch is in 2nd spin-orbital.
Needs 1/2 $\ell_+ s_-$ operator to give nonzero spin-orbit matrix element.

$$= \left\langle 3\beta\left|\frac{1}{2}\ell_+ s_-\right|2\alpha\right\rangle$$

$$= \hbar^2\zeta_{nf}\,\frac{1}{2}\left[3\cdot 4 - 2\cdot 3\right]^{1/2} = \hbar^2\zeta_{nf}\left(3/2\right)^{1/2}$$

zero for all singlet states

$$\mathbf{H}^{\mathrm{SO}}_{J=6} = \begin{matrix} \\ {}^1I_6 \\ {}^3H_6 \end{matrix}\begin{matrix} \begin{matrix} \downarrow \end{matrix} \\ \begin{pmatrix} 0 & (3/2)^{1/2} \\ (3/2)^{1/2} & 5/2 \end{pmatrix} \end{matrix}$$

$$\left\langle {}^3H_6\,6\left|\mathrm{H}^{\mathrm{SO}}\right|{}^3H_6\,6\right\rangle = \frac{\hbar^2}{2}\left[J(J+1) - L(L+1) - S(S+1)\right]\zeta\left(nf^2, {}^3H\right)$$

$$= \hbar^2 5\zeta\left(nf^2, {}^3H\right) = \hbar^2\left(5/2\right)\zeta_{nf}$$

---

for more complex configurations such as $(n\ell)^a(n'\ell')^b \rightarrow \varsigma_{n\ell}$ and $\varsigma_{n'\ell'}$: two ς parameters needed, 1 for each of the two open subshell orbitals.

But can use the value of $\varsigma_{n\ell}$ determined from some other configuration:
e.g. $\varsigma_{3d}$ from $3d^6 4s^2$ can be used to predict the 3d part of $\mathbf{H}^{\mathrm{SO}}$ in $3d^6\ 4s4p$. Unexpected *inter-relationships* between superficially unrelated observables. Small number of control parameters. Magic decoder!

---

**Hund's 3rd Rule**: lowest energy J level of lowest energy L – S term is J = |L – S| if subshell is less than ½ full, but it is J = L + S if subshell is more than ½ full, and J = S (no spin-orbit splitting) because L = 0 for a half-filled subshell. Sign of ζ(n,L,S) as diagnostic!

NEXT TIME!

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## Zeeman Effect

Read CTDL, pp. 1156-1178, preparation for 1-D solids

LAST TIME:

* computational tricks and inter-relationships among atomic and molecular constants
* $\mathbf{H}^{SO} = \zeta(N,L,S)\mathbf{L}\cdot\mathbf{S} \rightarrow$ a different $\zeta(N,L,S)$ constant for each L–S state in configuration N

or, from $\mathbf{H}^{SO} = \sum_i a(r_i)\boldsymbol{\ell}_i\mathbf{s}_i \rightarrow$ one $\zeta_{n\ell}$ parameter valid for all diagonal and off-diagonal intra-configurational matrix elements

$\zeta(N,L,S)$ a fit parameter
$\zeta_{n\ell}$ an orbital integral

micro→macro forms of operators

---

TODAY:

1. electrons↔holes
   shortcuts for $e^2/r_{ij}$ and $\mathbf{H}^{SO}$
2. Hund's 3rd Rule: the lowest L–S–J level of an electronic configuration
3. Zeeman effect. Landé g-factor via Wigner-Eckart Theorem. Another way to evaluate reduced matrix elements.
4. $\mathbf{H}^{Zeeman}$ in Slater determinantal basis set.

1. electrons ($e^-$)↔holes ($h^+$) shortcuts

| subshell | $n\ell^N$ | | | |
|---|---|---|---|---|
| ½ full | s | p | d | f |
| #$e^-$ | 1 | 3 | 5 | 7 |

for $p^5$ do we need to consider $5e^-$ or one $h^+$?

$p^5$ $^2P$ $M_L = 1, M_S = 1/2$ $\|1\alpha 1\beta 0\alpha 0\beta -1\alpha\|$

$(ph^+)^1$ $^2P$ $M_L = 1, M_S = 1/2$ $\|1\alpha\|$

Get same energy order of L-S states via $5e^-$ or $1h^+$ Slater determinant

Why? $e^2/r_{ij}$ expresses $e^- - e^-$ repulsion
$h^+ - h^+$ is also repulsion

so we get the same result for $e^2/r_{ij}$ by $e^-$ or $h^+$ algebra. All $F_k$, $G_k$ Slater-Condon parameters are > 0.

What about $p^5d^1$ vs. $p^1d^1$? Is the energy level order inverted? Hund's 1st and 2nd rules predict the same lowest state in $p^5d$ as in $(ph^+)^1\ d^1$.

What about $e^-$ vs. $h^+$ for $\mathbf{H}^{SO}$?

$$\mathbf{H}^{SO} = \zeta(N,L,S)\mathbf{L}\cdot\mathbf{S} = \sum_i a(r_i)\ell_i\mathbf{s}_i$$

Take $\Delta M_J = 0$, $M_J = J = L + S$ matrix element of both sides of the $\mathbf{H}^{SO}$ equation.

This is an extreme state from the $M_L$,$M_S$ box diagram. There is only one Slater determinant in this box.

$p^5\ ^2P$ vs. $p^1\ ^2P$

$$\hbar^2\zeta\left(p^5\ ^2P, M_L = 1, M_s = 1/2\right)\left(1\cdot\frac{1}{2}\right) = \hbar^2\zeta_{np}\left[1\cdot\frac{1}{2}+1\cdot\left(-\frac{1}{2}\right)+0+0+-1\cdot\frac{1}{2}\right]$$

$$= \hbar^2\zeta_{np}(-1)\left(\frac{1}{2}\right)$$

$\zeta(p^5\ ^2P) = -\zeta_{np}$

Now $p^1\ ^2P$

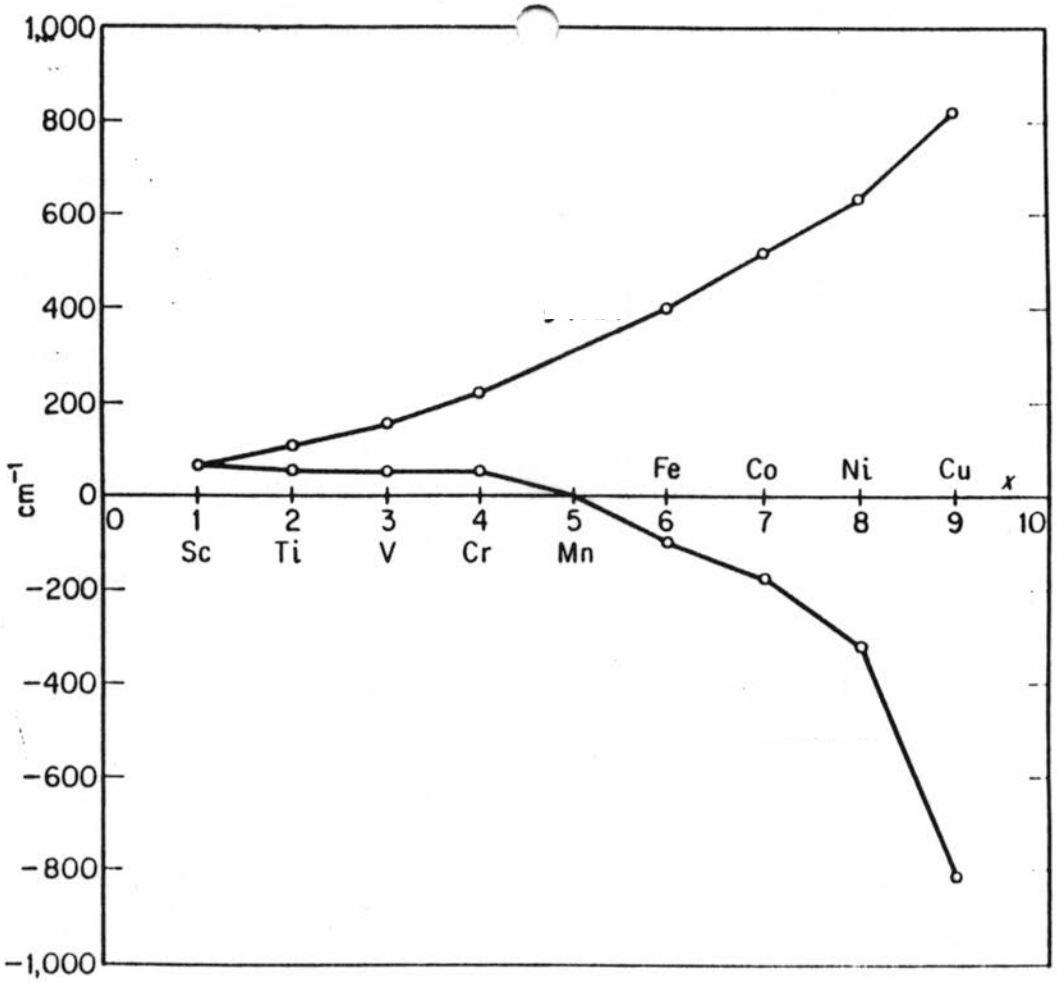
$$\hbar^2\zeta\left(p^1\ ^2P, M_L = 1, M_s = 1/2\right)\underbrace{L\cdot S}_{1\cdot\frac{1}{2}} = \hbar^2\zeta_{np}\left[1\cdot\frac{1}{2}\right]$$

$\zeta(p^1\ ^2P) = \zeta_{np}$

So we get a sign reversal in $\zeta(N\ ^2P)$ for $p^5$ vs. $p^1$.

So $e^-\leftrightarrow h^+$ gives no change in $e^2/r_{ij}$ expressed by an always positive $F_k$ and $G_k$. However, get sign reversed in $\mathbf{H}^{SO}$ for $\zeta(N,L,S)$ but not for $\zeta_{n\ell}$.

Figure 6-2 from Tinkham (page 187) for $L_{MAX}$, $S_{MAX}$ state of $3d^N\ 4s^2$ configuration




This figure is in the public domain.

**Fig. 6-2.** *Spin-orbit parameters in $3d^x$ transition elements. The splitting parameters $\zeta(LS)$ are computed as indicated in Table 6-2 and averaged over the various splittings. The data used are for the $3d^x4s^2$ configurations of the neutral atoms. (From Charlotte E. Moore, "Atomic Energy Levels," Natl. Bur. Standards, Circ. 467, vols. I and II, 1949 and 1952. A very similar figure appears in Condon and Shortley.)*

Note that $\zeta_{n\ell}$ is a one-e⁻ orbital integral and is expected to exhibit "periodic" variation. It increases with atomic number and this is expected for increasing $Z^{eff}$:

$$Z \longrightarrow Z + 1,\ Z^{eff} \longrightarrow Z^{eff} + 1 - 1/2$$

See Burns' rule paper on shielding included here: *J. Chem. Phys.* **41**, 1521 (1964).

General result for $\zeta$ (N,L,S):

$$\zeta(N,L,S) = \pm\frac{1}{2S_{MAX}}\zeta_{n\ell}$$

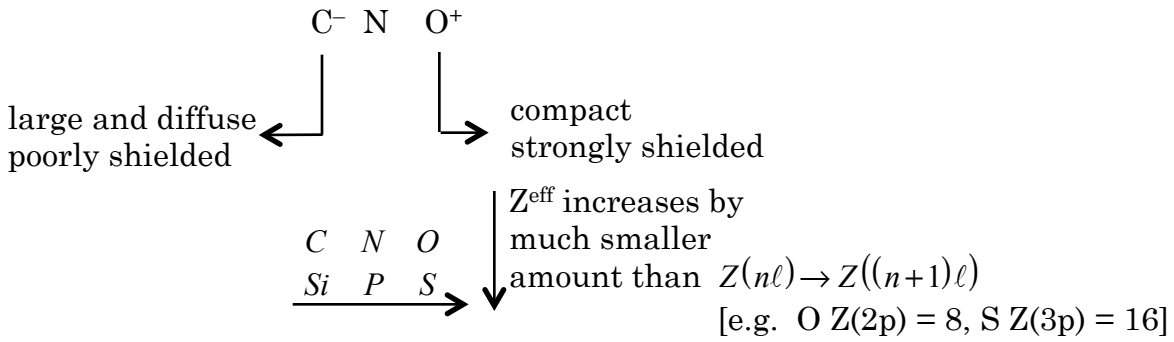
$+$ for $n < 2\ell+1$

$-$ for $n > 2\ell+1$

$\zeta(N,L,S) = 0$ for $n = 2\ell+1$

Burn's paper on Pages 5 and 6 has been removed due to copyright restrictions. See Burn, Gerald. "Atomic Shielding Parameters." *J. Chem. Phys*. **41** (1964): 1521–1522.

Isoelectronic species (5 valence electrons):




$Z^{eff}$ increases more slowly than Z

---

Hund's 3rd Rule

1st and 2nd Hund's rules (for most stable $L$-$S$ term) specify $S_{MAX}$ and then $L_{MAX}$ among all of the $L$-$S$ states of a configuration.

$L_{MAX} M_{L_{MAX}} S_{MAX} M_{S_{MAX}}$ is a single Slater determinant. It is also equal to

$$J = L_{MAX} + S_{MAX} \quad M_{J_{MAX}} = L_{MAX} + S_{MAX}$$

$$\zeta\left(n\ell^N, L_{MAX}, S_{MAX}\right) L_{MAX}, S_{MAX} = \zeta_{n\ell} \sum_i m_{\ell_i} m_{s_i}$$

$M_L$ $M_S$

$$\zeta\left(n\ell^N, L_{MAX}, S_{MAX}\right) = \zeta_{n\ell} \frac{\sum_i m_{\ell_i} m_{s_i}}{L_{MAX} S_{MAX}}$$

Shell $< \frac{1}{2}$ full $\qquad N < 2\ell + 1$

All spins are α

$$S_{MAX} = N/2$$

$$M_{L_{MAX}} = \ell + (\ell - 1) + \ldots (\ell - N + 1)$$

Sum includes sum of one $\ell$ for each of N terms, each additive term decreased by 0, 1, … $N-1$.

$$M_{L_{MAX}} = N\left(\ell - (N-1)/2\right)$$

$m_s = \alpha$ for all occupied spin-orbitals

$$\zeta\left(n\ell^N, L_{\text{MAX}}, S_{\text{MAX}}\right) = \zeta_{n\ell}\,\frac{\frac{1}{2}\sum_i m_{\ell_i}}{L_{\text{MAX}} S_{\text{MAX}}}$$

($M_L = L_{\text{MAX}}$)

$$= \zeta_{n\ell}\,\frac{\frac{1}{2}L_{\text{MAX}}}{L_{\text{MAX}}(N/2)} = \zeta_{n\ell}/N = \zeta_{n\ell}/2S_{\text{MAX}}$$

Similarly for shell that is more than half-full.

$$S_{\text{MAX}} = \frac{1}{2}\left\{(2\ell+1)\alpha \text{ spins} - \left[N-(2\ell+1)\right]\beta \text{ spins}\right\}$$
$$= (2\ell+1) - N/2$$
$$M_S = S_{\text{MAX}} = (2\ell+1) - N/2$$

($= 0$; $L_{\text{MAX}}$)

$$\zeta\left(n\ell^N, L_{\text{MAX}}, S_{\text{MAX}}\right) = \zeta_{n\ell}\,\frac{\left[\frac{1}{2}\sum_{(\alpha)} m_{\ell_i} - \frac{1}{2}\sum_{(\beta)} m_{\ell_i}\right]}{L_{\text{MAX}} S_{\text{MAX}}}$$

$$= -\frac{\zeta_{n\ell}}{2S_{\text{MAX}}} = -\frac{\zeta_{n\ell}}{(2\ell+1) - N/2}$$

So we have

$$\zeta\left(n\ell^N, L_{\text{MAX}}, S_{\text{MAX}}\right) = \frac{\zeta_{n\ell}}{2S_{\text{MAX}}} \qquad N < 2\ell+1$$

$$S_{\text{MAX}} = N/2 = \frac{1}{2}\ \#\ \text{of}\ e^- \qquad N = 2\ell+1$$

$$= -\frac{\zeta_{n\ell}}{2S_{\text{MAX}}} \qquad N > 2\ell+1$$

$$S_{\text{MAX}} = \left[(2\ell+1) - N/2\right] = \frac{1}{2}\ \left(\#\ \text{of}\ h^+\right)$$

Shell half full

$$\left.\begin{array}{c} L_{\text{MAX}} = 0 \\ S_{\text{MAX}} = N/2 \end{array}\right] {}^{N+1}S_{N/2}$$

No fine structure. Only a single J-Level, J = N/2.

3. Zeeman Effect.

Provides additional information for assignment of an L–S–J state.

$$\mathbf{H}^{\text{Zeeman}} = -\left(\mu_0/\hbar\right)\left(L_Z + 2S_Z\right)B_Z$$

$\mu_0$: Bohr Magneton, 1.399613 MHz/Gauss, 1 $cm^{-1}$ = 29979 MHz

$B_Z$: magnetic field (entirely along Z laboratory axis)

In the coupled representation, $|JM_JL_S\rangle$ matrix elements of $\mathbf{H}^{\text{Zeeman}}$ are awkward because it is necessary to transform from coupled to uncoupled $|LM_LSM_S\rangle$ basis set in order to evaluate matrix elements of $\mathbf{H}^{\text{Zeeman}}$.

This is a fast way to get a closed-form expression for matrix elements of $\mathbf{H}^{\text{Zeeman}}$ in the coupled basis set. It involves a clever way to evaluate reduced matrix elements using the Wigner-Eckart Theorem.

$\mathbf{H}^{\text{SO}}$ and $e^2/r_{ij}$ are rigorously diagonal in J (WHY?) but $\mathbf{H}^{\text{Zeeman}}$ is not (WHY?). [Answers depend on how you classify an operator relative to $\vec{\mathbf{J}}$.]

We have a battle between [$\mathbf{H}^{\text{SO}}$ and $e^2/r_{ij}$], which is diagonal in J, thus it tries to *defend* the coupled representation against $\mathbf{H}^{\text{Zeeman}}$, which is non-diagonal in J and tries to *destroy* the coupled representation.

We are interested in the weak field limit where

$$\left\langle J'\left|\mathbf{H}^{\text{Zeeman}}\right|J\right\rangle \ll \left|E^{(0)}_{J'} - E^{(0)}_{J}\right| \text{ due to the operators } \mathbf{H}^{\text{SO}} \text{ and } e^2/r_{ij}$$

Our special case, limited to $\Delta J = 0$ matrix elements, fails when $\zeta$(NLS) is small and $B_Z$ is large.

Wigner Eckart Theorem tells us

$$\langle JM'LS|\mathbf{L}|JMLS\rangle = \langle JLS||\mathbf{L}||JLS\rangle\langle JM'_J LS|\mathbf{J}|JM_J LS\rangle$$

Similarly for **S**

$$\langle JM'LS|\mathbf{S}|JMLS\rangle = \langle JLS||\mathbf{S}||JLS\rangle\langle JM'LS|\mathbf{J}|JMLS\rangle$$

but $J = L + S$

$$\langle|\mathbf{J}|\rangle = \underbrace{\left(\langle||\mathbf{L}||\rangle + \langle||\mathbf{S}||\rangle\right)}_{=1}\langle JM'LS|\mathbf{J}|JMLS\rangle$$

$$\underbrace{(1-\alpha) \qquad \alpha}$$

Thus $\langle|\mathbf{S}|\rangle = \alpha\langle|\mathbf{J}|\rangle$ or $\boxed{\mathbf{S} = \alpha\mathbf{J}}$! An operator replacement!

Now for the cleverness:

$$\mathbf{J} = \mathbf{L} + \mathbf{S}$$

$$\mathbf{J} - \mathbf{S} = \mathbf{L}$$

$$(\mathbf{J} - \mathbf{S})^2 = \mathbf{L}^2$$

$$\mathbf{J}^2 + \mathbf{S}^2 - 2\mathbf{J}\cdot\mathbf{S} = \mathbf{L}^2$$

$$\mathbf{J}\cdot\mathbf{S} = \frac{\mathbf{J}^2 + \mathbf{S}^2 - \mathbf{L}^2}{2}$$

Matrix elements of $\mathbf{J}^2$, $\mathbf{S}^2$, and $\mathbf{L}^2$ are explicitly evaluable in the coupled representation, $|JLSM_J\rangle$.

Thus:

$$\langle JM'LS|\mathbf{J}\cdot\mathbf{S}|JMLS\rangle = \delta_{M'M}\hbar^2\left[\frac{J(J+1)+S(S+1)-L(L+1)}{2}\right]$$

but $\mathbf{J}\cdot\mathbf{S} = \alpha\mathbf{J}^2$

$$\langle JM'LS|\alpha\mathbf{J}^2|JMLS\rangle = \delta_{m'm}\left[\hbar^2\alpha J(J+1)\right]$$

we can solve for $\alpha$

$$\alpha J(J+1) = \frac{J(J+1)+S(S+1)-L(L+1)}{2}$$

$$\boxed{\alpha = \frac{J(J+1)+S(S+1)-L(L+1)}{2J(J+1)}}$$

$$\mathbf{H}^{\text{Zeeman}} - \mu_0 B_z\left(\mathbf{L}_z + 2\mathbf{S}_z\right) = -\mu_0 B_z(1+\alpha)\mathbf{J}_z$$

$(1-\alpha)\mathbf{J}_z$ $\quad$ $\alpha\mathbf{J}_z$

$$\langle JMLS|\mathbf{H}^{\text{Zeeman}}|JMLS\rangle = -\mu_0 B_z \hbar M_J \underbrace{(1+\alpha)}_{g_J}$$

Zeeman energy level tuning rate:

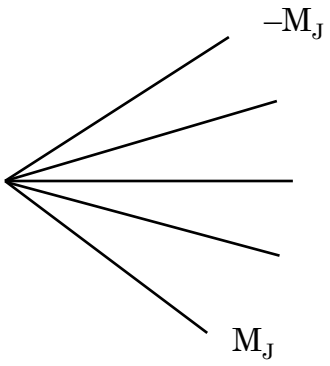
$$\frac{dE}{dB_z} = \frac{d}{dB_z}\left(-\mu_0 B_z M_J(1+\alpha)\right)$$

$$= -\mu_0 M_J g_J$$

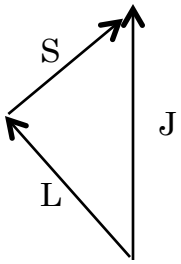
$$g_J = -\frac{1}{\mu_0}\frac{1}{M_J}\frac{dE}{dB_z}$$

Apply a magnetic field, each J level splits into $2J + 1$ $M_J$ components.

# of components tells us J, $g_J$ tells us something about L and S




tuning is linear in $B_z$




$J_{MAX} = L + S$ $L$ and $S$ parallel (magnetic moments add)

$J_{MIN} = |L - S|$ $L$ and $S$ anti-parallel (magnetic moments partially cancel)

e.g.

| for J = 3 | L = 0, S = 3 | L = 1, S = 2 | L = 2, S = 1 | L = 3, S = 0 |
|---|---|---|---|---|
| $g_J$ | 2.000 | 1.667 | 1.333 | 1.000 |
| or for L = 3, S = 1 | | J = 4 | J = 3 | J = 2 |
| $g_J$ | | 1.250 | 1.0833 | 0.667 |
| | | parallel | | anti-parallel |

To determine J: apply $B_Z$ field and count $M_J$ components

light Z polarized, $\Delta M_J = 0$

J′ set of equal $M'_J, M'_J - 1$ splittings
J″ another set of equal $M''_J, M''_J - 1$ splittings ] $g_{J'} \neq g_{J''}$

(′ for upper state, ″ for lower state)

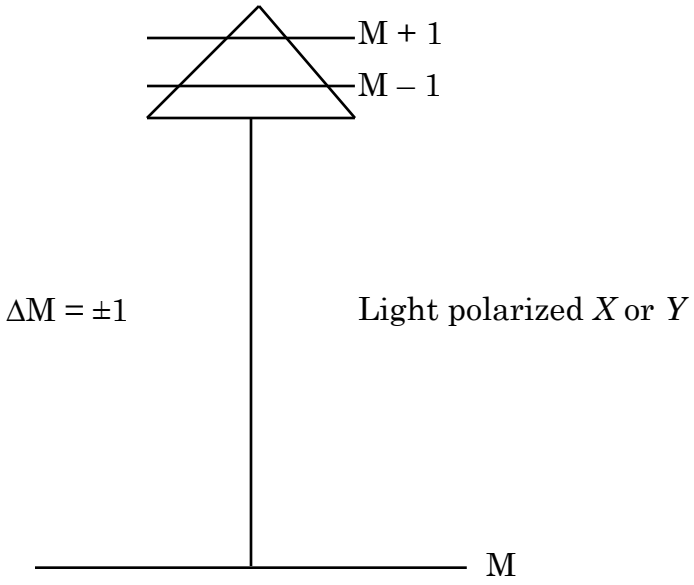
Alternatively, light polarized $X$ or $Y$ (perpendicular to $B_Z$)

get $M' - M'' = \pm 1$

If you know the upper and lower state patterns, you can disentangle the spectrum and get $g_{J'}$ and $g_{J''}$.

Use a pulsed laser

Excite




Get quantum beats ⟶ determine $g_{J'}$! For many M" components, get a set of Zeeman Quantum Beats for J' upper levels, but all upper level quantum beats are at the same frequency!

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You can show that $g_J$ is the same for $e^-$ and $h^+$.

You can use the hole Slater determinant representation to evaluate $\mathbf{H}^{Zeeman}$ for more than 1/2 filled shells.

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# Infinite 1-D Lattice I

CTDL, pages 1156-1168

LAST TIME:

hole ($\hbar^+$) vs. $e^-$ configurations — $\ell^N \leftrightarrow \ell^{2(2\ell+1)-N}$ [e.g. $f^{13} \leftrightarrow f^1$] for $N > 2\ell + 1$

$e^2/r_{ij}$ unchanged

$\zeta(\text{NLS}) \leftrightarrow -\zeta(\text{NLS})$ — [$\zeta_{n\ell}$ unchanged]

Hund's 3rd Rule (Lowest L-S term of $\ell^N$ only)

| | | |
|---|---|---|
| $N<2\ell+1$ | $E_{MIN}$ for $J = \lvert L-S \rvert$ | regular |
| $N=2\ell+1$ | ${}^{(2\ell+1)+1}S_{J=\frac{2\ell+1}{2}}$ | $S$ state: no fine structure |
| $N>2\ell+1$ | $E_{MIN}$ for $J = L+S$ | inverted |

Zeeman Effect

Wigner-Eckart Theorem used to define $g_J$

$$E^{Zeeman} = -\mu_0 M_J g_J B_Z$$

equal spacings

$g_J$ as L,S,J diagnostic

$$g_J = 1 + \frac{J(J+1) + S(S+1) - L(L+1)}{2J(J+1)}$$

Confirm by $\mathbf{H}^{Zeeman}$ in Slater determinantal basis

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TODAY:

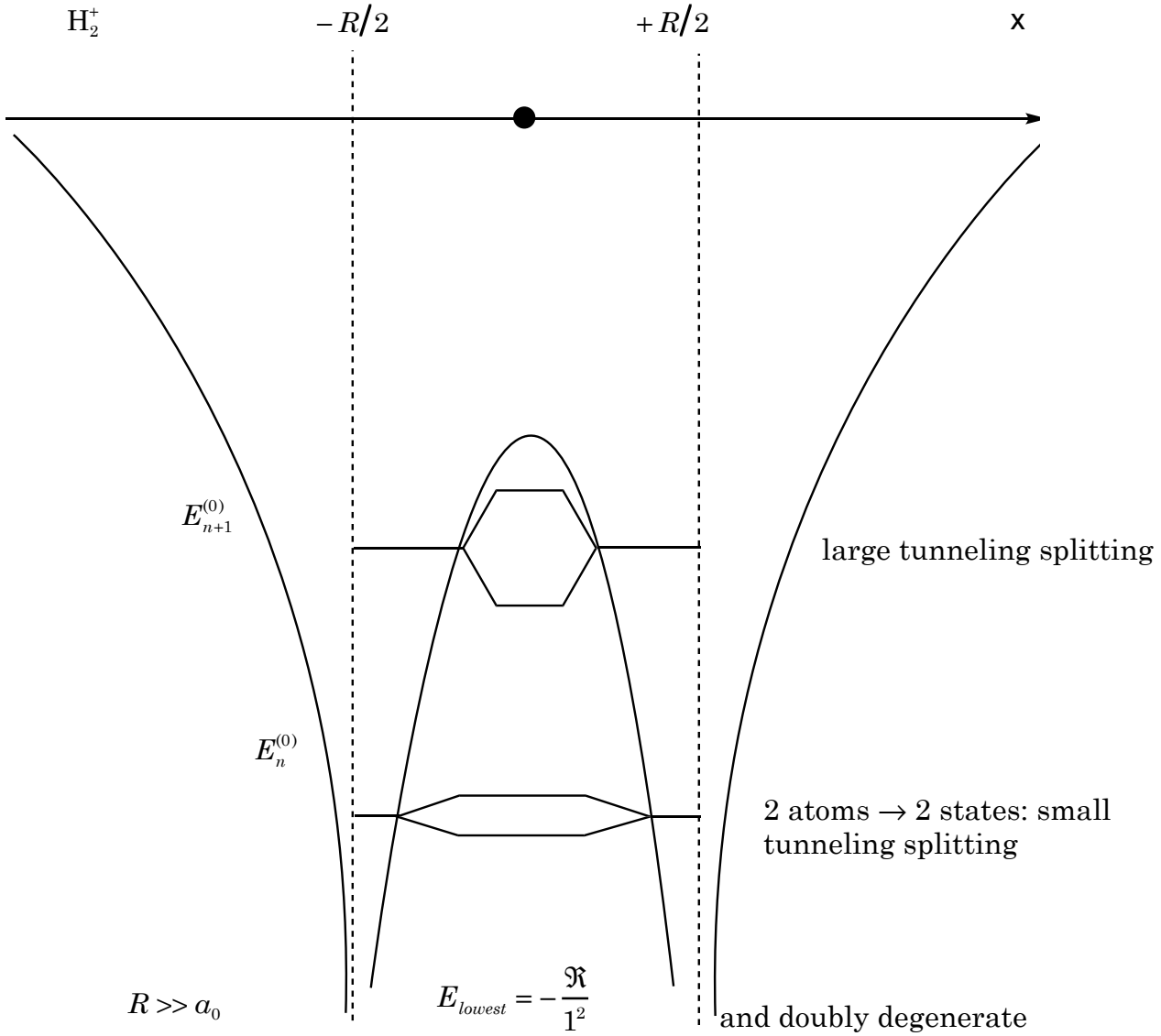
1. $H_2$ as example of localization, delocalization, tunneling
2. $\infty$ dimension secular equation for simplified 1-D lattice
3. eigenvectors by equal probability trick
4. restrict k to $|k| < \pi/\ell$ : 1st Brillouin Zone
5. $E(k) = E_0 - 2A\cos k\ell$ (are these all of the allowed states?)
6. Bloch functions $\psi_k(x) = e^{ikx}\mu_k(x)$

next lecture

7. wavepackets, motion, group velocity
8. transitions: energy bands and intensity profiles
9. conductivity

Start with $H_2^+$, a lattice with only 2 equivalent sites.

qualitative picture: atomic energy levels
tunneling between identical localized states:
is slow behind a high barrier (small splitting)
is fast behind a low barrier (large splitting)
levels → bands, of width related to tunneling rate




For exact degeneracy between left-well and right-well localized states, can choose any linear combination

Localized basis set: $\psi_{\text{localized}} = \psi_{\text{left}}^{(0)}$ or $\psi_{\text{right}}^{(0)}$

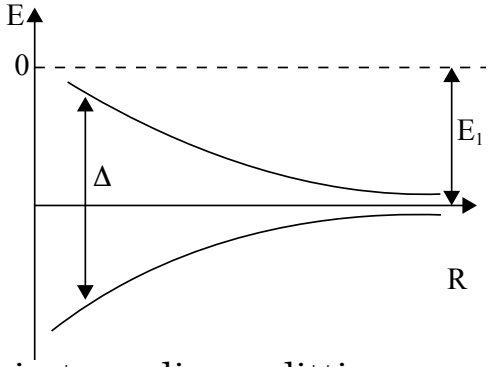
Delocalized basis set $\psi_{\textit{de}\text{localized}} = 2^{-1/2}\left[\psi_{\text{left}}^{(0)} \pm \psi_{\text{right}}^{(0)}\right]$

If initially in a localized state, tunneling rate depends on

* height (relative to $E_n^{(0)}$) of barrier
* width of barrier
* size of overlap between exponential tails of $\psi_{\text{left}}^{(0)}$ and $\psi_{\text{right}}^{(0)}$

clear that tunneling rate (i.e. reciprocal of adjacent level-splitting) increases

* as n↑ at constant R (internuclear separation)
* as R↓ at constant n

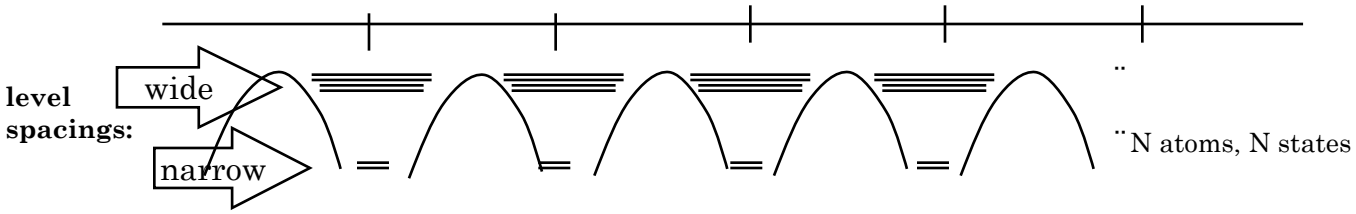



double degeneracy
at $R \to \infty$

Δ is tunneling splitting — gets larger as R↓. Less localization causes tunneling rate to increase. The tunneling rate is $\frac{\Delta E}{h}$ (which has units of 1/t).

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N ATOMS ALONG A STRAIGHT LINE




Each electronic state of the isolated atom becomes a band of states for ∞–atom lattice. Energy width of each band increases as the principal quantum number increases because atomic states require more room: $\langle r \rangle_n \propto a_0 n^2$. Tunneling gets faster. Greater sensitivity to the world outside a single atom.

Simplified model for ∞ 1–Dimensional Lattice: basis for qualitative insights and early-time predictions.

1. Each ion, called q, has **one** bound state, $|\nu_q\rangle$
   at $E_0 = \langle\nu_q|\mathbf{H}|\nu_q\rangle$ [diagonal element of **H**] (actually 2 spin-orbitals)

2. Only permit orbitals on *adjacent* ions to interact [important simplifying assumption], like Hückel theory.

3. symmetry: all ions are equally spaced, $x_{q+1} - x_q = \ell$, and all adjacent-orbital interaction matrix elements are identical

$\langle\nu_q|\mathbf{H}|\nu_{q+1}\rangle \equiv -A$ [off-diagonal elements of **H**]
[reasons for – A sign choice later.]

(|A| must increase as $\ell \to 0$)

$$\text{so } \mathbf{H} = \begin{pmatrix} E_0 & -A & & \mathbf{0} \\ -A & \ddots & \ddots & \\ & \ddots & E_0 & -A \\ \mathbf{0} & & -A & \ddots \\ & & & \ddots \end{pmatrix}$$

tridiagonal infinite matrix

Since this is infinite, we need a trick to diagonalize it.

try a general variational function

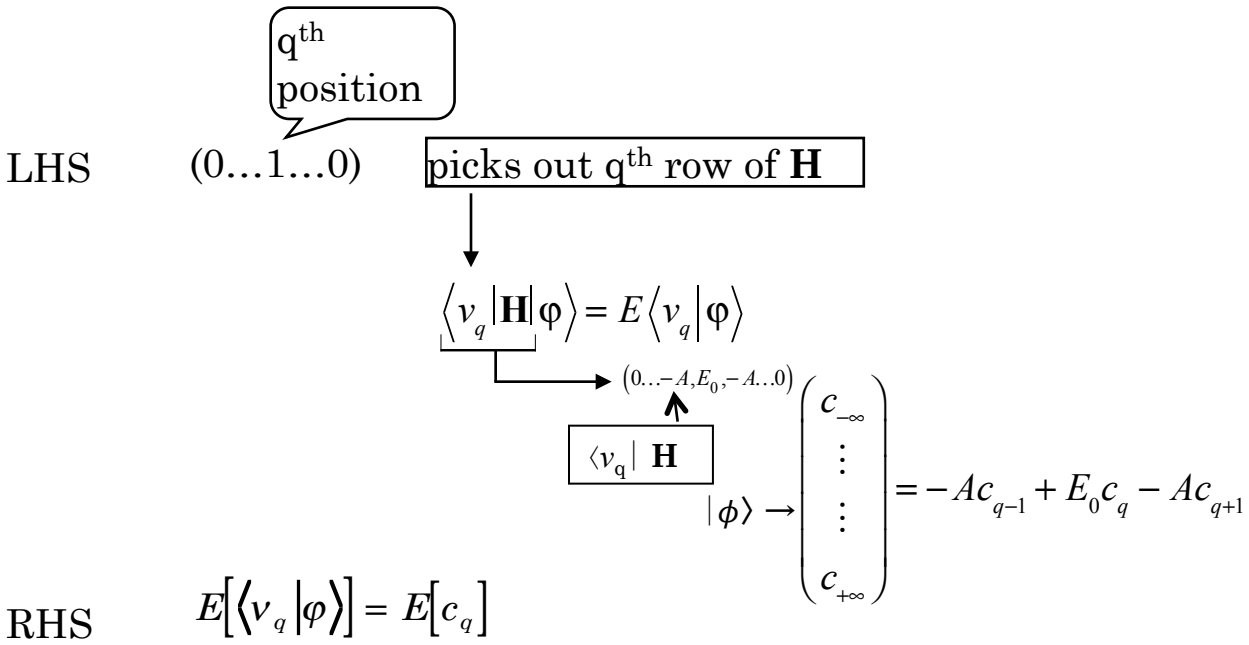
$$|\varphi\rangle = \sum_{q=-\infty}^{\infty} c_q |\nu_q\rangle$$

superposition of AO's each localized at each site

we get requirements on $c_q$ by plugging this into the Schrödinger equation

$$\mathbf{H}|\varphi\rangle = E|\varphi\rangle$$

left multiply by $\langle\nu_q|$




$$0 = c_q\left[E_0 - E\right] - c_{q-1}A - c_{q+1}A$$

comes from the assumed simple form of the model

TRICK: probability of finding $e^-$ on each lattice site should be the same for all sites (complex amplitudes might differ but the probabilities will be constant)

let $\boxed{c_q = e^{ikq\ell}}$ $\quad |c_q|^2 = 1$ for all $q$

This choice of $c_q$ is a good guess that is consistent with expectation of equal probabilities on each lattice site.

$\ell$ is the distance between adjacent atoms
$q$ is an integer
$q\ell$ is the coordinate of the q-th site: looks like an $e^{ikx}$ plane wave
$k$ is of dimension $\ell^{-1}$

This problem reduces to finding the allowed values of $k$.

The periodicity of the lattice provides the important result, that if $k$ is replaced by $k'$, where $k' = k + \frac{2\pi}{\ell}$, the wavefunction does not change (translational symmetry).

$$c'_q = e^{ik'q\ell} = e^{\left(ikq\ell + i\frac{2\pi}{\ell}\ell\right)} = e^{ikq\ell}\underbrace{e^{i2\pi q}}_{=1} = c_q$$

Since all distinguishable $|\varphi\rangle$ may be generated by choosing the value of $k$ in the interval $-\frac{\pi}{\ell} \le k < \frac{\pi}{\ell}$, restrict $k$ to this range; **this range of $k$ is called the "First Brillouin zone".**

Return to question about what happens when E is not in the range allowed by this range of $k$. Then $E$ is not in the 1st Brillouin Zone. Next lecture [get another part of the band structure using qualitative perturbation theory rather than a matrix diagonalization calculation].

Plug $c_q = e^{ikq\ell}$ into Schrödinger Equation

$$0 = c_q\left(E_0 - E\right) - A\left(c_{q+1} + c_{q-1}\right)$$

$$0 = e^{ikq\ell}\left(E_0 - E\right) - A\left(e^{ik(q+1)\ell} + e^{ik(q-1)\ell}\right)$$

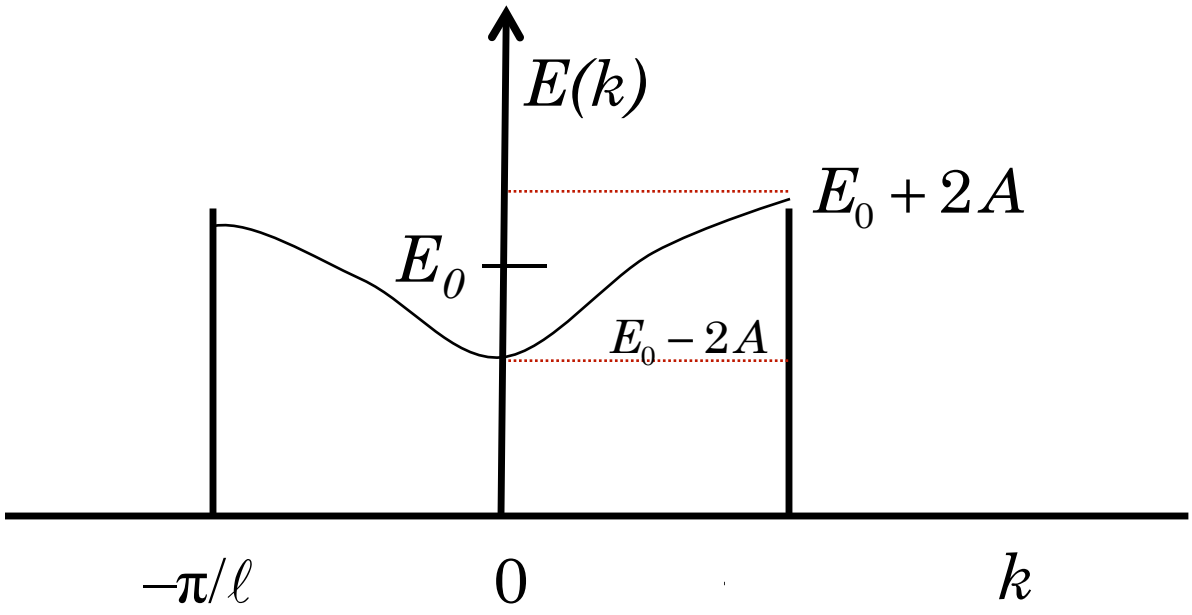
divide by $e^{ikq\ell}$ and rearrange

$$E = E_0 - A\underbrace{\left[e^{ik\ell} + e^{-ik\ell}\right]}_{2\cos k\ell}$$

This is the condition on $E,k$ that must be satisfied for all eigenfunctions of the Schrödinger equation!

$$\boxed{E = E_0 - 2A\cos k\ell}$$

$E$ varies continuously over finite interval $E_0 \pm 2A$




The choice $\left\langle \nu_q \left| \mathbf{H} \right| \nu_{q+1} \right\rangle = -A$ leads to a minimum E at $k = 0$.

Are these *all* of the allowed energy levels that arise from a single orbital at each lattice site? Apparently not — see Lecture #38. They are only half of the states. [One orbital per atom → two spin-orbitals per atom. Antisymmetrization gives another separate band.] like singlet vs. triplet.

We could repeat the calculation looking for a higher energy state at each site. Would get a broader band centered at higher energy.

---

Take a closer look at the spatial form of $\varphi_k(x) \equiv \langle x | \varphi_k \rangle$

$$\varphi_k(x) = \left\langle x \middle| \varphi_k \right\rangle = \sum_{q=-\infty}^{+\infty} e^{ikq\ell} \underbrace{\left\langle x \middle| \nu_q \right\rangle}_{\nu_q(x)}$$

treated as sum over localized functions

The goal is to replace the infinite sum by a single term:

This is called a Bloch function

show that: $\varphi_k(x) \sim \overbrace{e^{ikx} u_k(x)}$

plane wave (Free particle)

periodicity of lattice

Begin by requiring that $\varphi_k(x) = \sum_{q=-\infty}^{\infty} e^{ikq\ell} \nu_q(x)$

Translational symmetry imposes a relationship between $\nu_q(x)$ and $\nu_0(x)$

each $\nu_q(x)$ is localized at site $q$.

This is a single function rather than a sum of separate localized functions.

$$\nu_q(x) = \nu_0\left(x - q\ell\right)$$

$$\varphi_k(x) = \sum_{q=-\infty}^{\infty} e^{ikq\ell} \nu_0\left(x - q\ell\right)$$

shift $x$ by $-q\ell$ to get from site $q$ to site 0

$$\varphi_k(x+\ell) = \sum_{q=-\infty}^{\infty} e^{ikq\ell} \underbrace{\nu_0\left(x + \ell - q\ell\right)}_{=\nu_0(x-(q-1)\ell)}$$

$$= e^{ik\ell} \sum_{q=-\infty}^{\infty} e^{ik(q-1)\ell} \nu_0\left(x - (q-1)\ell\right)$$

Re-index the sum (replace q–1 by q) get original function multiplied by $e^{ik\ell}$

$$\varphi_k(x+\ell) = \underbrace{e^{ik\ell}}_{\text{translation by } \ell}\varphi_k(x)$$

This form of $\phi_k$ has all of the symmetry properties we will need. This form is sufficient to satisfy the symmetry requirements (boundary conditions). This means, instead of writing $\varphi_k(x)$ as sum over atom-localized $v_q(x)$'s, it is possible to write $\varphi_k(x)$ for all $k$ as a product of 2 factors.

$$\varphi_k(x) = e^{ikx}u_k(x)$$

this has the properties of the above sum form, but is more general

1st factor conveys translational symmetry of a plane wave with wavevector $k$, 2nd factor builds in translational symmetry of a lattice with spacing $\ell$. This is a more general expression that incorporates all of the properties of the original definition of $\varphi_k(x)$ as a sum over localized orbitals.

$$u_k(x+\ell) = u_k(x)$$

note that

$$\varphi_k(x+\ell) = e^{ikx}e^{ik\ell}u_k(x+\ell) = e^{ik\ell}\left[e^{ikx}u_k(x)\right]$$

$$= e^{ik\ell}\varphi_k(x)$$

as required.

Note also that $|\varphi_k(x+n\ell)|^2 = |\varphi_k(x)|^2$ implies that, as required, the $e^-$ has equal probability of being found on each site.

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# Infinite 1-D Lattice II

LAST TIME:

$\mathbf{H}_2^+$ localization↔tunneling: overlap !
bonding and antibonding orbitals

- Internuclear distance, $R$, vs. $a_0 n^2$ Bohr radius for n$^{th}$ orbit in H atom
- energy below top of barrier

TIGHT-BINDING (Kronig-Penney) Model (see Baym pp. 116-122)

1-D $\infty$ lattice: 1 state per ion!
tunneling only between nearest neighbors!
Infinite dimension **H** matrix

$$\mathbf{H} = \begin{pmatrix} E_0 & -A & \ddots & 0 \\ -A & E_0 & \ddots & 0 \\ 0 & \ddots & \ddots & \ddots \\ 0 & 0 & \ddots & \ddots \end{pmatrix}$$

$\langle \nu_q | \mathbf{H} | \varphi \rangle = E \langle \nu_q | \varphi \rangle$ left multiply both sides by $\langle \nu_q |$

$|\varphi\rangle = \sum_{q=-\infty}^{\infty} c_q |\nu_q\rangle$ Variational wavefunction. Minimize E.

$0 = c_q (E_0 - E) - A(c_{q-1} + c_{q+1})$ $\infty$ # of coupled linear equations, one for each q

Usually solve for $\{c_q\}$ by setting determinant of coefficients = 0, and then solve for E.
Can't do this because determinant is of $\infty$ dimension.

TRICK: expect equal probability of finding $e^-$ on each lattice site by analogy to the plane wave $e^{ikx}$, where probability density is uniform at all sites along x.
Try:

$c_q = e^{ikq\ell}$ $x_q = q\ell$

integer

distance between lattice sites

Notice that this is similar to free particle $e^{ikx}$, which seems rather strange because particle is never really free in the "tight-binding" model.

$|c_q|^2 = 1$

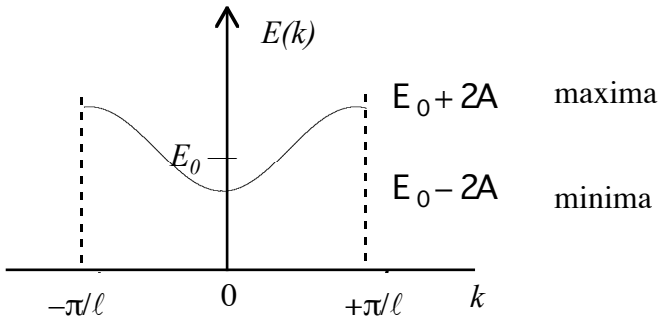
plug trial form for $c_q$ into $0 = c_q\left(E_0 - E\right) - A\left(c_{q-1} + c_{q+1}\right)$

$$0 = e^{ikq\ell}\left(E_0 - E\right) - Ae^{ikq\ell}\left(e^{-ik\ell} + e^{+ik\ell}\right)$$

divide through by $e^{ikq\ell}$

$0 = \left(E_0 - E\right)k - A2\cos k\ell$

$\boxed{E(k) = E_0 - 2A\cos k\ell}$ $E$ vs. $k$, we have achieved our goal!




E varies continuously over an interval of 4A, where A is the adjacent site interaction strength or the "**tunneling integral**"

What happens when we look at $k$ outside $-\pi/\ell \le k < \pi/\ell$ , which is called the "1st Brillouin Zone"

$c_k = e^{ikq\ell}$

$k' = k + \dfrac{2\pi}{\ell}$ (one additional allowed wavelength per lattice spacing $\ell$)

$$c_{k'} = e^{i\left(k+\frac{2\pi}{\ell}\right)q\ell} = e^{ikq\ell}\underbrace{e^{i2\pi q}}_{=1} = e^{ikq\ell}$$

wavefunction is unchanged!

So if $k$ goes outside 1st Brillouin Zone, we get the same $\Psi$, so we get the same E.!
Nothing new!
No point in allowing $k$ to vary more widely than $-\pi/\ell \le k \le \pi/\ell$.

Today's Lecture:

Unanswered Questions:

1. How many distinct orbitals are there in a band?

   N-atom periodic array. Try *Periodic Boundary* conditions:

   longest $\lambda = \ell N$
   shortest $\lambda = \ell$
   } N possible values of $\lambda = 2\pi / k$. N atoms, N values of $k$

   $0 \approx \frac{2\pi}{\ell N} \le |k| \le \frac{2\pi}{\ell}$ in N steps of $\frac{2\pi}{\ell}$

   infinite lattice: $\frac{-\pi}{\ell} < k < \frac{\pi}{\ell}$ contains all of the states generated from one state per atom.

2. What happens at $E > E_0 + 2A$?

   There is a gap – no states allowed
   What is the next higher state of each atom?
   Get a free particle (electron) if E > *work function* of the solid
   [energy required to remove $e^-$ from solid]

3. But we have *orbitals* not *states*! Two spin-orbitals per orbital.

   Antisymmetrization.
   Lowest band: all spins paired. G = 0 (exchange integral). No contribution from G term.
   $e^- - e^-$ repulsion raises overall E above that of the single-electronic ground state of each atom
   Therefore, the work function of the solid is less than that of the single-atom ionization potential.

4. How many $e^-$ does each atom contribute to $\psi$?

   alkali: $1e^- \rightarrow$ *half full* band
   alkaline earth: $2e^- \rightarrow$ *full* band

5. $\Psi(x,t)$: Motion

   phase velocity
   group velocity

6. Effective Mass

Now take a closer look at $\phi_k(x)$

$$\varphi(x) = \langle x \mid \varphi_k \rangle = \sum_{q=-\infty}^{+\infty} e^{ikq\ell} \underbrace{\langle x \mid \nu_q \rangle}_{\nu_q(x)}$$

$$\nu_q(x) = \nu_0(x - q\ell)$$ shift x by $-q\ell$ to get from site $q$ to site 0

$$\varphi_k(x) = \sum_q e^{ikq\ell} \nu_0(x - q\ell)$$

This makes it easy to see the effect of translation of the entire $\phi$ by $\ell$ (one lattice spacing)

$$\varphi_k(x+\ell) = \sum_q e^{ikq\ell} \nu_0(\overbrace{x - q\ell + \ell}^{-\ell(q-1)})$$

take factor $e^{ik\ell}$ out of summation:

$$= e^{ik\ell} \sum_{q=-\infty}^{\infty} e^{ik(q-1)\ell} \nu_0(x - (q-1)\ell)$$

re-index infinite summation (replace $q - 1$ by $q$ everywhere)

$$\varphi_k(x+\ell) = e^{ik\ell} \sum_q e^{ikq\ell} \nu_0(x - q\ell) = \underbrace{e^{ik\ell}}_{\text{translation of plane wave by } \ell} \varphi_k(x)$$

This implies that it is possible to write $\phi_k(x)$ in a more general form:

$$\phi_k(x) = e^{ikx} u_k(x)$$ **Bloch wave function**

where $u_k(x + \ell) = u_k(x)$ perodicity of $\ell$

$e^{ikx}$ expresses translational symmetry of plane wave with wavevector $k$
$u_k(x)$ expresses translational symmetry of lattice with spacing $\ell$

Now consider a localized time-dependent state : "wavepacket"
We are gong to build intuitive insight by comparison to free particle.

Recall free particle:

$$\Psi(x,t) = (2\pi)^{-1/2} \int dk \underbrace{g(k)}_{\substack{\text{envelope of } k \\ \text{centered at } k_0}} e^{i\left[kx - \underbrace{E(k)t/\hbar}_{\omega t}\right]}$$

Group velocity: motion of *stationary phase point* (we want it to be stationary with respect to $k$ near $k_0$)

$$0 = \frac{d}{dk}\left[kx - Et/\hbar\right]\Big|_{k=k_0}$$

$$x_{\text{center}}(t) = \left.\frac{dE}{dk}\right|_{k_0} t/\hbar \quad \text{position} \qquad \text{now take } \frac{d}{dt}$$

$$v_{center} = \left.\frac{dE}{dk}\right|_{k_0} \frac{1}{\hbar} \quad \text{velocity}$$

$$E = \frac{\hbar^2 k^2}{2m} \qquad \left[\frac{1}{2}mv^2 = \frac{p^2}{2m} = \frac{(\hbar k)^2}{2m}\right]$$

$$\left.\frac{dE}{dk}\right|_{k_0} = \frac{\hbar^2 k_0}{m}$$

$$v_{center} = v_G = \frac{1}{\hbar}\left[\left.\frac{dE}{dk}\right|_{k_0}\right] = \frac{\hbar k_0}{m}$$

group velocity

free particle relationship between! $v_G$ and ` k and m

use this to understand motion in a periodic lattice

*updated August 27, 2020 @ 1:45 PM*

Up to here we have been considering a free particle.

For 1–D lattice: the time-dependent wave function is

$$\left|\Psi(t)\right\rangle = (2\pi)^{-1/2}\int dk\, \underbrace{g(k)}_{\substack{\text{maximum}\\ \text{amplitude at } k_0}} e^{-iE(k)t/\hbar}\left|\varphi_k\right\rangle.$$

Instead of asking for the location of the stationary phase point, we now ask for the time dependent overlap of Ψ(t) with a specific lattice site, $|\nu_q\rangle$.

$$\left\langle \nu_q \middle| \Psi(t)\right\rangle = (2\pi)^{-1/2}\int dk\; g(k) e^{i\left[kq\ell - E(k)t/\hbar\right]}$$

because $\left|\varphi_k\right\rangle = \sum_{q=-\infty}^{\infty} e^{ikq\ell}\left|\nu_q\right\rangle$

$$\left(\text{same thing as } \phi_k(x) = \sum_{q=-\infty}^{\infty} e^{ikq\ell}\left\langle x \middle| \nu_q\right\rangle\right)$$

We can use either *state vector* or *wavefunction* picture.

and $\left\langle \nu_q\right|$ picks out only the $e^{ikq\ell}$ term from the sum over $q$,

because $\left\langle \nu_p \middle| \nu_q\right\rangle = \delta_{p,q}$.

Recall that $x = q\ell$, so we can think of $\left\langle \nu_q \middle| \Psi(t)\right\rangle$ as function of $x,t$.

We want the overlap of Ψ(t) with a particular lattice site, $\left|\nu_q\right\rangle$.

Ψ(t) moves and sequentially overlaps successive lattice sites.

meaningful only for regions of $x$ near $q\ell$

$$\left\langle \nu_q \middle| \Psi(t)\right\rangle = \chi(x,t) = (2\pi)^{-1/2}\int dk g(k) e^{i\left[kx - E(k)t/\hbar\right]}$$

real part · phase factor

Now ask for the stationary phase factor (near $x = 0, \pm\ell, \pm 2\ell, \ldots$) with respect to k:

$$0 = \frac{d}{dk}\left[ kx - E(k)t / \hbar \right]$$ center of wavepacket, $x_c$

$$x_c(t) = \left.\frac{dE}{dk}\right|_{k_0} t / \hbar$$ wavepacket is created at $t = 0$ centered at $k = k_0$

$$v_G = \frac{dx_c}{dt} = \left.\frac{dE}{dk}\right|_{k_0} \frac{1}{\hbar}$$ This is the velocity of center of wavepacket. Up to this point, everything is identical for free particle and motion in a periodic lattice.

$$E(k) = E_0 - 2A\cos k\ell$$ Now use the $E \leftrightarrow k$ relationship derived for periodic (tight binding) lattice.

$$\left.\frac{dE}{dk}\right|_{k_0} = 2A\ell \sin k_0\ell$$

$$v_G = \frac{2A\ell}{\hbar}\sin k_0\ell$$ This is quite different from the free particle plane wave result: $v_G = \frac{\hbar k_0}{m}$

Note that, when $k_0$ is at the bottom ($k = 0$, $E = E_0 - 2A$) or top ($k_0 = \pm\pi/\ell$, $E = E_0 + 2A$) of the band, $v_G = 0$.

Building of new intuition:

* $v_G \propto A$ as |A| (the adjacent site interaction) increases, it becomes easier to take a step and the wavepacket moves faster.

* $v_G \propto \ell$ (but A $\downarrow$ as $\ell \uparrow$)

because the tunneling rate decreases as $\ell$ increases but if A is kept constant as $\ell$ increases, each step is longer so velocity will be faster

* $v_G = 0$ when $k_0 = 0$ and when $k_0 = \pm\pi / \ell$

*minimum E of band*
**Not a surprise**
because expect
k = 0 → v = 0

*maximum E of band*
**Big surprise**.
Use concept of "**effective mass**" to rationalize.

$e^-$ cannot move if it is too close to either edge of the band

## "**Effective Mass**:" free vs. lattice

$$v_G = \frac{\hbar k_0}{m} \qquad v_G = \frac{2A\ell \sin k_0\ell}{\hbar} \approx \hbar k_0 \left[\frac{2A\ell^2}{\hbar^2}\right]$$

because at small $k_0\ell$

$\sin k_0\ell \approx k_0\ell$

Compare the terms and identify the reciprocal of the coefficient of ` $k_0$ as the "effective mass", *by analogy with the free particle*:

$$m_{\text{eff}} = \frac{\hbar^2}{2A\ell^2} \text{ at small } k_0\ell$$

* large interaction strength makes $m_{eff}$ small
* large $\ell$ makes $m_{eff}$ small (large jumps become possible)

*Next* : How do we show that $m_{\text{eff}}$ increases to ∞ at the k-edges of the band ($k = \pm\pi / \ell$)?

When $k_0$ is near $\pm\pi/\ell$

$$k_0 = \pm\left(\frac{\pi}{\ell} - \varepsilon\right) \quad \varepsilon \text{ is small}$$

$$\sin k_0\ell = \sin\left[\pm\left(\frac{\pi}{\ell} - \varepsilon\right)\ell\right] \approx \pm\varepsilon\ell$$

$$v_G = \hbar k_0\left[\frac{2A\ell}{\hbar^2 k_0}\sin k_0\ell\right] \approx \pm\hbar k_0\left[\underbrace{\frac{2A\ell}{\hbar^2 k_0}}_{1/m_{eff}}\varepsilon\ell\right] \qquad [v = p/m]$$

$$m_{eff} = \frac{\hbar^2 k_0}{2A\ell^2\varepsilon} \longrightarrow \infty \text{ as } \varepsilon \to 0$$

**This means that at the energy of the top of a filled band : no e⁻ transport is possible!**

1/2-Full band: $m_{eff} = \frac{\sqrt{2}}{2}\frac{\hbar^2}{A\ell^2}$ ($k = \pm\frac{\pi}{2\ell}$)

(slightly heavier than at bottom of band, $m_{eff} = \frac{\hbar^2}{2A\ell^2}$, $k_0 = 0$)

# Alternative approach to $m_{eff}$:

$E = p^2/2m$ for free particle

$\left(\frac{d^2E}{dp^2}\right)^{-1} = m$ use this to define $m_{eff}$

$E(k) = E_0 - 2A\cos k\ell$

$E(p) = E_0 - 2A\cos(p\ell/\hbar)$

$$\frac{d^2E}{dp^2} = \left(2A\ell^2/\hbar^2\right)\cos k\ell \qquad \cos k\ell = 1 - \tfrac{1}{2}(k\ell)^2 + \ldots$$

thus, at small $k\ell$ $\quad m_{eff} = \frac{\hbar^2}{2A\ell^2}$

We have an intuitive picture for the time-evolution of a localized wavepacket. The concept of effective mass guides our intuition.

$$v_G = \frac{\hbar k}{m_{\text{eff}}}$$

$$m_{\text{eff}} = \frac{\hbar^2}{2A\ell^2} \quad \text{at bottom of band}$$

$$m_{\text{eff}} = \frac{\hbar^2 k_0}{2A\ell^2 \varepsilon} \quad \to \infty \text{ near top of band (full band)}$$

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## One Dimensional Lattice: Weak-Coupling Limit

In Lectures 37 and 38 we considered the strong coupling limit, like tunneling in $H_2^+$. Now we will look at the periodic lattice as a perturbation on the free particle.

See Baym "Lectures on Quantum Mechanics" pages 237-241.

Each atom in a lattice is represented by a 1-D V(x) that could bind an unspecified number of electronic states:

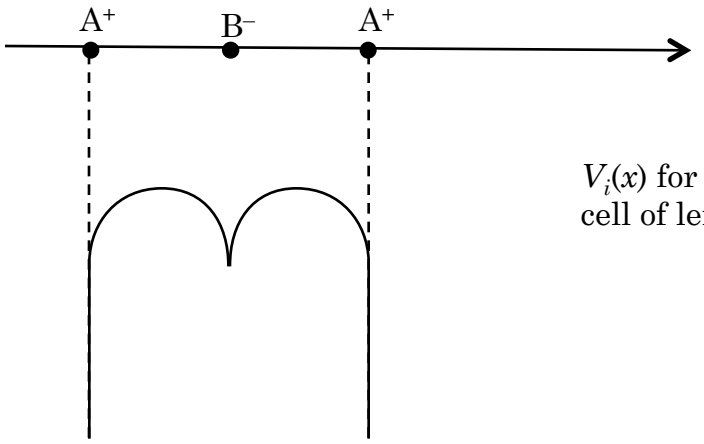
Now consider a lattice that could consist of two or more different types of atoms.

Periodic structure: repeated for each "unit cell", of length $\ell$.

Consider a finite lattice (N atoms), but impose a periodic (head-to-tail) boundary condition.

$L = N\ell$

For each unit cell:




This is an infinitely repeated finite interval: Fourier Series

$$V(x) = \sum_{n=-\infty}^{\infty} e^{iKnx} V_n$$

$$K = \frac{2\pi}{\ell} \quad \text{"reciprocal lattice vector"}$$

$V_n$ is the (possibly complex) Fourier coefficient of the part of $V(x)$ that *looks like* a free particle state with wave-vector $Kn$ (momentum $\hbar Kn$). Note that $Kn$ is larger than the largest $k$ (shortest $\lambda$) free-particle state that can be supported by a lattice of spacing $\ell$.

$$Kn = n\frac{2\pi}{\ell} \quad , \quad \text{first Brillouin Zone for } k\text{:}$$

$$-\frac{\pi}{\ell} \le k \le \frac{\pi}{\ell}$$

We will see that the lattice is able to exchange momentum in quanta of $\hbar nK$ with the free particle. In 3-D, $\vec{K}$ is a vector.

To solve for the effect of $V(x)$ on a free-particle, we use perturbation theory. The free particle basis states are weakly perturbed by the periodic lattice.

1. Define the basis set.

$$\left.\begin{aligned} \mathbf{H}^{(0)} &= \frac{\mathbf{p}^2}{2m} = -\frac{\hbar^2}{2m}\frac{d^2}{dx^2} + V^{(0)} \\ V^{(0)} &= \text{constant} \\ \psi_k^{(0)} &= L^{-1/2}e^{ikx} \end{aligned}\right\} \text{ free particle}$$

$$\left[\text{Normalization: } \int_0^L dx\left|\psi_k^{(0)}\right|^2 = L(1/L) = 1\right]$$

one particle per L

$$E_k^{(0)} = \frac{\hbar^2 k^2}{2m} + V^{(0)} \longleftarrow \text{ignore}$$

2. $$\mathbf{H}^{(1)} = \sum_{n=-\infty}^{\infty} e^{iKn}V_n$$

$$\text{Matrix elements: } H_{k'k}^{(1)} = \int_0^L dx\,[\overbrace{L^{-1/2}e^{-ik'x}}^{\psi_{k'}^{(0)*}}]\left[\sum_n e^{iKnx}V_n\right][L^{-1/2}e^{ikx}]$$

$$H_{k'k}^{(1)} = \frac{1}{L}\int_0^L dx \sum_n e^{ix(k+Kn-k')}V_n$$

integral = 0 if $k + Kn - k' \neq 0$

$$\therefore k' = k + Kn \quad \text{selection rule}$$

$$H_{k'k}^{(1)} = \frac{1}{L}L\sum_n V_n\delta_{k',k+Kn} = \sum_n V_n\delta_{k',k+Kn}$$

Must be careful about $H^{(1)}_{kk'}$ $\left(\text{relative to } H^{(1)}_{k'k}\right)$. Return to definition.

$$H^{(1)}_{kk'} = \frac{1}{L}\int_0^L dx \sum_n e^{ix(-k+Kn+k')} V_n = \sum_n V_n \delta_{k',k-Kn}$$

but Hermitian **H** requires $H^{(1)}_{kk'} = H^{(1)*}_{k'k}$

$$\therefore \sum_n V_n \delta_{k',k-Kn} = \sum_n V_n^* \delta_{k',k+Kn}$$

true if $V_n = V^*_{-n}$.

So now that we have the matrix elements of $\mathbf{H}^{(0)}$ and $\mathbf{H}^{(1)}$, the problem is essentially solved. All that remains is to plug into perturbation theory and arrange the results.

3. Solve for $\psi_k = \psi^{(0)}_k + \psi^{(1)}_k$

$$\psi^{(0)}_k = L^{-1/2} e^{ikx}$$

$\psi^{(0)}_{k'}$ ↙ $H^{(1)}_{k,k'}$

$$\psi^{(1)}_k = L^{-1/2} \sum_n{}' \frac{H^{(1)}_{kk'} e^{ik'x}}{E^{(0)}_k - E^{(0)}_{k'}} = L^{-1/2} \sum_n{}' \frac{V_n \delta_{k',k-Kn} e^{ik'x}}{E^{(0)}_k - E^{(0)}_{k-Kn}} \quad (\Sigma' \text{ means } k' \neq k)$$

$k' = k - Kn$

$$\psi^{(1)}_k = L^{-1/2} \sum_n{}' \frac{V_n e^{i(k-Kn)x}}{E^{(0)}_k - E^{(0)}_{k-Kn}}$$

← imposing Kronecker δ-fn restriction on $e^{ik'x}$

Now be careful to express $\psi^{(1)*}_k$ correctly.

$$\psi^{(1)*}_k = L^{-1/2} \sum_n{}' \frac{V^*_n e^{-i(k-Kn)x}}{E^{(0)}_k - E^{(0)}_{k-Kn}}$$

$V^*_n = V_{-n}$

$$\psi^{(1)*}_k = L^{-1/2} \sum_n{}' \frac{V_{-n} e^{-i(k-Kn)x}}{E^{(0)}_k - E^{(0)}_{k-Kn}} = L^{-1/2} \sum_{-n}{}' \frac{V_n e^{-i(k+Kn)x}}{E^{(0)}_k - E^{(0)}_{k+Kn}}$$

↑ replace n by –n

But n is just a dummy index and sum is $-\infty$ to $\infty$, so we can replace $-n$ by $n$.

4. Use $\psi_k$ and $\psi_k^*$ to compute $E_k = E_k^{(0)} + E_k^{(1)} + E_k^{(2)}$.

Rather than use the usual formula for $E^{(2)}$, go back to the $\lambda^n$ formulation of perturbation theory.

$$E_k = \lambda^0 E_k^{(0)} + \lambda^1 E_k^{(1)} + \lambda^2 E_k^{(2)} = \left\langle \psi_k \middle| \lambda^0 \mathbf{H}^{(0)} + \lambda^1 \mathbf{H}^{(1)} \middle| \psi_k \right\rangle$$

Retain terms only through $\lambda^2$

$$E_k = \frac{1}{L}\int_0^L dx \left[ \underbrace{e^{-ikx}}_{\psi_k^{(0)*}} + \lambda \overbrace{\sum_n{}' \frac{V_n e^{-i(k+Kn)x}}{E_k^{(0)} - E_{k+Kn}^{(0)}}}^{\psi_k^{(1)*}} \right] \left[ \underbrace{-\frac{\hbar^2}{2m}\frac{d^2}{dx^2}}_{\mathbf{H}^{(0)}} + \underbrace{\lambda \sum_m V_m e^{iKmx}}_{\mathbf{H}^{(1)}} \right]$$
$$\times \left[ \underbrace{e^{ikx}}_{\psi_k^{(0)}} + \underbrace{\lambda \sum_{n'}{}' \frac{V_{n'} e^{i(k-Kn')x}}{E_k^{(0)} - E_{k-Kn'}^{(0)}}}_{\psi_k^{(1)}} \right]$$

$$E_k^{(0)} = \lambda^0 \frac{1}{L}\left[ -\frac{\hbar^2}{2m}\left(-k^2\right) L \right] = \lambda^0 \frac{\hbar^2 k^2}{2m} \qquad \left[ \text{recall } \frac{d^2}{dx^2} e^{ikx} = -k^2 e^{ikx} \right]$$

$$E_k^{(1)} = \lambda^1 \frac{1}{L}\left[ \int dx\, e^{-ikx} \sum_m e^{iKmx} V_m e^{ikx} + 2 \text{ terms involving } \left( -\frac{\hbar^2}{2m}\frac{d^2}{dx^2} \right) \right],$$

and also retain one of the $\sum'$ sums (that excludes $n = 0$)

We will get to the $\lambda^2$ equation on the next page.

For $E_k^{(1)}$**:** for 1st term, only the $m = 0$ term in the sum gives a nonzero integral. For 2nd terms, need an $n$ or $n' = 0$ term from the sum, but these are excluded by the $\sum'$ sums.

$$E_k^{(1)} = \lambda^1 \frac{1}{L} L V_0 = \lambda^1 V_0$$

This is basically telling us the location of the bottom of the band relative to "vacuum".

Three ways to get $\lambda^2$ terms, two ways involve the $\mathbf{H}^{(1)}$ term, and one involves the $\mathbf{H}^{(0)}$ term. The $\mathbf{H}^{(0)}$ term requires $n = -n'$ ..

$$E_k^{(2)} = \frac{1}{L}\lambda^2 \left[ \int dx\, e^{-ikx} \sum_{m=-\infty}^{\infty} V_m e^{iKmx} \sum_{n'=-\infty}^{\infty}{}' \frac{V_{n'} e^{i(k-Kn')x}}{E_k^{(0)} - E_{k-Kn'}^{(0)}} \right.$$

$$\left. + \int dx \sum_{n\neq 0}{}' \frac{V_n e^{-i(k+Kn)x}}{E_k^{(0)} - E_{k+Kn}^{(0)}} \left( \sum_m V_m e^{iKm} \right) e^{ikx} \right]$$

needed to make integral non-zero

1st term $0 = -k + Km + k - Kn'$, requires m = n′

2nd term $0 = -k - Kn + Km + k$, requires m = n

$$E_k^{(2)} = \frac{1}{L}\lambda^2 \left[ \int dx \sum_m{}' \frac{V_m^2}{E_k^{(0)} - E_{k-Km}^{(0)}} + \int dx \sum_m{}' \frac{V_m^2}{E_k^{(0)} - E_{k+Km}^{(0)}} \right]$$

These are both the same sum, the $\int dx$ integral gives $L$. Now we simplify this.

$$E_k^{(2)} = 2\lambda^2 \sum_{n=-\infty}^{\infty}{}' \frac{V_n^2}{E_k^{(0)} - E_{k+Kn}^{(0)}}$$

Combine terms for $n$ and $-n$ and sum $\sum_{n=1}^{\infty}$

Solve for the energy denominator terms

$$E_k^{(0)} - E_{k\pm Kn}^{(0)} = \frac{\hbar^2}{2m}\left[k^2 - (k \pm Kn)^2\right] = -\frac{\hbar^2 Kn}{2m}\left[Kn \pm 2k\right]$$

Combine the energy denominators

$$\frac{1}{E_k^{(0)} - E_{k+Kn}^{(0)}} + \frac{1}{E_k^{(0)} - E_{k-Kn}^{(0)}} = \frac{4m}{\hbar^2}\frac{1}{K^2n^2 - 4k^2}$$

Assemble $E_k^{(2)}$

$$E_k^{(2)} = \frac{-8m}{\hbar^2}\sum_{n=1}^{\infty}\ \frac{V_n^2}{K^2n^2 - 4k^2}.$$

*But there are many zeroes in this denominator as n* goes from $1 \rightarrow \infty$.

Must use degenerate perturbation theory for *each* small denominator.

Recall, for the 2-level problem

$$\begin{pmatrix} E_k & V \\ V & E_{k'} \end{pmatrix} \longrightarrow E_{\pm} = \frac{E_k + E_{k'}}{2} \pm \left[\left(\frac{E_k - E_{k'}}{2}\right)^2 + V^2\right]^{1/2}$$

$$E_k = \frac{\hbar^2 k^2}{2m} + V_0 - \frac{8m}{\hbar^2}\sum_{n=1}^{\infty} \frac{V_n^2}{K^2 n^2 - 4k^2}$$

Must be negative near $k = 0$ because the lowest states are always pushed to lower energy.

zeroes at $k = \pm\frac{Kn}{2} = \pm\frac{2\pi}{2\ell}n = \pm\frac{n\pi}{\ell}$, except $n = 0$.

At $k = 0$, there are no nearby zeroes

$$\left.\frac{dE_k}{dk}\right|_{k=0} = \frac{\hbar^2 k}{m}$$ (minimum at $k = 0$)

$$\left.\frac{d^2 E_k}{dk^2}\right|_{k=0} = \frac{\hbar^2}{m}$$ (positive curvature)

just like free particle!

At $k = \pm\frac{K}{2}$, there are zeroes in the denominator, so there are gaps in the energy:

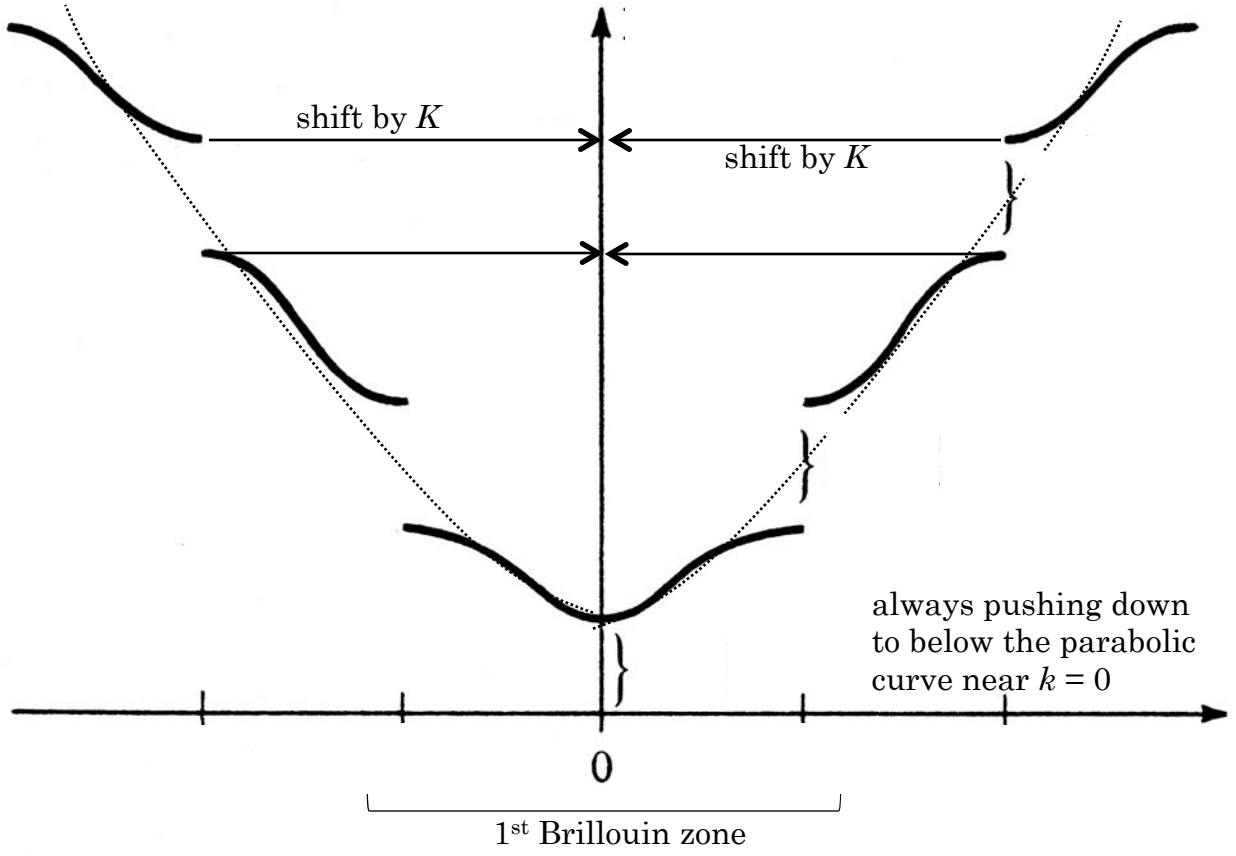
$2|V_1|$ at $k = \pm\frac{K}{2}$

$2|V_2|$ at $k = \pm K$

$\vdots$

$2|V_n|$ at $k = \pm\frac{nK}{2}$

What does this look like?



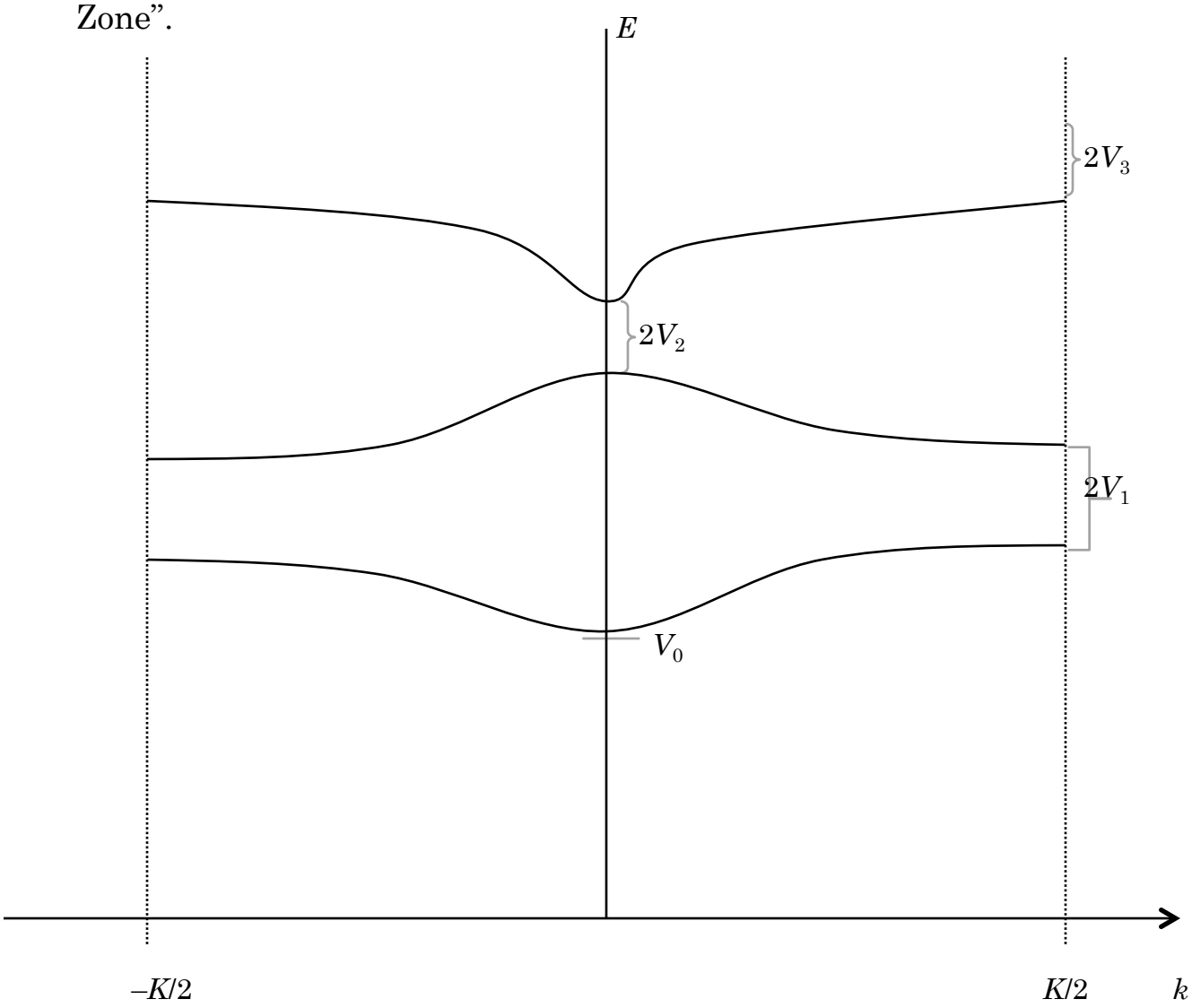

$$E = V_0 + \left( \frac{\hbar^2}{2m} \right) k^2$$

look at text Baym "Lectures on Quantum Mechanics," Benjamin (1981), page 240.

$k = \frac{\pi}{\ell}$ for the lowest energy segment of the $E_k(k)$ curve.

We know that all $\psi$'s have been generated within $-\frac{\pi}{2\ell} \leq k \leq \frac{\pi}{2\ell}$, but there are some different values of $E$ for the same $k$ at each discontinuity.

But we want to shift each of the segments by an integer times $K$ to the left or right so that each shift within the $-\frac{K}{2} \leq k \leq \frac{K}{2}$ "First Brilouin Zone".




$E$ vs. $k$ diagram. Curvature gives $m_{\text{eff}}$.

3-D diagram — gives much more information. Tells us where to find transitions allowed as a function of 3-D $\vec{k}$ vector in "reciprocal lattice" of lattice vector $\vec{K}$.

Scattering of free particle off lattice. Conservation of momentum in the sense $\vec{k}_{\text{final}} - \vec{k}_{\text{initial}} = \vec{K}$.

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